THE ARITHMETICA

BOOK I

PRELIMINARY

Dedication.

"Knowing, my most esteemed friend Dionysius, that you are anxious to learn how to investigate problems in numbers, I have tried, beginning from the foundations on which the science is built up, to set forth to you the nature and power subsisting in numbers.

"Perhaps the subject will appear rather difficult, inasmuch as it is not yet familiar (beginners are, as a rule, too ready to despair of success); but you, with the impulse of your enthusiasm and the benefit of my teaching, will find it easy to master; for eagerness to learn, when seconded by instruction, ensures rapid progress."

After the remark that "all numbers are made up of some multitude of units, so that it is manifest that their formation is subject to no limit," Diophantus proceeds to define what he calls the different "species" of numbers, and to describe the abbreviative signs used to denote them. These "species" are, in the first place, the various powers of the unknown quantity from the second to the sixth inclusive, the unknown quantity itself, and units.

Definitions.

A *square* ($=x^2$) is *δύναμις* ("power"), and its sign is a Δ with Y superposed, thus Δ^Y.

A *cube* ($=x^3$) is *κύβος*, and its sign K^Y.

A *square-square* ($=x^4$) is *δυναμοδύναμις*[1], and its sign is $\Delta^Y\Delta$.

A *square-cube* ($=x^5$) is *δυναμόκυβος*, and its sign ΔK^Y.

A *cube-cube* ($=x^6$) is *κυβόκυβος*, and its sign K^YK.

[1] The term *δυναμοδύναμις* was already used by Heron (*Metrica*, ed. Schöne, p. 48, 11, 19) for the fourth power of a side of a triangle.

"It is," Diophantus observes, "from the addition, subtraction or multiplication of these numbers or from the ratios which they bear to one another or to their own sides respectively that most arithmetical problems are formed"; and "each of these numbers... is recognised as an element in arithmetical inquiry."

"But the number which has none of these characteristics, but *merely has in it an indeterminate multitude of units* (πλῆθος μονάδων ἀόριστον) *is called* ἀριθμός, '*number*,' *and its sign is* ς [$=x$]."

"And there is also another sign denoting that which is invariable in determinate numbers, namely the unit, the sign being M with o superposed, thus $\overset{o}{M}$."

Next follow the definitions of the reciprocals, the names of which are derived from the names of the corresponding species themselves.

Thus

from	ἀριθμός [x]	we derive the term		ἀριθμοστόν [$=1/x$]
„	δύναμις [x^2]	„	„	δυναμοστόν [$=1/x^2$]
„	κύβος [x^3]	„	„	κυβοστόν [$=1/x^3$]
„	δυναμοδύναμις [x^4]	„	„	δυναμοδυναμοστόν [$=1/x^4$]
„	δυναμόκυβος [x^5]	„	„	δυναμοκυβοστόν [$=1/x^5$]
„	κυβόκυβος [x^6]	„	„	κυβοκυβοστόν [$=1/x^6$],

and each of these has the same sign as the corresponding original species, but with a distinguishing mark which Tannery writes in the form $^{\chi}$ above the line to the right.

Thus $\Delta^{Y\chi} = 1/x^2$, just as $\gamma^{\chi} = \frac{1}{3}$.

Sign of Subtraction (*minus*).

"*A minus multiplied by a minus makes a plus*[1]*; a minus multiplied by a plus makes a minus; and the sign of a minus is a truncated* Ψ *turned upside down, thus* ⋔."

Diophantus proceeds: "It is well that one who is beginning this study should have acquired practice in the addition, subtraction and multiplication of the various species. He should know how to add positive and negative terms with different coefficients to

[1] The literal rendering would be "A wanting multiplied by a wanting makes a forthcoming." The word corresponding to *minus* is λεῖψις ("wanting"): when it is used exactly as our *minus* is, it is in the dative λείψει, but there is some doubt whether Diophantus himself used this form (cf. p. 44 above). For the probable explanation of the sign, see pp. 42–44. The word for "forthcoming" is ὕπαρξις, from ὑπάρχω, to exist. Negative terms are λείποντα εἴδη, and positive ὑπάρχοντα.

other terms[1], themselves either positive or likewise partly positive and partly negative, and how to subtract from a combination of positive and negative terms other terms either positive or likewise partly positive and partly negative.

"Next, if a problem leads to an equation in which certain terms are equal to terms of the same species but with different coefficients, it will be necessary to subtract like from like on both sides, until one term is found equal to one term. If by chance there are on either side or on both sides any negative terms, it will be necessary to add the negative terms on both sides, until the terms on both sides are positive, and then again to subtract like from like until one term only is left on each side.

"This should be the object aimed at in framing the hypotheses of propositions, that is to say, to reduce the equations, if possible, until one term is left equal to one term; *but I will show you later how, in the case also where two terms are left equal to one term, such a problem is solved.*"

Diophantus concludes by explaining that, in arranging the mass of material at his disposal, he tried to distinguish, so far as possible, the different types of problems, and, especially in the elementary portion at the beginning, to make the more simple lead up to the more complex, in due order, such an arrangement being calculated to make the beginner's course easier and to fix what he learns in his memory. The treatise, he adds, has been divided into thirteen Books.

PROBLEMS

1. To divide a given number into two having a given difference.

Given number 100, given difference 40.

Lesser number required x. Therefore

$$2x + 40 = 100,$$

$$x = 30.$$

The required numbers are 70, 30.

2. To divide a given number into two having a given ratio.

Given number 60, given ratio 3 : 1.

Two numbers x, $3x$. Therefore $x = 15$.

The numbers are 45, 15.

[1] εἶδος, "species," is the word used by Diophantus throughout.

3. To divide a given number into two numbers such that one is a given ratio of the other *plus* a given difference[1].

Given number 80, ratio $3:1$, difference 4.

Lesser number x. Therefore the larger is $3x+4$, and $4x+4=80$, so that $x=19$.

The numbers are **61, 19.**

4. To find two numbers in a given ratio and such that their difference is also given.

Given ratio $5:1$, given difference 20.

Numbers $5x$, x. Therefore $4x=20$, $x=5$, and the numbers are **25, 5.**

5. To divide a given number into two numbers such that given fractions (not the same) of each number when added together produce a given number.

Necessary condition. The latter given number must be such that it lies between the numbers arising when the given fractions respectively are taken of the first given number.

First given number 100, given fractions $\frac{1}{3}$ and $\frac{1}{5}$, given sum of fractions 30.

Second part $5x$. Therefore first part $= 3(30-x)$.

Hence $90+2x=100$, and $x=5$.

The required parts are **75, 25.**

6. To divide a given number into two numbers such that a given fraction of the first exceeds a given fraction of the other by a given number.

Necessary condition. The latter number must be less than that which arises when that fraction of the first number is taken which exceeds the other fraction.

Given number 100, given fractions $\frac{1}{4}$ and $\frac{1}{6}$ respectively, given excess 20.

Second part $6x$. Therefore the first part is $4(x+20)$.

Hence $10x+80=100$, $x=2$, and the parts are **88, 12.**

[1] Literally "to divide an assigned number into two in a given ratio and difference (ἐν λόγῳ καὶ ὑπεροχῇ τῇ δοθείσῃ)." The phrase means the same, though it is not so clear, as Euclid's expression (*Data*, Def. 11 and *passim*) δοθέντι μείζων ἢ ἐν λόγῳ. According to Euclid's definition a magnitude is greater than a magnitude "by a given amount (more) than in a (certain) ratio" when the remainder of the first magnitude, after subtracting the given amount, has the said ratio to the second magnitude. This means that, if x, y are the magnitudes, d the given amount, and k the ratio, $x-d=ky$ or $x=ky+d$.

7. From the same (required) number to subtract two given numbers so as to make the remainders have to one another a given ratio.

Given numbers 100, 20, given ratio 3 : 1.

Required number x. Therefore $x - 20 = 3(x - 100)$, and
$x = 140$.

8. To two given numbers to add the same (required) number so as to make the resulting numbers have to one another a given ratio.

Necessary condition. The given ratio must be less than the ratio which the greater of the given numbers has to the lesser.

Given numbers 100, 20, given ratio 3 : 1.

Required number x. Therefore $3x + 60 = x + 100$, and
$x = 20$.

9. From two given numbers to subtract the same (required) number so as to make the remainders have to one another a given ratio.

Necessary condition. The given ratio must be greater than the ratio which the greater of the given numbers has to the lesser.

Given numbers 20, 100, given ratio 6 : 1.

Required number x. Therefore $120 - 6x = 100 - x$, and
$x = 4$.

10. Given two numbers, to add to the lesser and to subtract from the greater the same (required) number so as to make the sum in the first case have to the difference in the second case a given ratio.

Given numbers 20, 100, given ratio 4 : 1.

Required number x. Therefore $(20 + x) = 4(100 - x)$, and
$x = 76$.

11. Given two numbers, to add the first to, and subtract the second from, the same (required) number, so as to make the resulting numbers have to one another a given ratio.

Given numbers 20, 100, given ratio 3 : 1.

Required number x. Therefore $3x - 300 = x + 20$, and
$x = 160$.

12. To divide a given number twice into two numbers such that the first of the first pair may have to the first of the second pair a given ratio, and also the second of the second pair to the second of the first pair another given ratio.

Given number 100, ratio of greater of first parts to lesser of second 2 : 1, and ratio of greater of second parts to lesser of first parts 3 : 1.

x lesser of second parts.

The parts then are

$$\left.\begin{matrix} 2x \\ 100-2x \end{matrix}\right\} \text{ and } \left.\begin{matrix} 300-6x \\ x \end{matrix}\right\}.$$

Therefore $300 - 5x = 100$, $x = 40$, and

the parts are (80, 20), (60, 40).

13. To divide a given number thrice into two numbers such that one of the first pair has to one of the second pair a given ratio, the second of the second pair to one of the third pair another given ratio, and the second of the third pair to the second of the first pair another given ratio.

Given number 100, ratio of greater of first parts to lesser of second 3 : 1, of greater of second to lesser of third 2 : 1, and of greater of third to lesser of first 4 : 1.

x lesser of third parts.

Therefore greater of second parts $= 2x$, lesser of second $= 100 - 2x$, greater of first $= 300 - 6x$.

Hence lesser of first $= 6x - 200$, so that greater of third $= 24x - 800$.

Therefore $25x - 800 = 100$, $x = 36$, and

the respective divisions are (84, 16), (72, 28), (64, 36).

14. To find two numbers such that their product has to their sum a given ratio. [One is arbitrarily assumed.]

Necessary condition. The assumed value of one of the two must be greater than the number representing the ratio[1].

Ratio 3 : 1, x one of the numbers, 12 the other (> 3).

Therefore $12x = 3x + 36$, $x = 4$, and

the numbers are 4, 12.

15. To find two numbers such that each after receiving from the other a given number may bear to the remainder a given ratio.

Let the first receive 30 from the second, the ratio being then 2 : 1, and the second 50 from the first, the ratio being then 3 : 1; take $x + 30$ for the second.

[1] Literally "the number homonymous with the given ratio."

Therefore the first $= 2x - 30$, and

$$(x + 80) = 3\,(2x - 80).$$

Thus $x = 64$, and

the numbers are **98, 94**.

16. To find three numbers such that the sums of pairs are given numbers.

Necessary condition. Half the sum of the three given numbers must be greater than any one of them singly.

Let $(1) + (2) = 20$, $(2) + (3) = 30$, $(3) + (1) = 40$.

x the sum of the three. Therefore the numbers are

$$x - 30,\quad x - 40,\quad x - 20.$$

The sum $x = 3x - 90$, and $x = 45$.

The numbers are **15, 5, 25**.

17. To find four numbers such that the sums of all sets of three are given numbers.

Necessary condition. One-third of the sum of the four must be greater than any one singly.

Sums of threes 22, 24, 27, 20 respectively.

x the sum of all four. Therefore the numbers are

$$x - 22,\quad x - 24,\quad x - 27,\quad x - 20.$$

Therefore $4x - 93 = x$, $x = 31$, and

the numbers are **9, 7, 4, 11**.

18. To find three numbers such that the sum of any pair exceeds the third by a given number.

Given excesses 20, 30, 40.

$2x$ the sum of all three.

We have $(1) + (2) = (3) + 20$.

Adding (3) to each side, we have: twice $(3) + 20 = 2x$, and $(3) = x - 10$.

Similarly the numbers (1) and (2) are $x - 15$, $x - 20$ respectively.

Therefore $3x - 45 = 2x$, $x = 45$, and

the numbers are **30, 25, 35**.

[*Otherwise thus*[1]. As before, if the third number (3) is x,

$$(1) + (2) = x + 20.$$

Next, if we add the equations

$$\left.\begin{array}{l}(1) + (2) - (3) = 20 \\ (2) + (3) - (1) = 30\end{array}\right\},$$

[1] Tannery attributes the alternative solution of I. 18 (as of I. 19) to an old scholiast.

we have $(2) = \frac{1}{2}(20 + 30) = 25.$

Hence $(1) = x - 5.$

Lastly $(3) + (1) - (2) = 40,$

or $2x - 5 - 25 = 40.$

Therefore $x = 35.$

The numbers are 30, 25, 35.]

19. To find four numbers such that the sum of any three exceeds the fourth by a given number.

Necessary condition. Half the sum of the four given differences must be greater than any one of them.

Given differences 20, 30, 40, 50.

$2x$ the sum of the required numbers. Therefore the numbers are

$$x - 15,\ x - 20,\ x - 25,\ x - 10.$$

Therefore $4x - 70 = 2x$, and $x = 35$.

The numbers are 20, 15, 10, 25.

[*Otherwise thus*[1]. If the fourth number (4) is x,

$$(1) + (2) + (3) = x + 20.$$

Put $(2) + (3)$ equal to half the sum of the two excesses 20 and 30, *i.e.* 25 [this is equivalent to adding the two equations

$$(1) + (2) + (3) - (4) = 20,$$
$$(2) + (3) + (4) - (1) = 30].$$

It follows by subtraction that $(1) = x - 5$.

Next we add the equations beginning with (2) and (3) respectively, and we obtain

$$(3) + (4) = \tfrac{1}{2}(30 + 40) = 35,$$

so that $(3) = 35 - x.$

It follows that $(2) = x - 10.$

Lastly, since $(4) + (1) + (2) - (3) = 50,$

$$3x - 15 - (35 - x) = 50, \text{ and } x = 25.$$

The numbers are accordingly 20, 15, 10, 25.]

20. To divide a given number into three numbers such that the sum of each extreme and the mean has to the other extreme a given ratio.

Given number 100; and let $(1) + (2) = 3 \cdot (3)$ and $(2) + (3) = 4 \cdot (1)$.

[1] Tannery attributes the alternative solution of I. 19 (as of I. 18) to an old scholiast.

x the third number. Thus the sum of the first and second $= 3x$, and the sum of the three $= 4x = 100$.

Hence $x = 25$, and the sum of the first two $= 75$.

Let y be the first[1]. Therefore sum of second and third $= 4y$, $5y = 100$ and $y = 20$.

The required parts are **20, 55, 25.**

21. To find three numbers such that the greatest exceeds the middle number by a given fraction of the least, the middle exceeds the least by a given fraction of the greatest, but the least exceeds a given fraction of the middle number by a given number.

Necessary condition. The middle number must exceed the least by such a fraction of the greatest that, if its denominator[2] be multiplied into the excess of the middle number over the least, the coefficient of x in the product is greater than the coefficient of x in the expression for the middle number resulting from the assumptions made[3].

Suppose greatest exceeds middle by $\frac{1}{3}$ of least, middle exceeds least by $\frac{1}{3}$ of greatest, and least exceeds $\frac{1}{3}$ of middle by 10. [Diophantus assumes the three given fractions or submultiples to be one and the same.]

$x + 10$ the least. Therefore middle $= 3x$, and greatest $= 6x - 30$.

Hence, lastly, $6x - 30 - 3x = \frac{1}{3}(x + 10)$,

or $\qquad x + 10 = 9x - 90$, and $x = 12\frac{1}{2}$.

The numbers are **45, 37½, 22½.**

[1] As already remarked (p. 52), Diophantus does not use a second symbol for the second unknown, but makes ἀριθμός do duty for the second as well as for the first.

[2] "Denominator," literally the "number homonymous with the fraction," *i.e.* the denominator on the assumption that the fraction is, or is expressed as, a submultiple.

[3] Wertheim points out that this condition has reference, not to the general solution of the problem, but to the general applicability of the particular procedure which Diophantus adopts in his solution. Suppose X, Y, Z required such that $X - Y = Z/m$, $Y - Z = X/n$, $Z - a = Y/p$. Diophantus assumes $Z = x + a$, whence $Y = px$, $X = n(px - x - a)$. The condition states that $np - n > p$. If we solve for x by substituting the values of X, Y, Z in the first equation, we in fact obtain

$$m\{(np - n - p)x - na\} = x + a,$$

or

$$x(mnp - mn - mp - 1) = a(mn + 1).$$

In order that the value of x may be positive, we must have $mnp > mn + mp + 1$, that is,

$$np > n + p + \frac{1}{m}$$

or (if m, n, p are positive *integers*) $np > n + p$.

[Another solution[1].

Necessary condition. The given fraction of the greatest must be such that, when it is added to the least, the coefficient of x in the sum is less than the coefficient of x in the expression for the middle number resulting from the assumptions made[2].

Let the least number be $x + 10$, as before, and the given fraction $\frac{1}{3}$; the middle number is therefore $3x$.

Next, greatest = middle + $\frac{1}{3}$ (least) $= 3\frac{1}{3}x + 3\frac{1}{3}$.

Lastly, $3x = x + 10 + \frac{1}{3}(3\frac{1}{3}x + 3\frac{1}{3})$

$= 2\frac{1}{9}x + 11\frac{1}{9}$.

Therefore $x = 12\frac{1}{2}$, and

the numbers are, as before, 45, $37\frac{1}{2}$, $22\frac{1}{2}$.]

22. To find three numbers such that, if each give to the next following a given fraction of itself, in order, the results after each has given and taken may be equal.

Let first give $\frac{1}{3}$ of itself to second, second $\frac{1}{4}$ of itself to third, third $\frac{1}{5}$ of itself to first.

Assume first to be a number of x's divisible by 3, say $3x$, and second to be a number of *units* divisible by 4, say 4.

Therefore second after giving and taking becomes $x + 3$.

Hence the first also after giving and taking must become $x + 3$; it must therefore have taken $x + 3 - 2x$, or $3 - x$; $3 - x$ must therefore be $\frac{1}{5}$ of third, or third $= 15 - 5x$.

Lastly, $15 - 5x - (3 - x) + 1 = x + 3$,

or $13 - 4x = x + 3$, and $x = 2$.

The numbers are 6, 4, 5.

23. To find four numbers such that, if each give to the next following a given fraction of itself, the results may all be equal.

Let first give $\frac{1}{3}$ of itself to second, second $\frac{1}{4}$ of itself to third, third $\frac{1}{5}$ of itself to fourth, and fourth $\frac{1}{6}$ of itself to first.

Assume first to be a number of x's divisible by 3, say $3x$, and second to be a number of *units* divisible by 4, say 4.

[1] Tannery attributes this alternative solution, like the others of the same kind, to an ancient scholiast.

[2] Wertheim observes that the scholiast's necessary condition comes to the same thing as Diophantus' own.

The second after giving and taking becomes $x+3$.

Therefore first after giving x to second and receiving $\frac{1}{6}$ of fourth $=x+3$; therefore fourth
$$=6(x+3-2x)=18-6x.$$

But fourth after giving $3-x$ to first and receiving $\frac{1}{5}$ of third $=x+3$; therefore third $=30x-60$.

Lastly, third after giving $6x-12$ to fourth and receiving 1 from second $=x+3$.

That is, $24x-47=x+3$, and $x=\frac{50}{23}$.

The numbers are therefore $\frac{150}{23}$, 4, $\frac{120}{23}$, $\frac{114}{23}$; or, after multiplying by the common denominator, **150, 92, 120, 114.**

24. To find three numbers such that, if each receives a given fraction of the sum of the other two, the results are all equal.

Let first receive $\frac{1}{3}$ of (second + third), second $\frac{1}{4}$ of (third + first), and third $\frac{1}{5}$ of (first + second).

Assume first $=x$, and for convenience' sake (*τοῦ προχείρου ἕνεκεν*) take for sum of second and third a number of *units* divisible by 3, say 3.

Then sum of the three $=x+3$,

and first $+\frac{1}{3}$ (second + third) $=x+1$.

Therefore second $+\frac{1}{4}$ (third + first) $=x+1$;

hence 3 times second + sum of all $=4x+4$,

and therefore second $=x+\frac{1}{3}$.

Lastly, third $+\frac{1}{5}$ (first + second) $=x+1$,

or 4 times third + sum of all $=5x+5$,

and third $=x+\frac{1}{2}$.

Therefore $x+(x+\frac{1}{3})+(x+\frac{1}{2})=x+3$,

and $x=\frac{13}{12}$.

The numbers, after multiplying by the common denominator, are **13, 17, 19.**

25. To find four numbers such that, if each receives a given fraction of the sum of the remaining three, the four results are equal.

Let first receive $\frac{1}{3}$ of the rest, second $\frac{1}{4}$ of the rest, third $\frac{1}{5}$ of rest, and fourth $\frac{1}{6}$ of rest.

Assume first to be x and sum of rest a number of units divisible by 3, say 3.

Then sum of all $=x+3$.

Now first $+\frac{1}{3}$ (second + third + fourth) $=x+1$.

Therefore second + $\frac{1}{4}$ (third + fourth + first) $= x + 1$,
whence 3 times second + sum of all $= 4x + 4$,
and therefore second $= x + \frac{1}{3}$.
Similarly third $= x + \frac{1}{2}$,
and fourth $= x + \frac{3}{5}$.
Adding, we have $4x + \frac{43}{30} = x + 3$,
and $x = \frac{47}{90}$.

The numbers, after multiplying by a common denominator, are 47, 77, 92, 101.

26. Given two numbers, to find a third number which, when multiplied into the given numbers respectively, makes one product a square and the other the side of that square.

Given numbers 200, 5; required number x.
Therefore $200x = (5x)^2$, and
$x = 8$.

27. To find two numbers such that their sum and product are given numbers.

Necessary condition. The square of half the sum must exceed the product by a square number. ἔστι δὲ τοῦτο πλασματικόν[1].

Given sum 20, given product 96.
$2x$ the difference of the required numbers.
Therefore the numbers are $10 + x$, $10 - x$.
Hence $100 - x^2 = 96$.
Therefore $x = 2$, and
the required numbers are 12, 8.

28. To find two numbers such that their sum and the sum of their squares are given numbers.

Necessary condition. Double the sum of their squares must exceed the square of their sum by a square. ἔστι δὲ καὶ τοῦτο πλασματικόν[1].

[1] There has been controversy as to the meaning of this difficult phrase. Xylander, Bachet, Cossali, Schulz, Nesselmann, all discuss it. Xylander translated it by "effictum aliunde." Bachet of course rejects this, and, while leaving the word untranslated, maintains that it has an active rather than a passive signification; it is, he says, not something "made up" (effictum) but something "a quo aliud quippiam effingi et plasmari potest," "from which something else can be made up," and this he interprets as meaning that from the conditions to which the term is applied, combined with the solutions of the respective problems in which it occurs, the rules for solving mixed quadratics can be evolved. Of the two views I think Xylander's is nearer the mark. πλασματικόν should apparently mean "of the nature of a πλάσμα," just as δραματικόν means something connected with or suitable for a drama; and πλάσμα means something

Given sum 20, given sum of squares 208.
Difference $2x$.
Therefore the numbers are $10+x$, $10-x$.
Thus $200+2x^2=208$, and $x=2$.

The required numbers are **12, 8.**

29. To find two numbers such that their sum and the difference of their squares are given numbers.

Given sum 20, given difference of squares 80.
Difference $2x$.
The numbers are therefore $10+x$, $10-x$.

Hence $$(10+x)^2-(10-x)^2=80,$$ or $40x=80$, and $x=2$.

The required numbers are **12, 8.**

30. To find two numbers such that their difference and product are given numbers.

Necessary condition. Four times the product together with the square of the difference must give a square. *ἔστι δὲ καὶ τοῦτο πλασματικόν.*

Given difference 4, given product 96.
$2x$ the sum of the required numbers.
Therefore the numbers are $x+2$, $x-2$; accordingly $x^2-4=96$, and $x=10$.

The required numbers are **12, 8.**

31. To find two numbers in a given ratio and such that the sum of their squares also has to their sum a given ratio.

Given ratios $3:1$ and $5:1$ respectively.
Lesser number x.
Therefore $10x^2=5\cdot 4x$, whence $x=2$, and
the numbers are **2, 6.**

32. To find two numbers in a given ratio and such that the sum of their squares also has to their difference a given ratio.

Given ratios $3:1$ and $10:1$.
Lesser number x, which is then found from the equation $10x^2=10\cdot 2x$.
Hence $x=2$, and
the numbers are **2, 6.**

"formed" or "moulded." Hence the expression would seem to mean "this is of the nature of a formula," with the implication that the formula is not difficult to make up or discover. Nesselmann, like Xylander, gives it much this meaning, translating it "das lässt sich aber bewerkstelligen." Tannery translates πλασματικόν by "formativum."

33. To find two numbers in a given ratio and such that the difference of their squares also has to their sum a given ratio.

Given ratios 3 : 1 and 6 : 1.

Lesser number x, which is found to be 3.

The numbers are 3, 9.

34. To find two numbers in a given ratio and such that the difference of their squares also has to their difference a given ratio.

Given ratios 3 : 1 and 12 : 1.

Lesser number x, which is found to be 3.

The numbers are 3, 9.

Similarly by the same method can be found two numbers in a given ratio and (1) such that their product is to their sum in a given ratio, or (2) such that their product is to their difference in a given ratio.

35. To find two numbers in a given ratio and such that the square of the lesser also has to the greater a given ratio.

Given ratios 3 : 1 and 6 : 1 respectively.

Lesser number x, which is found to be 18.

The numbers are 18, 54.

36. To find two numbers in a given ratio and such that the square of the lesser also has to the lesser itself a given ratio.

Given ratios 3 : 1 and 6 : 1.

Lesser number x, which is found to be 6.

The numbers are 6, 18.

37. To find two numbers in a given ratio and such that the square of the lesser also has to the sum of both a given ratio.

Given ratios 3 : 1 and 2 : 1.

Lesser number x, which is found to be 8.

The numbers are 8, 24.

38. To find two numbers in a given ratio and such that the square of the lesser also has to the difference between them a given ratio.

Given ratios 3 : 1 and 6 : 1.

Lesser number x, which is found to be 12.

The numbers are 12, 36.

Similarly can be found two numbers in a given ratio and

(1) such that the square of the greater also has to the lesser a given ratio, or

(2) such that the square of the greater also has to the greater itself a given ratio, or

(3) such that the square of the greater also has to the sum or difference of the two a given ratio.

39. Given two numbers, to find a third such that the sums of the several pairs multiplied by the corresponding third number give three numbers in arithmetical progression.

Given numbers 3, 5.

Required number x.

The three products are therefore $3x+15$, $5x+15$, $8x$.

Now $3x+15$ must be either the middle or the least of the three, and $5x+15$ either the greatest or the middle.

(1) $5x+15$ greatest, $3x+15$ least.

Therefore $5x+15+3x+15=2.8x$, and

$$x=\frac{15}{4}.$$

(2) $5x+15$ greatest, $3x+15$ middle.

Therefore $(5x+15)-(3x+15)=3x+15-8x$, and

$$x=\frac{15}{7}.$$

(3) $8x$ greatest, $3x+15$ least.

Therefore $8x+3x+15=2(5x+15)$, and

$$x=15.$$

BOOK II

[The first five problems of this Book are mere repetitions of problems in Book I. They probably found their way into the text from some ancient commentary. In each case the ratio of one required number to the other is assumed to be 2 : 1. The enunciations only are here given.]

1. To find two numbers such that their sum is to the sum of their squares in a given ratio [cf. I. 31].

2. To find two numbers such that their difference is to the difference of their squares in a given ratio [cf. I. 34].

3. To find two numbers such that their product is to their sum or their difference in a given ratio [cf. I. 34].

4. To find two numbers such that the sum of their squares is to their difference in a given ratio [cf. I. 32].

5. To find two numbers such that the difference of their squares is to their sum in a given ratio [cf. I. 33].

6[1]. To find two numbers having a given difference and such that the difference of their squares exceeds their difference by a given number.

Necessary condition. The square of their difference must be less than the sum of the said difference and the given excess of the difference of the squares over the difference of the numbers.

Difference of numbers 2, the other given number 20.

Lesser number x. Therefore $x+2$ is the greater, and

$$4x+4=22.$$

Therefore $x=4\frac{1}{2}$, and

the numbers are $4\frac{1}{2}$, $6\frac{1}{2}$.

7[1]. To find two numbers such that the difference of their squares is greater by a given number than a given ratio of their difference[2]. [*Difference assumed.*]

Necessary condition. The given ratio being $3:1$, the square of the difference of the numbers must be less than the sum of three times that difference and the given number.

Given number 10, difference of required numbers 2.

Lesser number x. Therefore the greater is $x+2$, and

$$4x+4=3\cdot 2+10.$$

Therefore $x=3$, and

the numbers are 3, 5.

8. To divide a given square number into two squares[3].

[1] The problems II. 6, 7 also are considered by Tannery to be interpolated from some ancient commentary.

[2] Here we have the identical phrase used in Euclid's *Data* (cf. note on p. 132 above): the difference of the squares is *τῆς ὑπεροχῆς αὐτῶν δοθέντι ἀριθμῷ μείζων ἢ ἐν λόγῳ*, literally "greater than their difference by a given number (more) than in a (given) ratio," by which is meant "greater by a given number than a given proportion or fraction of their difference."

[3] It is to this proposition that Fermat appended his famous note in which he enunciates what is known as the "great theorem" of Fermat. The text of the note is as follows:

"On the other hand it is impossible to separate a cube into two cubes, or a

Given square number 16.

x^2 one of the required squares. Therefore $16-x^2$ must be equal to a square.

Take a square of the form[1] $(mx-4)^2$, m being any integer and 4 the number which is the square root of 16, *e.g.* take $(2x-4)^2$, and equate it to $16-x^2$.

Therefore $4x^2-16x+16=16-x^2$,

or $5x^2=16x$, and $x=\frac{16}{5}$.

The required squares are therefore $\frac{256}{25}$, $\frac{144}{25}$.

9. To divide a given number which is the sum of two squares into two other squares[2].

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biquadrate into two biquadrates, or generally *any power except a square into two powers with the same exponent.* I have discovered a truly marvellous proof of this, which however the margin is not large enough to contain."

Did Fermat really possess a proof of the general proposition that $x^m+y^m=z^m$ cannot be solved in rational numbers where m is any number >2? As Wertheim says, one is tempted to doubt this, seeing that, in spite of the labours of Euler, Lejeune-Dirichlet, Kummer and others, a general proof has not even yet been discovered. Euler proved the theorem for $m=3$ and $m=4$, Dirichlet for $m=5$, and Kummer, by means of the higher theory of numbers, produced a proof which only excludes certain particular values of m, which values are rare, at all events among the smaller values of m; thus there is no value of m below 100 for which Kummer's proof does not serve. (I take these facts from Weber and Wellstein's *Encyclopädie der Elementar-Mathematik*, I_2, p. 284, where a proof of the formula for $m=4$ is given.)

It appears that the Göttingen Academy of Sciences has recently awarded a prize to Dr A. Wieferich, of Münster, for a proof that the equation $x^p+y^p=z^p$ cannot be solved in terms of positive integers not multiples of p, if 2^p-2 is not divisible by p^2. "This surprisingly simple result represents the first advance, since the time of Kummer, in the proof of the last Fermat theorem" (*Bulletin of the American Mathematical Society*, February 1910).

Fermat says ("Relation des nouvelles découvertes en la science des nombres," August 1659, *Oeuvres*, II. p. 433) that he proved that *no cube is divisible into two cubes* by a variety of his method of *infinite diminution* (*descente infinie* or *indéfinie*) different from that which he employed for other negative or positive theorems; as to the other cases, see Supplement, sections I., II.

[1] Diophantus' words are: "I form the square from any number of ἀριθμοί *minus* as many units as there are in the side of 16." It is implied throughout that m must be so chosen that the result may be *rational* in Diophantus' sense, *i.e.* rational and positive.

[2] Diophantus' solution is substantially the same as Euler's (*Algebra*, tr. Hewlett, Part II. Art. 219), though the latter is expressed more generally.

Required to find x, y such that

$$x^2+y^2=f^2+g^2.$$

If $x \gtreqless f$, then $y \lesseqgtr g$.

Put therefore $$x=f+pz, \quad y=g-qz:$$

Given number $13 = 2^2 + 3^2$.

As the roots of these squares are 2, 3, take $(x+2)^2$ as the first square and $(mx-3)^2$ as the second (where m is an integer), say $(2x-3)^2$.

Therefore $(x^2+4x+4)+(4x^2+9-12x)=13$,

or $$5x^2+13-8x=13.$$

Therefore $x=\frac{8}{5}$, and

the required squares are $\frac{324}{25}$, $\frac{1}{25}$.

10. To find two square numbers having a given difference.

Given difference 60.

Side of one number x, side of the other x *plus* any number the square of which is not greater than 60, say 3.

Therefore $$(x+3)^2-x^2=60;$$

$x=8\frac{1}{2}$, and

the required squares are $72\frac{1}{4}$, $132\frac{1}{4}$.

11. To add the same (required) number to two given numbers so as to make each of them a square.

(1) Given numbers 2, 3; required number x.

Therefore $\left.\begin{matrix} x+2 \\ x+3 \end{matrix}\right\}$ must both be squares.

This is called a double-equation (διπλοϊσότης).

To solve it, *take the difference between the two expressions and resolve it into two factors*[1]; in this case let us say 4, $\frac{1}{4}$.

Then *take either*

(*a*) *the square of half the difference between these factors and equate it to the lesser expression,*

or (*b*) *the square of half the sum and equate it to the greater.*

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hence $$2fpz+p^2z^2-2gqz+q^2z^2=0,$$

and $$z=\frac{2gq-2fp}{p^2+q^2},$$

so that $$x=\frac{2gpq+f(q^2-p^2)}{p^2+q^2},\quad y=\frac{2fpq+g(p^2-q^2)}{p^2+q^2},$$

in which we may substitute all possible numbers for p, q.

[1] Here, as always, the factors chosen must be suitable factors, *i.e.* such as will lead to a "rational" result, in Diophantus' sense.

In this case (a) the square of half the difference is $\frac{225}{64}$.

Therefore $x+2=\frac{225}{64}$, and $x=\frac{97}{64}$, the squares being $\frac{225}{64}$, $\frac{289}{64}$.

Taking (b) the square of half the sum, we have $x+3=\frac{289}{64}$, which gives the same result.

(2) To avoid a double-equation[1],

first find a number which when added to 2, or to 3, gives a square.

Take *e.g.* the number x^2-2, which when added to 2 gives a square.

Therefore, since this same number added to 3 gives a square,

$$x^2+1=\text{a square}=(x-4)^2, \text{ say,}$$

the number of units in the expression (in this case 4) being so taken that the solution may give $x^2>2$.

Therefore $x=\frac{15}{8}$, and

the required number is $\frac{97}{64}$, as before.

12. To subtract the same (required) number from two given numbers so as to make both remainders squares.

Given numbers 9, 21.

Assuming $9-x^2$ as the required number, we satisfy one condition, and the other requires that $12+x^2$ shall be a square.

Assume as the side of this square x *minus* some number the square of which >12, say 4.

Therefore $\quad (x-4)^2=12+x^2$,

and $\quad x=\frac{1}{2}$.

The required number is then $8\frac{3}{4}$.

[Diophantus does not reduce to lowest terms, but says $x=\frac{4}{8}$ and then subtracts $\frac{16}{64}$ from 9 or $\frac{576}{64}$.]

[1] This is the same procedure as that of Euler, who does not use double-equations. Euler (*Algebra*, tr. Hewlett, Part II. Art. 214) solves the problem

$$\left.\begin{aligned}x+4&=u^2\\x+7&=v^2\end{aligned}\right\}.$$

Suppose $\quad x+4=p^2$;

therefore $\quad x=p^2-4$, and $x+7=p^2+3$.

Suppose that $\quad p^2+3=(p+q)^2$;

therefore $\quad p=(3-q^2)/2q$.

Thus $\quad x=(9-22q^2+q^4)/4q^2$,

or, if we take a fraction r/s instead of q,

$$x=(9s^4-22r^2s^2+r^4)/4r^2s^2.$$

13. From the same (required) number to subtract two given numbers so as to make both remainders squares.

Given numbers 6, 7.

(1) Let x be the required number.

Therefore $\left.\begin{array}{l} x-6 \\ x-7 \end{array}\right\}$ are both squares.

The difference is 1, which is the product of, say, 2 and $\frac{1}{2}$; and, by the rule for solving a double equation,

$$x-7=\tfrac{9}{16}, \text{ and } x=\frac{121}{16}.$$

(2) To avoid a double-equation, seek a number which exceeds a square by 6, say x^2+6.

Therefore x^2-1 must also be a square $=(x-2)^2$, say.

Therefore $x=\frac{5}{4}$, and

the required number is $\dfrac{121}{16}$.

14. To divide a given number into two parts and to find a square which when added to each of the two parts gives a square number.

Given number 20.

Take two numbers[1] such that the sum of their squares <20, say 2, 3.

[1] Diophantus implies here that the two numbers chosen *must* be such that the sum of their squares <20. Tannery pointed out (*Bibliotheca Mathematica*, 1887, p. 103) that this is not so and that the condition actually necessary to ensure a real solution in Diophantus' sense is something different. We have to solve the equations

$$x+y=a, \quad z^2+x=u^2, \quad z^2+y=v^2.$$

We assume $u=z+m$, $v=z+n$, and, eliminating x, y, we obtain

$$z=\frac{a-(m^2+n^2)}{2(m+n)}.$$

In order that z may be positive, we must have $m^2+n^2<a$; but z need not be positive in order to satisfy the above equations. What is really required is that x, y shall both be positive.

Now from the above we derive

$$x-y=(u^2-v^2)=2z(m-n)+m^2-n^2$$
$$=\frac{(m-n)(a+2mn)}{m+n}.$$

Solving for x, y, we have

$$x=\frac{m(a+mn-n^2)}{m+n}, \quad y=\frac{n(a+mn-m^2)}{m+n}.$$

If, of the two assumed numbers, $m>n$, the condition necessary to secure that x, y shall both be positive is $a+mn>m^2$.

Add x to each and square.

We then have

$$\left.\begin{array}{l} x^2+4x+4 \\ x^2+6x+9 \end{array}\right\},$$

and, if $\left.\begin{array}{l} 4x+4 \\ 6x+9 \end{array}\right\}$ are respectively subtracted, the remainders are the same square.

Let then x^2 be the required square, and we have only to make $\left.\begin{array}{l} 4x+4 \\ 6x+9 \end{array}\right\}$ the required parts of 20.

Thus $$10x+13=20,$$

and $$x=\tfrac{7}{10}.$$

The required parts are then $\left(\frac{68}{10}, \frac{132}{10}\right)$, and the required square is $\frac{49}{100}$.

15. To divide a given number into two parts and to find a square which, when each part is respectively subtracted from it, gives a square.

Given number 20.

Take $(x+m)^2$ for the required square[1], where m^2 is not greater than 20,

e.g. take $(x+2)^2$.

This leaves a square if $\left.\begin{array}{ll} \text{either} & 4x+4 \\ \text{or} & 2x+3 \end{array}\right\}$ is subtracted.

Let these then be the parts of 20.

[1] Here again the implied condition, namely that m^2 is not greater than 20, is not necessary; the condition necessary for a real solution is something different.

The equations to be solved are $x+y=a$, $z^2-x=u^2$, $z^2-y=v^2$.

Diophantus here puts $(\xi+m)^2$ for z^2, so that, if $x=2m\xi+m^2$, the second equation is satisfied. Now $(\xi+m)^2-y$ must also be a square, and if this square is equal to $(\xi+m-n)^2$, say, we must have

$$y=2n\xi+2mn-n^2.$$

Therefore, since $x+y=a$,

$$2(m+n)\xi+m^2+2mn-n^2=a,$$

whence $$\xi=\frac{a-m^2+n^2-2mn}{2(m+n)},$$

and it follows that

$$x=\frac{m(a-mn+n^2)}{m+n}, \quad y=\frac{n(a-mn+m^2)}{m+n}.$$

If $m>n$, it is necessary, in order that x, y may both be positive, that $a+n^2>mn$, which is the true condition for a real solution.

Therefore $6x+7=20$, and $x=\frac{13}{6}$.

The required parts are therefore $\left(\frac{76}{6}, \frac{44}{6}\right)$, and the required square is $\frac{625}{36}$.

16. To find two numbers in a given ratio and such that each when added to an assigned square gives a square.

Given square 9, given ratio $3:1$.

If we take a square of side $(mx+3)$ and subtract 9 from it, the remainder may be taken as one of the numbers required.

Take, *e.g.*, $(x+3)^2-9$, or x^2+6x, for the lesser number.

Therefore $3x^2+18x$ is the greater number, and $3x^2+18x+9$ must be made a square $=(2x-3)^2$, say.

Therefore $x=30$, and

the required numbers are 1080, 3240.

17. To find three numbers such that, if each give to the next following a given fraction of itself and a given number besides, the results after each has given and taken may be equal[1].

First gives to second $\frac{1}{5}$ of itself $+6$, second to third $\frac{1}{6}$ of itself $+7$, third to first $\frac{1}{7}$ of itself $+8$.

Let first and second be $5x$, $6x$ respectively.

When second has taken $x+6$ from first it becomes $7x+6$, and when it has given $x+7$ to third it becomes $6x-1$.

But first when it has given $x+6$ to second becomes $4x-6$; and this too when it has taken $\frac{1}{7}$ of third $+8$ must become $6x-1$.

Therefore $\frac{1}{7}$ of third $+8=2x+5$, and

$$\text{third}=14x-21.$$

Next, third after receiving $\frac{1}{6}$ of second $+7$ and giving $\frac{1}{7}$ of itself $+8$ must become $6x-1$.

Therefore $13x-19=6x-1$, and $x=\frac{18}{7}$.

The required numbers are $\frac{90}{7}, \frac{108}{7}, \frac{105}{7}$.

[1] Tannery is of opinion that the problems II. 17 and 18 have crept into the text from an ancient commentary to Book I. to which they would more appropriately belong. Cf. I. 22, 23.

18. To divide a given number into three parts satisfying the conditions of the preceding problem[1].

Given number 80.

Let first give to second $\frac{1}{5}$ of itself $+ 6$, second to third $\frac{1}{6}$ of itself $+ 7$, and third to first $\frac{1}{7}$ of itself $+ 8$.

[What follows in the text is not a solution of the problem but an alternative solution of the preceding. The first two numbers are assumed to be $5x$ and 12, and the numbers found are $\frac{170}{19}, \frac{228}{19}, \frac{217}{19}$.]

19. To find three squares such that the difference between the greatest and the middle has to the difference between the middle and the least a given ratio.

Given ratio 3 : 1.

Assume the least square $= x^2$, the middle $= x^2 + 2x + 1$.

Therefore the greatest $= x^2 + 8x + 4 =$ square $= (x + 3)^2$, say.

Thus $x = 2\frac{1}{2}$, and

the squares are $30\frac{1}{4}$, $12\frac{1}{4}$, $6\frac{1}{4}$.

20. To find two numbers such that the square of either added to the other gives a square[2].

[1] Though the solution is not given in the text, it is easily obtained from the *general* solution of the preceding problem, which again, at least with our notation, is easy.

Let us assume, with Wertheim, that the numbers required in II. 17 are $5x$, $6y$, $7z$. Then by the conditions of the problem

$$4x - 6 + z + 8 = 5y - 7 + x + 6 = 6z - 8 + y + 7,$$

from which two equations we can find x, z in terms of y.

In fact $\quad x = (26y - 18)/19$ and $z = (17y - 3)/19$,

and the general solution is

$$5(26y - 18)/19, \quad 6y, \quad 7(17y - 3)/19.$$

$\left[\text{In his solution Diophantus assumes } x = y, \text{ whence } y = \frac{18}{7}\right]$.

Now, to solve II. 18, we have only to equate the sum of the three expressions to 80, and so find y.

We have

$$y(5 \cdot 26 + 6 \cdot 19 + 7 \cdot 17) - 5 \cdot 18 - 7 \cdot 3 = 80 \cdot 19, \quad y = \frac{1631}{363};$$

and the required numbers are

$$\frac{9440}{363}, \quad \frac{9786}{363}, \quad \frac{9814}{363}.$$

[2] Euler (*Algebra*, Part II. Art. 239) solves this problem more generally thus.

Required to find x, y such that $x^2 + y$ and $y^2 + x$ are squares.

If we begin by supposing $x^2 + y = p^2$, so that $y = p^2 - x^2$, and then substitute the value of y in terms of x in the second expression, we must have

$$p^4 - 2p^2x^2 + x^4 + x = \text{square}.$$

But, as this is difficult to solve, let us suppose instead that

$$x^2 + y = (p - x)^2 = p^2 - 2px + x^2,$$

Assume for the numbers x, $2x+1$, which by their form satisfy one condition.

The other condition gives

$$4x^2+5x+1=\text{square}=(2x-2)^2, \text{ say.}$$

Therefore $x=\frac{3}{13}$, and

the numbers are $\frac{3}{13}, \frac{19}{13}$.

21. To find two numbers such that the square of either *minus* the other number gives a square.

$x+1$, $2x+1$ are assumed, satisfying one condition.

The other condition gives

$$4x^2+3x=\text{square}=9x^2, \text{ say.}$$

Therefore $x=\frac{3}{5}$, and

the numbers are $\frac{8}{5}, \frac{11}{5}$.

22. To find two numbers such that the square of either added to the sum of both gives a square.

Assume x, $x+1$ for the numbers. Thus one condition is satisfied.

It remains that

$$x^2+4x+2=\text{square}=(x-2)^2, \text{ say.}$$

Therefore $x=\frac{1}{4}$, and

the numbers are $\frac{1}{4}, \frac{5}{4}$.

[Diophantus has $\frac{2}{8}$, $\frac{10}{8}$.]

23. To find two numbers such that the square of either *minus* the sum of both gives a square.

Assume x, $x+1$ for the numbers, thus satisfying one condition.

Then $x^2-2x-1=\text{square}=(x-3)^2$, say.

Therefore $x=2\frac{1}{2}$, and

the numbers are $2\frac{1}{2}, 3\frac{1}{2}$.

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and that $$y^2+x=(q-y)^2=q^2-2qy+y^2.$$

It follows that $$y+2px=p^2,$$ $$x+2qy=q^2,$$

whence $$x=\frac{2qp^2-q^2}{4pq-1}, \quad y=\frac{2pq^2-p^2}{4pq-1}.$$

Suppose, for example, $p=2$, $q=3$, and we have $x=\frac{15}{23}$, $y=\frac{32}{23}$; and so on. We must of course choose p, q such that x, y are both positive. Diophantus' solution is obtained by putting $p=-1$, $q=3$.

24. To find two numbers such that either added to the square of their sum gives a square.

Since x^2+3x^2, x^2+8x^2 are both squares, let the numbers be $3x^2$, $8x^2$ and their sum x.

Therefore $121x^4=x^2$, whence $11x^2=x$, and $x=\frac{1}{11}$.

The numbers are therefore $\frac{3}{121}$, $\frac{8}{121}$.

25. To find two numbers such that the square of their sum *minus* either number gives a square.

If we subtract 7 or 12 from 16, we get a square.

Assume then $12x^2$, $7x^2$ for the numbers, and $16x^2$ for the square of their sum.

Hence $19x^2=4x$, and $x=\frac{4}{19}$.

The numbers are $\frac{192}{361}$, $\frac{112}{361}$.

26. To find two numbers such that their product added to either gives a square, and the sides of the two squares added together produce a given number.

Let the given number be 6.

Since $x(4x-1)+x$ is a square, let x, $4x-1$ be the numbers.

Therefore $4x^2+3x-1$ is a square, and the side of this square must be $6-2x$ [since $2x$ is the side of the first square and the sum of the sides of the square is 6].

Since $$4x^2+3x-1=(6-2x)^2,$$

we have $x=\frac{37}{27}$, and

the numbers are $\frac{37}{27}$, $\frac{121}{27}$.

27. To find two numbers such that their product *minus* either gives a square, and the sides of the two squares so arising when added together produce a given number.

Let the given number be 5.

Assume $4x+1$, x for the numbers, so that one condition is satisfied.

Also $$4x^2-3x-1=(5-2x)^2.$$

Therefore $x=\frac{26}{17}$, and

the numbers are $\frac{26}{17}$, $\frac{121}{17}$.

28. To find two square numbers such that their product added to either gives a square.

Let the numbers[1] be x^2, y^2.

Therefore $\left.\begin{matrix} x^2y^2+y^2 \\ x^2y^2+x^2 \end{matrix}\right\}$ are both squares.

To make the first expression a square we make x^2+1 a square, putting

$$x^2+1=(x-2)^2, \text{ say.}$$

Therefore $x=\frac{3}{4}$, and $x^2=\frac{9}{16}$.

We have now to make $\frac{9}{16}(y^2+1)$ a square [and y must be different from x].

Put $$9y^2+9=(3y-4)^2, \text{ say,}$$

and $y=\frac{7}{24}$.

Therefore the numbers are $\frac{9}{16}, \frac{49}{576}$.

29. To find two square numbers such that their product *minus* either gives a square.

Let x^2, y^2 be the numbers.

Then $\left.\begin{matrix} x^2y^2-y^2 \\ x^2y^2-x^2 \end{matrix}\right\}$ are both squares.

A solution of $x^2-1=$(a square) is $x^2=\frac{25}{16}$.

We have now to solve

$$\tfrac{25}{16}y^2-\tfrac{25}{16}=\text{a square.}$$

Put $$y^2-1=(y-4)^2, \text{ say.}$$

Therefore $y=\frac{17}{8}$, and

the numbers are $\frac{289}{64}, \frac{100}{64}$.

30. To find two numbers such that their product $\pm$ their sum gives a square.

Now $m^2+n^2\pm 2mn$ is a square.

Put 2, 3, say, for m, n respectively, and of course

$$2^2+3^2\pm 2.2.3 \text{ is a square.}$$

Assume then product of numbers $=(2^2+3^2)x^2$ or $13x^2$, and sum $=2.2.3x^2$ or $12x^2$.

The product being $13x^2$, let x, $13x$ be the numbers.

Therefore their sum $14x=12x^2$, and $x=\frac{7}{6}$.

The numbers are therefore $\frac{7}{6}, \frac{91}{6}$.

[1] Diophantus does not use two unknowns, but assumes the numbers to be x^2 and 1 until he has found x. Then he uses the same unknown (x) to find what he had first taken to be unity, as explained above, p. 52. The same remark applies to the next problem.

31. To find two numbers such that their sum is a square and their product $\pm$ their sum gives a square.

$2 \cdot 2m \cdot m =$ a square, and $(2m)^2 + m^2 \pm 2 \cdot 2m \cdot m =$ a square.

If $m = 2$, $4^2 + 2^2 \pm 2 \cdot 4 \cdot 2 = 36$ or 4.

Let then the product of the numbers be $(4^2 + 2^2)x^2$ or $20x^2$, and their sum $2 \cdot 4 \cdot 2x^2$ or $16x^2$, and take $2x$, $10x$ for the numbers.

Then $\qquad 12x = 16x^2$, and $x = \frac{3}{4}$.

The numbers are $\frac{6}{4}, \frac{30}{4}$.

32. To find three numbers such that the square of any one of them added to the next following gives a square.

Let the first be x, the second $2x + 1$, and the third $2(2x + 1) + 1$ or $4x + 3$, so that two conditions are satisfied.

The last condition gives $(4x + 3)^2 + x =$ square $= (4x - 4)^2$, say.

Therefore $x = \frac{7}{57}$, and

the numbers are $\frac{7}{57}, \frac{71}{57}, \frac{199}{57}$.

33. To find three numbers such that the square of any one of them *minus* the next following gives a square.

Assume $x + 1$, $2x + 1$, $4x + 1$ for the numbers, so that two conditions are satisfied.

Lastly, $\quad 16x^2 + 7x =$ square $= 25x^2$, say,

and $\qquad x = \frac{7}{9}$.

The numbers are $\frac{16}{9}, \frac{23}{9}, \frac{37}{9}$.

34. To find three numbers such that the square of any one added to the sum of all three gives a square.

$\{\frac{1}{2}(m - n)\}^2 + mn$ is a square. Take a number separable into two factors (m, n) in three ways, say 12, which is the product of $(1, 12)$, $(2, 6)$ and $(3, 4)$.

The values then of $\frac{1}{2}(m - n)$ are $5\frac{1}{2}$, 2, $\frac{1}{2}$.

Take $5\frac{1}{2}x$, $2x$, $\frac{1}{2}x$ for the numbers, and for their sum $12x^2$.

Therefore $8x = 12x^2$, and $x = \frac{2}{3}$.

The numbers are $\frac{11}{3}, \frac{4}{3}, \frac{1}{3}$.

[Diophantus says $\frac{4}{6}$, and $\frac{22}{6}, \frac{8}{6}, \frac{2}{6}$.]

35. To find three numbers such that the square of any one *minus* the sum of all three gives a square.

$\{\frac{1}{2}(m+n)\}^2 - mn$ is a square. Take, as before, a number divisible into factors in three ways, as 12.

Let then $6\frac{1}{2}x$, $4x$, $3\frac{1}{2}x$ be the numbers, and their sum $12x^2$.

Therefore $14x = 12x^2$, and $x = \frac{7}{6}$.

The numbers are $\frac{45\frac{1}{2}}{6}, \frac{28}{6}, \frac{24\frac{1}{2}}{6}$.

BOOK III

1. To find three numbers such that, if the square of any one of them be subtracted from the sum of all three, the remainder is a square[1].

Take two squares x^2, $4x^2$; the sum is $5x^2$.

If then we take $5x^2$ as the sum of the three numbers, and x, $2x$ as two of them, we satisfy two conditions.

Next divide 5, which is the sum of two squares, into two other squares $\frac{4}{25}$, $\frac{121}{25}$ [II. 9], and assume $\frac{2}{5}x$ for the third number.

Therefore $x + 2x + \frac{2}{5}x = 5x^2$, and $x = \frac{17}{25}$.

The numbers are $\frac{17}{25}, \frac{34}{25}, \frac{34}{125}$.

[Diophantus writes $\frac{85}{125}$ for x and $\frac{85}{125}, \frac{170}{125}, \frac{34}{125}$ for the numbers.]

2. To find three numbers such that the square of the sum of all three added to any one of them gives a square.

Let the square of the sum of all three be x^2, and the numbers $3x^2$, $8x^2$, $15x^2$.

Hence $26x^2 = x$, $x = \frac{1}{26}$, and

the numbers are $\frac{3}{676}, \frac{8}{676}, \frac{15}{676}$.

3. To find three numbers such that the square of the sum of all three *minus* any one of them gives a square.

Sum of all three $4x$, its square $16x^2$, the numbers $7x^2$, $12x^2$, $15x^2$.

Then $34x^2 = 4x$, $x = \frac{2}{17}$, and

the numbers are $\frac{28}{289}, \frac{48}{289}, \frac{60}{289}$.

[1] The fact that the problems III. 1–4 are very like II. 34, 35 makes Tannery suspect that they have found their way into the text from some ancient commentary.

4. To find three numbers such that, if the square of their sum be subtracted from any one of them, the remainder is a square.

Sum x, numbers $2x^2$, $5x^2$, $10x^2$.

Then $17x^2 = x$, $x = \frac{1}{17}$, and

the numbers are $\frac{2}{289}$, $\frac{5}{289}$, $\frac{10}{289}$.

5. To find three numbers such that their sum is a square and the sum of any pair exceeds the third by a square.

Let the sum of the three be $(x+1)^2$; let first + second = third + 1, so that third $= \frac{1}{2}x^2 + x$; let second + third = first $+ x^2$, so that first $= x + \frac{1}{2}$.

Therefore second $= \frac{1}{2}x^2 + \frac{1}{2}$.

It remains that first + third = second + a square.

Therefore $2x$ = square = 16, say, and $x = 8$.

The numbers are $8\frac{1}{2}$, $32\frac{1}{2}$, 40.

Otherwise thus[1].

First find three squares such that their sum is a square.

Find *e.g.* what square number + 4 + 9 gives a square, that is, 36;

4, 36, 9 are therefore squares with the required property.

Next find three numbers such that the sum of each pair = the third + a given number; in this case suppose

first + second − third = 4,
second + third − first = 9,
third + first − second = 36.

This problem has already been solved [I. 18].

The numbers are respectively 20, $6\frac{1}{2}$, $22\frac{1}{2}$.

[1] We should naturally suppose that this alternative solution, like others, was interpolated. But we are reluctant to think so because the solution is so elegant that it can hardly be attributed to a scholiast. If the solution is not genuine, we have here an illustration of the truth that, however ingenious they are, Diophantus' solutions are not always the best imaginable (Loria, *Le scienze esatte nell' antica Grecia*, Libro v. pp. 138–9). In this case the more elegant solution is the alternative one. Generalised, it is as follows. We have to find x, y, z so that

$$\left.\begin{array}{r} -x+y+z=\text{a square} \\ x-y+z=\text{a square} \\ x+y-z=\text{a square} \end{array}\right\},$$

and also $$x+y+z=\text{a square}.$$

We have only to equate the first three expressions to squares a^2, b^2, c^2 such that $a^2+b^2+c^2$ = a square, k^2 say, since the sum of the first three expressions is itself $x+y+z$.

The solution is then

$$x=\frac{1}{2}(b^2+c^2),\quad y=\frac{1}{2}(c^2+a^2),\quad z=\frac{1}{2}(a^2+b^2).$$

6. To find three numbers such that their sum is a square and the sum of any pair is a square.

Let the sum of all three be x^2+2x+1, sum of first and second x^2, and therefore the third $2x+1$; let sum of second and third be $(x-1)^2$.

Therefore the first $=4x$, and the second $=x^2-4x$.

But first + third = square,

that is, $6x+1=\text{square}=121$, say.

Therefore $x=20$, and

the numbers are **80, 320, 41.**

[An alternative solution, obviously interpolated, is practically identical with the above except that it takes the square 36 as the value of $6x+1$, so that $x=\frac{35}{6}$, and the numbers are $\frac{140}{6}$ $=\frac{840}{36}, \frac{385}{36}, \frac{456}{36}$.]

7. To find three numbers in A.P. such that the sum of any pair gives a square.

First find three square numbers in A.P. and such that half their sum is greater than any one of them. Let x^2, $(x+1)^2$ be the first and second of these; therefore the third is $x^2+4x+2=(x-8)^2$, say.

Therefore $x=\frac{62}{20}$ or $\frac{31}{10}$;

and we may take as the numbers 961, 1681, 2401.

We have now to find three numbers such that the sums of pairs are the numbers just found.

The sum of the three $=\frac{5043}{2}=2521\frac{1}{2}$, and

the three numbers are **$120\frac{1}{2}$, $840\frac{1}{2}$, $1560\frac{1}{2}$.**

8. Given one number, to find three others such that the sum of any pair of them added to the given number gives a square, and also the sum of the three added to the given number gives a square.

Given number 3.

Suppose first required number + second $=x^2+4x+1$,

second + third $=x^2+6x+6$,

sum of all three $=x^2+8x+13$.

Therefore third $=4x+12$, second $=x^2+2x-6$, first $=2x+7$.

Also first + third + 3 = a square,

that is, $6x+22=\text{square}=100$, suppose.

Hence $x=13$, and

the numbers are **33, 189, 64.**

9. Given one number, to find three others such that the sum of any pair of them *minus* the given number gives a square, and also the sum of the three *minus* the given number gives a square.

Given number 3.

Suppose first of required numbers + second $= x^2 + 3$,

second + third $= x^2 + 2x + 4$,

sum of the three $= x^2 + 4x + 7$.

Therefore third $= 4x + 4$, second $= x^2 - 2x$, first $= 2x + 3$.

Lastly, first + third $- 3 = 6x + 4 =$ a square $= 64$, say.

Therefore $x = 10$, and

(23, 80, 44) is a solution.

10. To find three numbers such that the product of any pair of them added to a given number gives a square.

Let the given number be 12. Take a square (say 25) and subtract 12. Take the difference (13) for the product of the first and second numbers, and let these numbers be $13x$, $1/x$ respectively.

Again subtract 12 from another square, say 16, and let the difference (4) be the product of the second and third numbers.

Therefore the third number $= 4x$.

The third condition gives $52x^2 + 12 =$ a square; now $52 = 4 . 13$, and 13 is not a square; but, if it were a square, the equation could easily be solved[1].

Thus we must find two numbers to replace 13 and 4 such that their product is a square, while either + 12 is also a square.

Now the product is a square if both are squares; hence we must find two squares such that either $+ 12 =$ a square.

"This is easy[2] and, as we said, it makes the equation easy to solve."

The squares 4, $\frac{1}{4}$ satisfy the condition.

[1] The equation $52x^2 + 12 = u^2$ can in reality be solved as it stands, by virtue of the fact that it has one obvious solution, namely $x = 1$. Another solution is found by substituting $y + 1$ for x, and so on. Cf. pp. 69, 70 above. The value $x = 1$ itself gives (13, 1, 4) as a solution of the problem.

[2] The method is indicated in II. 34. We have to find two pairs of squares differing by 12. (*a*) If we put $12 = 6 . 2$, we have

$$\left\{\frac{1}{2}(6-2)\right\}^2 + 12 = \left\{\frac{1}{2}(6+2)\right\}^2,$$

and 16, 4 are squares differing by 12, or 4 is a square which when added to 12 gives a square. (*b*) If we put $12 = 4 . 3$, we find $\left\{\frac{1}{2}(4-3)\right\}^2$ or $\frac{1}{4}$ to be a square which when added to 12 gives a square.

Retracing our steps, we now put $4x$, $1/x$ and $x/4$ for the numbers, and we have to solve the equation

$$x^2 + 12 = \text{square} = (x+3)^2, \text{ say.}$$

Therefore $x = \frac{1}{2}$, and

$\left(2, 2, \frac{1}{8}\right)$ is a solution[1].

11. To find three numbers such that the product of any pair *minus* a given number gives a square.

Given number 10.

Put product of first and second $=$ a square $+ 10 = 4 + 10$, say, and let first $= 14x$, second $= 1/x$.

Let product of second and third $=$ a square $+ 10 = 19$, say; therefore third $= 19x$.

By the third condition, $266x^2 - 10$ must be a square; but 266 is not a square[2].

Therefore, as in the preceding problem, we must find two squares each of which exceeds a square by 10.

The squares $30\frac{1}{4}$, $12\frac{1}{4}$ satisfy these conditions[3].

Putting now $30\frac{1}{4}x$, $1/x$, $12\frac{1}{4}x$ for the numbers, we have, by the third condition, $370\frac{9}{16}x^2 - 10 =$ square [for $370\frac{9}{16}$ Diophantus writes $370\frac{1}{2}\frac{1}{16}$];

therefore $5929x^2 - 160 = \text{square} = (77x - 2)^2$, say.

Therefore $x = \frac{41}{77}$, and

the numbers are $\frac{1240\frac{1}{4}}{77}, \frac{77}{41}, \frac{502\frac{1}{4}}{77}$.

[1] Euler (*Algebra*, Part II. Art. 232) has an elegant solution of this problem in whole numbers. Let it be required to find x, y, z such that $xy+a$, $yz+a$, $zx+a$ are all squares. Suppose $xy+a=p^2$, and make $z=x+y+q$;

therefore $\quad xz+a=x^2+xy+qx+a=x^2+qx+p^2$,

and $\quad yz+a=xy+y^2+qy+a=y^2+qy+p^2$;

and the right hand expressions are both squares if $q=\pm 2p$, so that $z=x+y\pm 2p$.

We can therefore take any value for p such that $p^2>a$, split p^2-a into factors, take those factors respectively for the values of x and y, and so find z.

E.g. suppose $a=12$ and $p^2=25$, so that $xy=13$; let $x=1$, $y=13$, and we have $z=14\pm 10=24$ or 4, and (1, 13, 4), (1, 13, 24) are solutions.

[2] As a matter of fact, the equation $266x^2-10=u^2$ can be solved as it stands, since it has one obvious solution, namely $x=1$. (Cf. pp. 69, 70 above and note on preceding problem, p. 159.) The value $x=1$ gives (14, 1, 19) as a solution of the problem.

[3] Tannery brackets the passage in the text in which these squares are found, on the ground that, as the solution was not given in the corresponding place of III. 10, there was no necessity to give it here. 10 and 1 being factors of 10,

$$\left\{\frac{1}{2}(10-1)\right\}^2+10=\left\{\frac{1}{2}(10+1)\right\}^2;$$

thus $30\frac{1}{4}$ is a square which exceeds a square by 10. Similarly $\left\{\frac{1}{2}(5+2)\right\}^2$ or $12\frac{1}{4}$ is such a square. The latter is found in the text by putting $m^2-10=\text{square}=(m-2)^2$, whence $m=3\frac{1}{2}$, and $m^2=12\frac{1}{4}$.

12. To find three numbers such that the product of any two added to the third gives a square.

Take a square and subtract part of it for the third number; let x^2+6x+9 be one of the sums, and 9 the third number.

Therefore product of first and second $=x^2+6x$; let first $=x$, so that second $=x+6$.

By the two remaining conditions

$$\left.\begin{array}{l} 10x+54 \\ 10x+\ \ 6 \end{array}\right\} \text{ are both squares.}$$

Therefore we have to find two squares differing by 48; "this is easy and can be done in an infinite number of ways."

The squares 16, 64 satisfy the condition. Equating these squares to the respective expressions, we obtain $x=1$, and

the numbers are 1, 7, 9.

13. To find three numbers such that the product of any two *minus* the third gives a square.

First x, second $x+4$; therefore product $=x^2+4x$, and we assume third $=4x$.

Therefore, by the other conditions,

$$\left.\begin{array}{l} 4x^2+15x \\ 4x^2-x-4 \end{array}\right\} \text{ are both squares.}$$

The difference $=16x+4=4(4x+1)$, and we put

$$\{\tfrac{1}{2}(4x+5)\}^2=4x^2+15x.$$

Therefore $x=\frac{25}{20}$, and

the numbers are $\frac{25}{20}, \frac{105}{20}, \frac{100}{20}$.

14. To find three numbers such that the product of any two added to the square of the third gives a square[1].

[1] Wertheim gives a more general solution, as follows. If we take as the required numbers $X=\frac{1}{4}ax$, $Y=ax+b^2$, $Z=\frac{1}{4}b^2$, two conditions are already satisfied, namely $XY+Z^2=$ a square, and $YZ+X^2=$ a square.

It only remains to satisfy the condition $ZX+Y^2=$ a square, or

$$a^2x^2+\frac{33}{16}ab^2x+b^4=\text{a square.}$$

Put $$a^2x^2+\frac{33}{16}ab^2x+b^4=(ax+kb^2)^2,$$

and $$x=\frac{16b^2(k^2-1)}{a(33-32k)},$$

where k remains undetermined.

First x, second $4x+4$, third 1. Two conditions are thus satisfied.

The third condition gives

$$x+(4x+4)^2 = \text{a square} = (4x-5)^2, \text{ say.}$$

Therefore $x=\frac{9}{78}$, and

the numbers (omitting the common denominator) are 9, 328, 73.

15. To find three numbers such that the product of any two added to the sum of those two gives a square[1].

[*Lemma.*] The product of the squares of any two consecutive numbers added to the sum of the said squares gives a square[2].

Let 4, 9 be two of the required numbers, x the third.

Therefore $\left.\begin{matrix}10x+9\\5x+4\end{matrix}\right\}$ are both squares.

The difference $=5x+5=5(x+1)$.

Equating the square of half the sum of the factors to $10x+9$, we have

$$\{\tfrac{1}{2}(x+6)\}^2 = 10x+9.$$

Therefore $x=28$, and (4, 9, 28) is a solution.

[1] The problem can of course be solved more elegantly, with our notation, thus. (The same remark applies to the next problem, III. 16.)

If x, y, z are the required numbers, $xy+x+y$, etc. are to be squares. We may therefore write the conditions in the form

$$(y+1)(z+1) = \text{a square } +1,$$
$$(z+1)(x+1) = \text{a square } +1,$$
$$(x+1)(y+1) = \text{a square } +1.$$

Assuming a^2, b^2, c^2 for the respective squares, and putting $\xi=x+1$, $\eta=y+1$, $\zeta=z+1$, we have to solve

$$\eta\zeta=a^2+1,$$
$$\zeta\xi=b^2+1,$$
$$\xi\eta=c^2+1.$$

[This is practically the same problem as that in the Lemma to Dioph. V. 8.]

Multiplying the second and third equations and dividing by the first, we have

$$\xi=\sqrt{\{(b^2+1)(c^2+1)/(a^2+1)\}},$$

with similar expressions for η, ζ.

x, y, z are these expressions *minus* 1 respectively. a^2, b^2, c^2 must of course be so chosen that the resulting values of ξ, η, ζ may be rational. Cf. Euler, *Commentationes arithmeticae*, II. p. 577.

[2] In fact, $a^2(a+1)^2+a^2+(a+1)^2=\{a(a+1)+1\}^2$.

Otherwise thus[1].

Assume first number to be x, second 3.

Therefore $4x+3=\text{square}=25$ say, whence $x=5\frac{1}{2}$, and $5\frac{1}{2}$, 3 satisfy one condition.

[1] This alternative solution would appear to be undoubtedly genuine.

Diophantus has solved the equations

$$\left.\begin{aligned} yz+y+z&=u^2\\ zx+z+x&=v^2\\ xy+x+y&=w^2\end{aligned}\right\}.$$

Fermat shows how to solve the corresponding problem with *four* numbers instead of three. He uses for this purpose Diophantus' solution of V. 5, namely the problem of finding x^2, y^2, z^2, such that

$$\left.\begin{aligned} &y^2z^2+x^2=r^2, && z^2x^2+y^2=s^2, && x^2y^2+z^2=t^2\\ &y^2z^2+y^2+z^2=u^2, && z^2x^2+z^2+x^2=v^2, && x^2y^2+x^2+y^2=w^2\end{aligned}\right\}.$$

Diophantus finds $\left(\frac{25}{9}, \frac{64}{9}, \frac{196}{9}\right)$ as a solution of the latter problem. Fermat takes these as the first three of the four numbers which are to satisfy the condition that the product of any two plus the sum of those two gives a square, and assumes x for the fourth. Three relations out of six are already satisfied, and the other three require

$$\left.\begin{aligned} &\frac{25}{9}x+x+\frac{25}{9}, \quad \text{or} \quad \frac{34x}{9}+\frac{25}{9}\\ &\frac{64}{9}x+x+\frac{64}{9}, \quad \text{or} \quad \frac{73x}{9}+\frac{64}{9}\\ &\frac{196}{9}x+x+\frac{196}{9}, \quad \text{or} \quad \frac{205x}{9}+\frac{196}{9}\end{aligned}\right\}$$

to be made squares: a "triple-equation" to be solved by Fermat's method. (See the Supplement, section V.)

Fermat does not give the solution, but I had the curiosity to work it out in order to verify to what enormous numbers the method of the triple-equation leads, even in such comparatively simple cases.

We may of course neglect the denominator 9 and solve the equations

$$\begin{aligned} 34x+25&=u^2,\\ 73x+64&=v^2,\\ 205x+196&=w^2.\end{aligned}$$

The method gives

$$x=-\frac{45981859840968844787200}{631629004828419699201},$$

the denominator being equal to $(25132230399)^2$.

Verifying the correctness of the solution, we find that, in fact,

$$\begin{aligned} \frac{34}{25}x+1&=\left(\frac{2505136897}{25132230399}\right)^2,\\ \frac{73}{64}x+1&=\left(\frac{10351251901}{25132230399}\right)^2,\\ \frac{205}{196}x+1&=\left(\frac{12275841601}{25132230399}\right)^2.\end{aligned}$$

Strictly speaking, as the value found for x is negative, we ought to substitute $y-A$ for it (where $-A$ is the value found) in the three equations and start afresh. The portentous numbers which would thus arise must be left to the imagination.

Let the third be x, while $5\frac{1}{2}$, 3 are the first two.

Therefore $\left.\begin{matrix} 4x+3 \\ 6\frac{1}{2}x+5\frac{1}{2} \end{matrix}\right\}$ must both be squares;

but, since the coefficients in one expression are respectively greater than those in the other, but neither of the ratios of corresponding coefficients is that of a square to a square, our suppositions will not serve the purpose; we cannot solve by our method.

Hence (to replace $5\frac{1}{2}$, 3) we must find two numbers such that their product + their sum = a square, and the ratio of the numbers increased by 1 respectively is the ratio of a square to a square.

Let these be y and $4y+3$, which satisfy the latter condition; and, in order that the other may be satisfied, we must have

$$4y^2+8y+3 = \text{square} = (2y-3)^2, \text{ say.}$$

Therefore $$y = \tfrac{3}{10}.$$

Assume now $\frac{3}{10}$, $4\frac{1}{5}$, x for the three numbers.

Therefore $\left.\begin{matrix} 5\frac{1}{5}x+4\frac{1}{5} \\ \frac{13}{10}x+\frac{3}{10} \end{matrix}\right\}$ are both squares,

or, if we multiply by 25 and 100 respectively,

$$\left.\begin{matrix} 130x+105 \\ 130x+\ \ 30 \end{matrix}\right\} \text{ are both squares.}$$

The difference is $75 = 3 \cdot 25$, and the usual method of solution gives $x = \frac{7}{10}$.

The numbers are $\frac{3}{10}$, $\frac{42}{10}$, $\frac{7}{10}$.

16. To find three numbers such that the product of any two *minus* the sum of those two gives a square.

Put x for the first, and any number for the second; we then fall into the same difficulty as in the last problem.

We have to find two numbers such that

(*a*) their product *minus* their sum = a square, and

(*b*) when each is diminished by 1, the remainders have the ratio of squares.

Now $4y+1$, $y+1$ satisfy the latter condition.

The former (*a*) requires that

$$4y^2-1 = \text{square} = (2y-2)^2, \text{ say,}$$

which gives $y = \frac{5}{8}$.

Assume then $\frac{13}{8}$, $\frac{28}{8}$, x for the numbers.

Therefore $\left.\begin{matrix} 2\frac{1}{2}x - 3\frac{1}{2} \\ \frac{5}{8}x - \frac{13}{8} \end{matrix}\right\}$ are both squares,

or, if we multiply by 4, 16 respectively,

$$\left.\begin{matrix} 10x - 14 \\ 10x - 26 \end{matrix}\right\} \text{ are both squares.}$$

The difference is $12 = 2 \cdot 6$, and the usual method gives $x = 3$.

The numbers are $\frac{13}{8}$, $3\frac{1}{2} = \frac{28}{8}$, $3 = \frac{24}{8}$.

17. To find two numbers such that their product added to both or to either gives a square.

Assume x, $4x - 1$ for the numbers, since

$$x(4x - 1) + x = 4x^2, \text{ a square.}$$

Therefore also $\left.\begin{matrix} 4x^2 + 3x - 1 \\ 4x^2 + 4x - 1 \end{matrix}\right\}$ are both squares.

The difference is $x = 4x \cdot \frac{1}{4}$, and we find

$$x = \tfrac{65}{224}.$$

The numbers are $\frac{65}{224}$, $\frac{36}{224}$.

18. To find two numbers such that their product *minus* either, or *minus* the sum of both, gives a square[1].

[1] With this problem should be compared that in paragraph 42 of Part I. of the *Inventum Novum* of Jacobus de Billy (*Oeuvres de Fermat*, III. pp. 351–2), where three conditions correspond to those of the above problem, and there is a fourth in addition. The problem is to find ξ, η $(\xi > \eta)$ such that

$$\left.\begin{matrix} \xi - \xi\eta \\ \eta - \xi\eta \\ \xi + \eta - \xi\eta \\ \xi - \eta - \xi\eta \end{matrix}\right\} \text{ are all squares.}$$

Suppose $\eta = x$, $\xi = 1 - x$; the first two conditions are thus satisfied. The other two give

$$x^2 - x + 1 = u^2,$$
$$x^2 - 3x + 1 = v^2.$$

Separating the difference $2x$ into the factors $2x$, 1, we put, as usual,

$$\left(x + \frac{1}{2}\right)^2 = x^2 - x + 1,$$

whence $x = \frac{3}{8}$, and the numbers are $\frac{5}{8}$, $\frac{3}{8}$.

To find another value of x by means of the value thus found, we put $y + \frac{3}{8}$ in place of x in the double-equation, whence

$$y^2 - \frac{1}{4}y + \frac{49}{64} = u^2,$$

$$y^2 - \frac{9}{4}y + \frac{1}{64} = v^2.$$

Multiplying the lower expression by 49, we can solve in the usual way. Our expressions

Assume $x+1$, $4x$ for the numbers, since

$$4x(x+1)-4x = \text{a square}.$$

Therefore also $\left.\begin{array}{l}4x^2+3x-1\\4x^2-x-1\end{array}\right\}$ are both squares.

The difference is $4x=4x.1$, and we find

$$x=1\tfrac{1}{4}.$$

The numbers are $2\frac{1}{4}$, 5.

19. To find four numbers such that the square of their sum *plus* or *minus* any one singly gives a square.

Since, in any right-angled triangle,

(sq. on hypotenuse) $\pm$ (twice product of perps.) = a square,

we must seek four right-angled triangles [in rational numbers] having the same hypotenuse,

or we must find a square which is divisible into two squares in four different ways; and "we saw how to divide a square into two squares in an infinite number of ways." [II. 8.]

Take right-angled triangles in the smallest numbers, (3, 4, 5) and (5, 12, 13); and multiply the sides of

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are now $y^2-\frac{1}{4}y+\frac{49}{64}$ and $49y^2-\frac{441}{4}y+\frac{49}{64}$, and the difference between them is $48y^2-110y$. The solution next mentioned by De Billy was clearly obtained by separating this difference into factors such that, when the square of half their difference is equated to $y^2-\frac{1}{4}y+\frac{49}{64}$, the absolute terms cancel out. The factors are $\frac{440}{7}y$, $\frac{42}{55}y-\frac{7}{4}$, and we put

$$\left\{\left(\frac{220}{7}-\frac{21}{55}\right)y+\frac{7}{8}\right\}^2=y^2-\frac{1}{4}y+\frac{49}{64}.$$

This gives $y=-\frac{4045195}{71362992}$, whence $x=\frac{22715927}{71362992}$, and the numbers are

$$\frac{48647065}{71362992},\ \frac{22715927}{71362992}.$$

A solution in smaller numbers is obtained by separating $48y^2-110y$ into factors such that the terms in x^2 in the resulting equation cancel out. The factors are $6y$, $8y-\frac{55}{3}$, and we put

$$\left(y-\frac{55}{6}\right)^2=y^2-\frac{1}{4}y+\frac{49}{64},$$

whence $y=\frac{47959}{10416}$, and $x=\frac{47959}{10416}+\frac{3}{8}=\frac{51865}{10416}$.

This would give a negative value for $1-x$; but, owing to the symmetry of the original double-equation in x, since $x=\frac{51865}{10416}$ satisfies it, so does $x=\frac{10416}{51865}$; hence the numbers are $\frac{10416}{51865}$ and $\frac{41449}{51865}$: a solution also mentioned by De Billy.

Cf. note on IV. 23.

the first by the hypotenuse of the second and *vice versa.*

This gives the triangles (39, 52, 65) and (25, 60, 65); thus 65^2 is split up into two squares in *two* ways.

Again, 65 is "naturally" divided into two squares in two ways, namely into 7^2+4^2 and 8^2+1^2, "which is due to the fact that 65 is the product of 13 and 5, each of which numbers is the sum of two squares."

Form now a right-angled triangle[1] from 7, 4. The sides are $(7^2-4^2, 2.7.4, 7^2+4^2)$ or (33, 56, 65).

Similarly, forming a right-angled triangle from 8, 1, we obtain $(2.8.1, 8^2-1^2, 8^2+1^2)$ or 16, 63, 65.

Thus 65^2 is split into two squares in *four* ways.

Assume now as the sum of the numbers $65x$ and

as first number $2.39.52x^2=4056x^2$,
„ second „ $2.25.60x^2=3000x^2$,
„ third „ $2.33.56x^2=3696x^2$,
„ fourth „ $2.16.63x^2=2016x^2$,

the coefficients of x^2 being four times the areas of the four right-angled triangles respectively.

The sum $12768x^2=65x$, and $x=\frac{65}{12768}$.

The numbers are

$$\frac{17136600}{163021824},\ \frac{12675000}{163021824},\ \frac{15615600}{163021824},\ \frac{8517600}{163021824}.$$

20. To divide a given number into two parts and to find a square which, when either of the parts is subtracted from it, gives a square[2].

Given number 10, required square x^2+2x+1.

Put for one of the parts $2x+1$, and for the other $4x$.

The conditions are therefore satisfied if

$$6x+1=10.$$

Therefore $x=1\frac{1}{2}$;

the parts are (4, 6) and the square $6\frac{1}{4}$.

[1] If there are two numbers p, q, to "form a right-angled triangle" from them means to take the numbers p^2+q^2, p^2-q^2, $2pq$. These are the sides of a right-angled triangle, since

$$(p^2+q^2)^2=(p^2-q^2)^2+(2pq)^2.$$

[2] This problem and the next are the same as II. 15, 14 respectively. It may therefore be doubted whether the solutions here given are genuine, especially as interpolations from ancient commentaries occur most at the beginning and end of Books.

21. To divide a given number into two parts and to find a square which, when added to either of the parts, gives a square.

Given number 20, required square $x^2 + 2x + 1$.

If to the square there be added either $2x + 3$ or $4x + 8$, the result is a square.

Take $2x + 3$, $4x + 8$ as the parts of 20, and $6x + 11 = 20$, whence $x = 1\frac{1}{2}$.

Therefore the parts are (6, 14) and the square $6\frac{1}{4}$.

BOOK IV

1. To divide a given number into two cubes such that the sum of their sides is a given number[1].

Given number 370, given sum of sides 10.

Sides of cubes $5 + x$, $5 - x$, satisfying one condition.

Therefore $30x^2 + 250 = 370$, $x = 2$,

and the cubes are 7^3, 3^3, or 343, 27.

2. To find two numbers such that their difference is a given number, and also the difference of their cubes is a given number.

Difference 6, difference of cubes 504.

Numbers $x + 3$, $x - 3$.

Therefore $18x^2 + 54 = 504$, $x^2 = 25$, and $x = 5$.

The sides of the cubes are 8, 2 and the cubes 512, 8.

3. To multiply one and the same number into a square and its side respectively so as to make the latter product a cube and the former product the side of the cube.

Let the square be x^2. Its side being x, let the number be $8/x$.

Hence the products are $8x$, 8, and

$$(8x)^3 = 8.$$

Therefore $2 = 8x$, $x = \frac{1}{4}$, and the number to be multiplied is 32.

The square is $\frac{1}{16}$ and its side $\frac{1}{4}$.

[1] It will be observed that Diophantus chooses, as his given numbers, numbers such as will make the resulting "pure" quadratic equation give a "rational" value for x. If the given numbers are $2a$, $2b$, respectively, we assume $b+x$, $b-x$ as the sides of the cubes, and we have

$$2b^3 + 6bx^2 = 2a,$$

so that $x^2 = (a - b^3)/3b$; x is therefore "irrational" unless $(a - b^3)/3b$ is a square. In Diophantus' hypothesis a is taken as 185, and b as 5, and the condition is satisfied. He shows therefore incidentally that he knew how to find two numbers a, b such that $(a - b^3)/3b$ is a square (Loria, *Le scienze esatte nell' antica Grecia*, Libro V. pp. 129–30).

A similar remark applies to the next problem, IV. 2.

4. To add the same number to a square and its side respectively and make them the same[1] [*i.e.* make the first product a square of which the second product is the side].

Square x^2, with side x.

Let the number added to x^2 be such as to make a square say $3x^2$.

Therefore $3x^2+x=$ side of $4x^2=2x$, and $x=\frac{1}{3}$.

The square is $\frac{1}{9}$, its side $\frac{1}{3}$, and the number $\frac{1}{3}$.

5. To add the same number to a square and its side and make them the opposite[2].

Square x^2, the number a square number of times x^2 *minus* x, say $4x^2-x$.

Hence $5x^2-x=$ side of $4x^2=2x$, and $x=\frac{3}{5}$.

The square is $\frac{9}{25}$, its side $\frac{3}{5}$, and the number $\frac{21}{25}$.

6. To add the same square number to a cube and a square and make them the same.

Let the cube be x^3 and the square any square number of times x^2, say $9x^2$.

We want now a square which when added to $9x^2$ makes a square. Take two factors of 9, say 9 and 1, subtract 1 from 9, take half the difference and square. This gives 16.

Therefore $16x^2$ is the square to be added.

Next, $x^3+16x^2=$ a cube $=8x^3$, say; and $x=\frac{16}{7}$.

The cube is therefore $\frac{4096}{343}$, the square $\frac{2304}{49}$, and the added square number $\frac{4096}{49}$.

[1] In this and the following enunciations I have kept closely to the Greek partly for the purpose of showing Diophantus' mode of expression and partly for the brevity gained thereby.

In Prop. 4 to "make them the same" means what I have put in brackets; to "make them the opposite" in Prop. 5 means to make the first product a side of which the second product is the square.

[2] Nesselmann solves the problem generally, thus (Notes in *Zeitschrift für Math. u. Physik*, XXXVII. (1892), Hist. litt. Abt. p. 162).

$x^2+y=\sqrt{(x+y)}$; therefore $x^4+2x^2y+y^2=x+y$, or $y^2-(1-2x^2)y=x-x^4$.

Solving for y, we obtain, as one of the solutions,

$$y=\frac{1}{2}-x^2+\sqrt{\left(\frac{1}{4}+x-x^2\right)}.$$

To make the expression under the radical a square we put $\frac{1}{4}+x-x^2=\left(mx-\frac{1}{2}\right)^2$, whence $x=\frac{m+1}{m^2+1}$, $y=\frac{m^4+m^3-m-1}{(m^2+1)^2}$. Diophantus' solution corresponds to $m=2$.

7. To add the same square number to a cube and a square respectively and make them the opposite.

For brevity call the cube (1), the second square (2) and the added square (3).

Now, since (2) + (3) = a cube, suppose (2) + (3) = (1).

Since $a^2+b^2 \pm 2ab$ is a square, suppose $(1)=(a^2+b^2)$, $(3)=2ab$, so that the condition that (1) + (3) = square is satisfied.

But (3) is a square, and, in order that $2ab$ may be a square, we put $a=x$, $b=2x$.

Suppose then $(1)=x^2+(2x)^2=5x^2$, $(3)=2 \cdot x \cdot 2x=4x^2$; therefore $(2)=x^2$, by subtraction.

But $5x^2$ is a cube; therefore $x=5$,

and the cube (1) = 125, the square (2) = 25, the square (3) = 100.

Otherwise thus.

Let (2) + (3) = (1).

Then, since (1) + (3) = a square, we have to find two squares such that their sum + one of them = a square.

Let the first of these squares be x^2, the second 4.

Therefore $2x^2+4=\text{square}=(2x-2)^2$, say; thus $x=4$, and the squares are 16, 4.

Assume now $(2)=4x^2$, $(3)=16x^2$.

Therefore $20x^2$ is a cube, so that $x=20$;

the cube (1) is 8000, the square (2) is 1600, and the added square (3) is 6400.

8. To add the same number to a cube and its side and make them the same[1].

Added number x, cube $8x^3$, say. Therefore second sum $=3x$, and this must be the side of $8x^3+x$.

That is, $8x^3+x=27x^3$, and $19x^3=x$, or $19x^2=1$.

[1] Nesselmann (*op. cit.* p. 163) gives a more general solution.

We have $x^3+y=(x+y)^3$, whence $1=3x^2+3xy+y^2$.

Solving for y, we find

$$y=-\frac{3}{2}x \pm \sqrt{\left(1-\frac{3}{4}x^2\right)}=\frac{1}{2}\{-3x \pm \sqrt{(4-3x^2)}\}.$$

Lastly, putting $4-3x^2=\left(2-\frac{m}{n}x\right)^2$, we find $x=\frac{4mn}{3n^2+m^2}$, $\sqrt{(4-3x^2)}=\pm\frac{2m^2-6n^2}{3n^2+m^2}$, and $y=\frac{-6mn \pm (m^2-3n^2)}{3n^2+m^2}$. If the positive sign be taken, then, in order that y may always be positive, m/n must be $>3+\sqrt{12}$; Diophantus' solution corresponds to $m=7$, $n=1$.

But 19 *is not a square.* Hence we must find, to replace it, some square number. Now $19x^3$ arises from $27x^3 - 8x^3$, where 27 is the cube of 3, and 8 the cube of 2. And the $3x$ comes from the assumed side $2x$, by increasing the coefficient by unity.

Thus we must find *two consecutive numbers such that their cubes differ by a square.*

Let them be $y, y+1$.

Therefore $3y^2 + 3y + 1 = \text{square} = (1 - 2y)^2$, say, and $y = 7$.

Going back to the beginning, we assume added number $= x$, side of cube $= 7x$.

The side of the new cube is then $8x$, and

$$343x^3 + x = 512x^3.$$

Therefore $x^2 = \frac{1}{169}$, and $x = \frac{1}{13}$.

The cube is $\frac{343}{2197}$, its side $\frac{7}{13}$, and the added number $\frac{1}{13}$.

9. To add the same number to a cube and its side and make them the opposite[1].

Suppose the cube is $8x^3$, its side being $2x$, and the added number is $27x^3 - 2x$. (The coefficients 8, 27 are chosen as cube numbers.)

Therefore $35x^3 - 2x = \text{side of cube } 27x^3 = 3x$, or $35x^2 = 5$.

This gives no rational value.

But $35 = 27 + 8$, and $5 = 3 + 2$.

Therefore we have to find two cubes such that their sum has to the sum of their sides the ratio of a square to a square[2].

Let sum of sides = any number, 2 say, and side of first cube $= z$, so that the side of the other cube is $2 - z$.

[1] Nesselmann (*op. cit.* p. 163) solves as follows. The equation being $x + y = (x^3 + y)^3$, put $y = z - x^3$, and the equation becomes $x + z - x^3 = z^3$, or $x^3 + z^3 = x + z$.

Dividing by $x + z$, we have $x^2 - xz + z^2 = 1$.

Solving for x, we obtain $x = \frac{1}{2}\{z \pm \sqrt{(4 - 3z^2)}\}$.

To make $4 - 3z^2$ a square, equate it to $\left(\frac{m}{n}z - 2\right)^2$; therefore $z = \frac{4mn}{m^2 + 3n^2}$, so that $x = \frac{2mn \pm (m^2 - 3n^2)}{m^2 + 3n^2}$, and $y = z - x^3$. If the positive sign be taken, Diophantus' solution corresponds to $m = 2$, $n = 1$.

[2] It will be observed that here and in the next problem Diophantus makes no use of the fact that

$$(x^3 + y^3)/(x + y) = x^2 - xy + y^2.$$

Cf. note on IV. 11 below.

Therefore $8 - 12z + 6z^2$ must be twice a square.

That is, $4 - 6z + 3z^2 = \text{square} = (2 - 4z)^2$, say; $z = \frac{10}{13}$, and the sides are $\frac{10}{13}$, $\frac{16}{13}$.

Neglecting the denominator and the factor 2 in the numerators, we take 5, 8 for the sides.

Starting afresh, we put for the cube $125x^3$ and for the number to be added $512x^3 - 5x$; we thus get

$$637x^3 - 5x = 8x, \text{ and } x = \tfrac{1}{7}.$$

The cube is $\frac{125}{343}$, its side $\frac{5}{7}$, and the added number $\frac{267}{343}$.

10. To find two cubes the sum of which is equal to the sum of their sides.

Let the sides be $2x$, $3x$.

This gives $35x^3 = 5x$; but *this equation gives an irrational result.*

We have therefore, as in the last problem, to find two cubes the sum of which has to the sum of their sides the ratio of a square to a square[1].

These are found, as before, to be 5^3, 8^3.

Assuming then $5x$, $8x$ as the sides of the required cubes, we obtain the equation $637x^3 = 13x$, and $x = \frac{1}{7}$.

The cubes are $\frac{125}{343}$, $\frac{512}{343}$.

[1] Here, as in the last problem, Diophantus could have solved his auxiliary problem of making $(x^3+y^3)/(x+y)$ a square by making $x^2 - xy + y^2$ a square in the same way as in Lemma 1. to v. 7 he makes $x^2 + xy + y^2$ a square.

The original problem, however, of solving

$$x^3 + y^3 = x + y$$

can be more directly and generally solved thus. Dividing out by $(x+y)$, we must have

$$x^2 - xy + y^2 = 1.$$

This can be solved by the method shown in the note to the preceding problem.

Alternatively, we may (with Wertheim) put $x^2 - xy + y^2 = (x+ky)^2$, and at the same time $1 = \pm(x+ky)$.

Thus we have to solve the equations

$$\left.\begin{array}{l} x(1+2k) = y(1-k^2) \\ x + ky = \pm 1 \end{array}\right\},$$

which give $$x = \pm\frac{1-k^2}{1+k+k^2}, \quad y = \pm\frac{1+2k}{1+k+k^2},$$

where k remains undetermined.

Diophantus' solution is obtained by taking the positive sign and putting $k = \frac{1}{4}$ or by taking the negative sign and putting $k = -\frac{3}{2}$.

11. To find two cubes such that their difference is equal to the difference of their sides.

Assume $2x$, $3x$ as the sides.

This gives $19x^3 = x$, and x is irrational.

We have therefore to find two cubes such that their difference has to the difference of their sides the ratio of a square to a square[1]. Let them be $(z+1)^3$, z^3, so that the difference of the sides may be a square, namely 1.

Therefore $3z^2 + 3z + 1 = \text{square} = (1 - 2z)^2$, say, and $z = 7$.

Starting afresh, assume $7x$, $8x$ as the sides; therefore $169x^3 = x$, and $x = \frac{1}{13}$.

The sides of the two cubes are therefore $\frac{7}{13}$, $\frac{8}{13}$.

[1] Nesselmann (*Die Algebra der Griechen*, pp. 447–8) comments on the fact that Diophantus makes no use here of the formula $(x^3 - y^3)/(x-y) = x^2 + xy + y^2$, although he must of course have known it (it is indeed included in Euclid's much more general summation of a geometrical progression, IX. 35). To solve the auxiliary problem in IV. 11 he had only to solve the equation

$$x^2 + xy + y^2 = \text{a square},$$

which equation he does actually solve in his Lemma I. to V. 7.

The whole problem can be more simply and generally solved thus. We are to have

$$x^3 - y^3 = x - y,$$

or

$$x^2 + xy + y^2 = 1.$$

Nesselmann's method of solution (cf. note on IV. 9) gives $x = \frac{1}{2}\{-y \pm \sqrt{(4 - 3y^2)}\}$, and hence $y = \frac{4mn}{m^2 + 3n^2}$, $x = \frac{-2mn \pm (m^2 - 3n^2)}{m^2 + 3n^2}$. Diophantus' solution is obtained by putting $m = 1$, $n = 2$ and taking the lower sign.

Wertheim's method (see note on preceding problem) gives in this case

$$x = \pm \frac{1 - k^2}{1 - k + k^2}, \quad y = \pm \frac{2k - 1}{1 - k + k^2},$$

where k is undetermined.

If we take the negative sign and put $k = -3$, we obtain Diophantus' solution.

Bachet in his notes to IV. 10, 11 solves the problems represented by

$$x^3 \pm y^3 = m(x \pm y)$$

subject to the condition that m is either a square or the third part of a square. His method corresponds to that of Diophantus. He does not divide out by $x \pm y$, and he reduces the problem to the subsidiary one of finding ξ, η such that the ratio of $\xi^3 \pm \eta^3$ to $\xi \pm \eta$ is the ratio of a square to a square. His assumptions for the "sides," ξ, η, are of the same kind as those made by Diophantus; in the first problem he assumes x, $6 - x$ and in the second x, $x + 2$. In fact, it being given that $(x^3 \pm y^3)/(x \pm y) = a$, Bachet assumes $x \pm y = z$ and thus obtains

$$3x^2 - 3xz + z^2 = a,$$

which equation can easily be solved by Diophantus' method if a is a square or triple of a square.

Fermat observes that the διορισμός of Bachet is incorrect because not general. It should be added that the number (m) may also be the product of a square number into a prime number of the form $3n + 1$, as 7, 13, 19, 37 etc. or into any number which has no factors except 3 and prime numbers of the form $3n + 1$, as 21, 91 etc. "The proof and the solution are to be obtained by my method."

12. To find two numbers such that the cube of the greater + the less = the cube of the less + the greater[1].

Assume $2x$, $3x$ for the numbers.

Therefore $27x^3 + 2x = 8x^3 + 3x$, or $19x^2 = x$, and x is irrational.

But 19 is the difference of two cubes, and 1 the difference of their sides. Therefore, as in the last problem, we have to find two cubes such that their difference has to the difference of their sides the ratio of a square to a square[2].

The sides of these cubes are found, as before, to be 7, 8.

Starting afresh, we assume $7x$, $8x$ for the numbers; then $343x^3 + 8x = 512x^3 + 7x$, and $x = \frac{1}{13}$.

The numbers are $\frac{7}{13}$, $\frac{8}{13}$.

13. To find two numbers such that either, or their sum, or their difference added to unity gives a square.

Take for the first number any square less 1; let it be, say, $9x^2 + 6x$. But the second + 1 = a square; and first + second + 1 also = a square. Therefore we must find a square such that the sum of that square and $9x^2 + 6x$ = a square.

Take factors of the difference $9x^2 + 6x$, say $9x + 6$, x; the square of half the difference between these factors $= 16x^2 + 24x + 9$.

Therefore, if we put for the second number this expression *minus* 1, or $16x^2 + 24x + 8$, three conditions are satisfied.

The remaining condition gives difference + 1 = square, or $7x^2 + 18x + 9$ = square $= (3 - 3x)^2$, say.

Therefore $x = 18$, and (3024, 5624) is a solution.

14. To find three square numbers such that their sum is equal to the sum of their differences.

Sum of differences = (greatest) − (middle) + (middle) − (least) + (greatest) − least = twice difference of greatest and least.

This is equal to the sum of all three, by hypothesis.

Let the least square be 1, the greatest $x^2 + 2x + 1$;

[1] This problem will be seen to be identical with the preceding problem.

[2] See note, p. 173.

therefore twice difference of greatest and least = sum of the three $= 2x^2 + 4x$.

But least + greatest $= x^2 + 2x + 2$, so that

$$\text{middle} = x^2 + 2x - 2.$$

Hence $x^2 + 2x - 2 = \text{square} = (x-4)^2$, say, and $x = \frac{9}{5}$.

$$\text{The squares are } \left(\frac{196}{25}, \frac{121}{25}, 1\right) \text{ or } (196, 121, 25).$$

15. To find three numbers such that the sum of any two multiplied into the third is a given number.

Let (first + second) × third = 35, (second + third) × first = 27 and (third + first) × second = 32.

Let the third be x.

Therefore (first + second) $= 35/x$.

Assume first $= 10/x$, second $= 25/x$; then

$$\left.\begin{aligned} \frac{250}{x^2} + 10 &= 27 \\ \frac{250}{x^2} + 25 &= 32 \end{aligned}\right\}.$$

These equations are inconsistent; but they would not be if $25 - 10$ *were equal to* $32 - 27$ *or* 5.

Therefore we have to divide 35 into two parts (to replace 25 and 10) such that their difference is 5. The parts are 15, 20. [Cf. I. 1.]

We take therefore $15/x$ for the first number, $20/x$ for the second, and we have

$$\left.\begin{aligned} \frac{300}{x^2} + 15 &= 27 \\ \frac{300}{x^2} + 20 &= 32 \end{aligned}\right\}.$$

Therefore $x = 5$, and $(3, 4, 5)$ is a solution[1].

[1] As Loria says (*Le scienze esatte nell' antica Grecia*, Libro V. p. 131), this method of the "false hypothesis," though somewhat indirect, would not be undeserving of a place in a modern textbook.

Here again, as in IV. 1, 2, Diophantus tacitly chooses, for his given numbers, numbers which will make the resulting "pure" quadratic equation give a rational value for x.

We may put the solution more generally thus. We have to solve the equations

$$(y+z)\,x = a, \quad (z+x)\,y = b, \quad (x+y)\,z = c.$$

Diophantus takes z for his principal unknown and, writing the third equation in the form $x + y = c/z$, he assumes $x = \alpha/z$, $y = \beta/z$, where α, β have to be determined. One equation connecting α, β is $\alpha + \beta = c$. Next, substituting the values of x, y in the first two equations, we have

$$\frac{\alpha\beta}{z^2} + \alpha = a, \quad \frac{\alpha\beta}{z^2} + \beta = b,$$

16. To find three numbers such that their sum is a square, while the sum of the square of each added to the next following number gives a square.

Let the middle number be any number of x's, say $4x$; we have therefore to find what square $+4x$ gives a square. Split $4x$ into two factors, say $2x$, 2, and take the square of half their difference, $(x-1)^2$. This is the square required.

Thus the first number is $x-1$.

Again, $16x^2+$ third number $=$ a square. Therefore, if we subtract $16x^2$ from a square, we shall have the third number. Take as the side of this square the side of $16x^2$, or $4x$, *plus* 1.

Therefore third number $=(4x+1)^2-16x^2=8x+1$.

Now the sum of the three numbers $=$ a square; therefore $13x=$ a square $=169y^2$, say[1].

The numbers are then $13y^2-1$, $52y^2$, $104y^2+1$.

Lastly, $(\text{third})^2+\text{first}=$ a square.

Therefore $\qquad 10816y^4+221y^2=$ a square,

or $10816y^2+221=$ a square $=(104y+1)^2$, say.

Therefore $y=\frac{220}{208}=\frac{55}{52}$,

and $\left(\frac{36621}{2704}, \frac{157300}{2704}, \frac{317304}{2704}\right)$ is a solution.

17. To find three numbers such that their sum is a square, while the square on any one *minus* the next following also gives a square.

The solution is precisely similar to the last.

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whence it follows that $\alpha-\beta=a-b$. From this condition and $\alpha+\beta=c$, we obtain

$$\alpha=\frac{1}{2}(a-b+c), \quad \beta=\frac{1}{2}(-a+b+c).$$

Thus
$$z=\sqrt{\left(\frac{\alpha\beta}{a-\alpha}\right)}=\sqrt{\left\{\frac{(a-b+c)(-a+b+c)}{2(a+b-c)}\right\}},$$

$$x=\frac{\alpha}{z}=\sqrt{\left\{\frac{(a-b+c)(a+b-c)}{2(-a+b+c)}\right\}}, \quad y=\frac{\beta}{z}=\sqrt{\left\{\frac{(-a+b+c)(a+b-c)}{2(a-b+c)}\right\}}.$$

Now x, y, z must all be rational, and this is the case if

$$-a+b+c=2qr, \quad a-b+c=2rp, \quad a+b-c=2pq,$$

where p, q, r are any integers.

This gives $\qquad a=p(q+r), \quad b=q(r+p), \quad c=r(p+q);$

a fact which can hardly have been unknown to Diophantus, since his values $a=27$, $b=32$, $c=35$ correspond to the values $p=3$, $q=4$, $r=5$ (Loria, *loc. cit.*).

[1] Diophantus uses the same unknown ς for y as for x, writing actually καὶ γίνεται ὁ ς Δ^Y $\overline{\iota\gamma}$, literally "and x becomes $13x^2$."

The middle number is assumed to be $4x$. The square which exceeds this by a square is $(x+1)^2$, and we therefore take $x+1$ for the first number.

For the third number we take $16x^2-(4x-1)^2$ or $8x-1$.

The sum of the numbers being a square,

$$13x = \text{a square} = 169y^2, \text{ say.}$$

The numbers are then $13y^2+1$, $52y^2$, $104y^2-1$.

Lastly, since $(\text{third})^2 - \text{first} = \text{a square}$,

$$10816y^4 - 221y^2 = \text{a square},$$

or $10816y^2 - 221 = \text{a square} = (104y-1)^2$, say.

Thus $y = \frac{111}{104}$,

and $\left(\frac{170989}{10816}, \frac{640692}{10816}, \frac{1270568}{10816}\right)$ is a solution.

18. To find two numbers such that the cube of the first added to the second gives a cube, and the square of the second added to the first gives a square.

First number x. Therefore second is a cube number *minus* x^3, say $8-x^3$.

Therefore $x^6 - 16x^3 + 64 + x = \text{a square} = (x^3+8)^2$, say, whence $32x^3 = x$, or $32x^2 = 1$.

This gives an irrational result; x would however be rational if 32 were a square.

But 32 comes from 4 times 8. We must therefore substitute for 8 in our assumptions a cube which when multiplied by 4 gives a square. If y^3 is the cube, $4y^3 = \text{a square} = 16y^2$ say; whence $y = 4$.

Thus we must assume x, $64-x^3$ for the numbers.

Therefore $x^6 - 128x^3 + 4096 + x = \text{a square} = (x^3+64)^2$, say; whence $256x^3 = x$, and $x = \frac{1}{16}$.

The numbers are $\frac{1}{16}$, $\frac{262143}{4096}$.

19. To find three numbers indeterminately[1] such that the product of any two increased by 1 is a square.

Take for the product of first and second some square *minus* 1, say x^2+2x; this satisfies one condition.

Let second $= x$, so that first $= x+2$.

Now product of second and third $+ 1 = \text{a square}$; let the

[1] The expression is ἐν τῷ ἀορίστῳ, which is defined at the end of the problem to mean in terms of one unknown (and units), so that the conditions of the problem are satisfied whatever value is given to the unknown.

square be $(3x+1)^2$, so that product of second and third $= 9x^2 + 6x$;

therefore third $= 9x + 6$.

Also product of third and first $+ 1 =$ a square; therefore $9x^2 + 24x + 13 =$ a square.

Now, if 13 *were a square, and the coefficient of x were twice the product of the side of this square and the side of the coefficient of x^2, the problem would be solved indeterminately.*

But 13 comes from $2.6+1$, the 2 in this from twice 1, and the 6 from twice 3. Therefore we want two coefficients (to replace 1, 3) such that the product of their doubles $+ 1 =$ a square, or four times their product $+ 1 =$ a square.

Now four times the product of any two numbers *plus* the square of their difference gives a square. Thus the requirement is satisfied by taking as coefficients any two consecutive numbers, since the square of their difference is 1. [The assumption of two consecutive numbers for the coefficients simultaneously satisfies the second of the two requirements indicated in the italicised sentence above.]

Beginning again, we take $(x+1)^2 - 1$ for the product of first and second and $(2x+1)^2 - 1$ for the product of second and third.

Let the second be x, so that first $= x + 2$, third $= 4x + 4$.

[Then product of first and third $+ 1 = 4x^2 + 12x + 9$, and the third condition is satisfied.]

Thus the required indeterminate solution[1] is

$$(x+2,\ x,\ 4x+4).$$

[1] The result obtained by Diophantus really amounts to the more general solution

$$a^2x + 2a,\quad x,\quad (a+1)^2x + 2(a+1).$$

With this solution should be compared that of Euler (*Algebra*, Part II. Art. 231).

I. To determine x, y, z so that

$$xy+1,\ yz+1,\ zx+1 \text{ are all squares.}$$

Suppose $zx + 1 = p^2,\quad yz + 1 = q^2,$

so that $x = (p^2-1)/z,\quad y = (q^2-1)/z.$

Therefore $$xy + 1 = \frac{(p^2-1)(q^2-1)}{z^2} + 1 = \text{a square},$$

or $$(p^2-1)(q^2-1) + z^2 = \text{a square}$$
$$= (z-r)^2, \text{ say;} \qquad \text{[Euler has } (z+r)^2\text{]}$$

whence $$z = \frac{r^2 - (p^2-1)(q^2-1)}{2r},$$

where any numbers may be substituted for p, q, r.

20. To find four numbers such that the product of any two increased by unity is a square.

For the product of first and second take a square *minus* 1, say $(x+1)^2 - 1 = x^2 + 2x$.

Let first $= x$, so that second $= x + 2$.

For example, if $r = pq + 1$, we shall have

$$z = \frac{(p+q)^2}{2(pq+1)} \text{ and } x = \frac{2(pq+1)(p^2-1)}{(p+q)^2}, \quad y = \frac{2(pq+1)(q^2-1)}{(p+q)^2}.$$

II. But, *if whole numbers are required*, we put $xy + 1 = p^2$, and assume $z = x + y + q$.

We then have $\quad xz + 1 = x^2 + xy + qx + 1 = x^2 + qx + p^2$,

and $\quad yz + 1 = xy + y^2 + qy + 1 = y^2 + qy + p^2$.

These expressions are both squares if $q = \pm 2p$.

Thus a solution is obtained from $xy = p^2 - 1$ combined with either

$$z = x + y + 2p, \text{ or } z = x + y - 2p.$$

We take a certain value for p^2, split $p^2 - 1$ into two factors, take these factors for the values of x, y respectively, and so find z.

For example, let $p = 3$, so that $p^2 - 1 = 8$; if we make $x = 2$, $y = 4$, we find $z =$ either 12 or 0; and in this case $x = 2$, $y = 4$, $z = 12$ is the solution.

If we put $p^2 = (\xi + 1)^2$, we have $xy = \xi^2 + 2\xi$; and if we put $x = \xi + 2$, $y = \xi$, we have

$$z = \xi + 2 + \xi \pm 2(\xi + 1) = 4\xi + 4 \text{ or } 0.$$

The solution is then $(\xi + 2, \xi, 4\xi + 4)$, as in Diophantus.

Fermat in his note on this problem shows how to find three numbers satisfying not only the conditions of the problem but three more also, namely that each of the numbers shall itself when increased by 1 give a square, *i.e.* to solve the equations

$$\left.\begin{array}{l} \eta\zeta + 1 = r^2, \quad \zeta\xi + 1 = s^2, \quad \xi\eta + 1 = t^2, \\ \xi + 1 = u^2, \quad \eta + 1 = v^2, \quad \zeta + 1 = w^2. \end{array}\right\}$$

Solve, he says, the present problem of Diophantus in such a way that the terms independent of x in the first and third of the numbers obtained by his method shall be such as when increased by 1 give a square. It is easy to find a value for a such that $2a + 1$ and $2(a+1) + 1$ are both squares. Fermat takes the value $2a = \frac{13}{36}$, which satisfies the conditions, and the general expressions for the three numbers sought are therefore

$$\frac{169}{5184}x + \frac{13}{36}, \quad x, \quad \frac{7225}{5184}x + \frac{85}{36}.$$

Each of these has, when increased by 1, to become a square, that is, we have to solve the triple-equation

$$\left.\begin{array}{r} \frac{169}{5184}x + \frac{49}{36} = u^2 \\ x + 1 = v^2 \\ \frac{7225}{5184}x + \frac{121}{36} = w^2 \end{array}\right\}.$$

Fermat does not give the solution; but it is effected as follows.

Multiplying the third expression by 36 and the first by $\frac{121}{49} \cdot 36$ (in order that the absolute terms in the two may be equal), we have to solve

For the product of first and third take $(2x+1)^2-1$, or $4x^2+4$, the coefficient of x being the number next following the coefficient (1) taken in the first case, for the reason shown in the last problem;

thus third number $=4x+4$.

Similarly take $(3x+1)^2-1$, or $9x^2+6x$, for the product of first and fourth; therefore fourth $=9x+6$.

And product of third and fourth $+1$

$$=(4x+4)(9x+6)+1=36x^2+60x+25,$$

which is a square[1].

$$\left.\begin{aligned} x+1&=v^2\\ \left(\frac{143}{7\cdot 12}\right)^2 x+121&=u'^2\\ \left(\frac{85}{12}\right)^2 x+121&=w'^2\end{aligned}\right\}.$$

In order to solve by the method of the triple-equation, we make $x+1$ a square by putting $x=y^2+2y$.

Substitute this value in the other two expressions, and for convenience multiply each by 144; this gives

$$\left.\begin{aligned}\left(\frac{143}{7}\right)^2(y^2+2y)+(132)^2&=\text{a square}\\ (85)^2(y^2+2y)+(132)^2&=\text{a square}\end{aligned}\right\}.$$

The difference $=(y^2+2y)\left(85+\frac{143}{7}\right)\left(85-\frac{143}{7}\right)$

$$=\frac{738}{7}y\left(\frac{452}{7}y+\frac{2\cdot 452}{7}\right).$$

The square of half the difference of the factors equated to the smaller expression gives

$$\left(\frac{143}{7}y-\frac{452}{7}\right)^2=\left(\frac{143}{7}\right)^2(y^2+2y)+(132)^2;$$

whence $y=-\frac{324736}{85085}$; and we find that

$$x=y^2+2y=\frac{50193144576}{7239457225}.$$

It is easily verified that

$$\left(\frac{13}{12}\right)^2x+7^2=\left(\frac{643149}{85085}\right)^2 \text{ and } \left(\frac{85}{12}\right)^2x+11^2=\left(\frac{1842375}{85085}\right)^2,$$

so that the value of x satisfies the three equations.

The numbers satisfying Fermat's six conditions are then

$$\frac{169}{5184}x+\frac{13}{36}=\frac{100604981}{171348100},\quad x=\frac{50193144576}{7239457225},\ \text{and}\ \frac{7225}{5184}x+\frac{85}{36}=\frac{48192621}{4008004}.$$

[1] This results from the fact that, if we have three numbers x, y, z such that

$$xy+1=(mx+1)^2 \text{ and } xz+1=\{(m+1)x+1\}^2,$$

then
$$yz+1=\{m(m+1)x+(2m+1)\}^2.$$

Lastly, product of second and fourth $+1=9x^2+24x+13$;

therefore $9x^2+24x+13=\text{square}=(3x-4)^2$, say;

which gives $x=\frac{1}{16}$.

All the conditions are now satisfied[1],

and $\frac{1}{16}, \frac{33}{16}, \frac{68}{16}, \frac{105}{16}$ is the solution[2].

[1] The remaining condition was: product of second and third $+1=$ a square. That this is satisfied also follows from the general property stated in the last note. In fact

$$(x+2)(4x+4)+1=4x^2+12x+9,$$

which is a square.

[2] With this solution should be compared Euler's solution (*Algebra*, Part II. Art. 233) of the problem of finding x, y, z, v such that the six expressions

$$xy+a, \quad yz+a, \quad zx+a, \quad xv+a, \quad yv+a, \quad zv+a$$

are all squares. The solution follows the method adopted to solve the corresponding problem with three unknowns x, y, z only. See note on III. 10 above.

If we begin by supposing $xy+a=p^2$, and take $z=x+y+2p$, the second and third expressions become squares (*vide* note on III. 10, p. 160).

If we further suppose $v=x+y-2p$, the fourth and fifth expressions also become squares (*vide* the same note).

Consequently we have only to secure that the sixth expression $zv+a$ shall be a square; that is,

$$x^2+2xy+y^2-4p^2+a=\text{a square},$$

or (since $xy+a=p^2$) $\quad x^2-2xy+y^2-3a=\text{a square}.$

Suppose that $\quad (x-y)^2-3a=(x-y-q)^2;$

therefore $\quad x-y=(q^2+3a)/2q,$

or $\quad x=y+\dfrac{q^2+3a}{2q}.$

Consequently $\quad p^2=xy+a=y^2+\dfrac{q^2+3a}{2q}y+a.$

If we put $p=y+r$, we have

$$2ry+r^2=\frac{q^2+3a}{2q}y+a,$$

and $\quad y=\dfrac{2qr^2-2aq}{q^2-4qr+3a},$

from which p, x, and therefore z, v also, are found in terms of q, r, where q, r may have any values provided that x, y, z, v are all positive.

Euler observed that this method is not suited for finding integral solutions, and, pursuing the matter further, he gave the following very elegant solution of Diophantus' actual problem (the case where $a=1$) in integers ("Miscellanea analytica" in *Commentationes arithmeticae*, II. pp. 45–6).

Six conditions have to be satisfied. If x, y, z, v are the required numbers, let $x=m$, $y=n$, where m, n are any integers such that $mn+1=l^2$.

Then put $z=m+n+2l$, and three conditions are already satisfied, for

$$xy+1=mn+1=l^2, \text{ by hypothesis},$$

$$xz+1=m(m+n+2l)+1=(l+m)^2,$$

$$yz+1=n(m+n+2l)+1=(l+n)^2.$$

The three conditions remaining to be satisfied are

$$xv+1=mv+1=\text{a square},$$

$$yv+1=nv+1=\text{a square},$$

$$zv+1=(m+n+2l)v+1=\text{a square}.$$

Let us make the continued product of these expressions a square.

21. To find three numbers in proportion and such that the difference of any two is a square.

Assume x for the least, $x+4$ for the middle (in order that the difference of middle and least may be a square), $x+13$ for the greatest (in order that difference of greatest and middle may be a square).

This product will be found to be

$$1+2(m+n+l)v+\{(m+n+l)^2-1\}v^2+mn(m+n+2l)v^3.$$

Let us equate this to $\left\{1+(m+n+l)v-\frac{1}{2}v^2\right\}^2$, in order that the terms in v, v^2 as well as the absolute term may vanish; therefore

$$mn(m+n+2l)=-(m+n+l)+\frac{1}{4}v,$$

whence
$$\begin{aligned}\frac{1}{4}v&=(m+n+l)+mn(m+n+2l)\\&=(mn+1)(m+n+l)+lmn\\&=l^2(m+n+l)+lmn\\&=l(l+m)(l+n),\end{aligned}$$

and therefore
$$v=4l(l+m)(l+n).$$

It is true that we have only made the product of the three expressions $mv+1$, $nv+1$, $(m+n+2l)v+1$ a square; but, as the value of v has turned out to be an integral number, so that all three formulae are prime to one another, we may conclude that each of the expressions is a square.

The solution is therefore

$$x=m,\quad y=n,\quad z=m+n+2l,\quad v=4l(l+m)(l+n),$$

where $mn+1=l^2$.

In fact, while three of the conditions have been above shown to be satisfied, we find, as regards the other three, that

$$xv+1=4lm(l+m)(l+n)+1=(2l^2+2lm-1)^2,$$
$$yv+1=4ln(l+m)(l+n)+1=(2l^2+2ln-1)^2,$$
$$zv+1=4l(m+n+2l)(l+m)(l+n)+1=(4l^2+2lm+2ln-1)^2.$$

It is to be observed that l may be either positive or negative.

Ex. Let $m=3$, $n=8$, so that $l=\pm 5$.

If $l=+5$, the solution is 3, 8, 21, 2080; if $l=-5$, the solution is 3, 8, 1, 120.

Fermat shows how to solve this problem, alternatively, by means of the "triple-equation."

Take *three* numbers with the required property, *e.g.* 3, 1, 8. Let x be the fourth, and we have then to satisfy the conditions

$$3x+1=u^2,\quad x+1=v^2,\quad 8x+1=w^2.$$

Put $x=y^2+2y$, so as to make the second expression a square, and then substitute the value of x in the other two. We have then the double-equation

$$3(y^2+2y)+1=u^2,$$
$$8(y^2+2y)+1=w^2.$$

The difference $=5(y^2+2y)=5y(y+2)$.

We put then
$$(3y+1)^2=8(y^2+2y)+1,$$

whence $y=10$, and $x=y^2+2y=120$, which value satisfies the triple-equation.

The four numbers are then 3, 1, 8, 120, which solution is identical with one of those obtained by Euler as above.

If now 13 were a square, we should have an indeterminate solution satisfying three of the conditions.

We must therefore replace 13 by a square which is the sum of two squares. Any rational right-angled triangle will furnish what is wanted, say 3, 4, 5;

we therefore put for the numbers x, $x+9$, $x+25$.

The fourth condition gives

$$x(x+25)=(x+9)^2, \text{ and } x=\tfrac{81}{7}.$$

Thus $\frac{81}{7}, \frac{144}{7}, \frac{256}{7}$ is a solution.

22. To find three numbers such that their solid content[1] added to any one of them gives a square.

Assume continued product x^2+2x, first number 1, second number $4x+9$, so that two conditions are satisfied.

The third number is then $(x^2+2x)/(4x+9)$.

This cannot be divided out unless $x^2 : 4x = 2x : 9$ or, alternately, $x^2 : 2x = 4x : 9$; but it could be done if 4 were half of 9.

Now $4x$ comes from $6x-2x$, and the $6x$ in this from twice $3x$; the 9 comes from 3^2.

Therefore we have to find a number m to replace 3 such that $2m-2=\frac{1}{2}m^2$: thus $m^2=4m-4$, whence[2] $m=2$.

We put therefore for the second number $(x+2)^2-(x^2+2x)$, or $2x+4$; the third number is then

$$(x^2+2x)/(2x+4) \text{ or } \tfrac{1}{2}x.$$

Lastly, the third condition requires

$$x^2+2x+\tfrac{1}{2}x=\text{a square}=4x^2, \text{ say.}$$

Therefore $x=\frac{5}{6}$,

and $\left(1, \frac{34}{6}, \frac{2\frac{1}{2}}{6}\right)$ is a solution[3].

[1] ὁ ἐξ αὐτῶν στερεός, "the solid (number formed) from them" = the continued product of the three numbers.

[2] Observe the solution of a mixed quadratic.

[3] Fermat gives a solution which avoids the necessity for the auxiliary problem.

Let the solid content be x^2-2x, the first number 1, and the second number $2x$; two conditions are thus satisfied.

The third number is now x^2-2x divided by $2x \cdot 1$, or $\frac{1}{2}x-1$; and the third condition gives

$$x^2-\frac{3}{2}x-1=\text{a square.}$$

Now x must be greater than 2; we therefore put

$$x^2-\frac{3}{2}x-1=(x-m)^2,$$

where m is greater than 2.

23. To find three numbers such that their solid content *minus* any one gives a square[1].

First number x, solid content x^3+x; therefore product of second and third $=x+1$.

Let the second be 1, so that the third is $x+1$.

The two remaining conditions require that

$$\left.\begin{array}{r} x^2+x-1 \\ x^2-1 \end{array}\right\} \text{shall both be squares. [Double-equation.]}$$

The difference $=x=\frac{1}{2}\,.\,2x$, say;

thus $(x+\frac{1}{4})^2=x^2+x-1$, and $x=\frac{17}{8}$.

The numbers are $\left(\frac{17}{8}, 1, \frac{25}{8}\right)$.

24. To divide a given number into two parts such that their product is a cube *minus* its side.

Given number 6. First part x; therefore second $=6-x$, and $6x-x^2=$ a cube *minus* its side.

Form a cube from a side of the form $mx-1$, say $2x-1$, and equate $6x-x^2$ to this cube *minus* its side.

Therefore $8x^3-12x^2+4x=6x-x^2$.

[1] A remarkable problem of this kind (in respect of the apparent number of conditions satisfied) is given by De Billy in the *Inventum Novum*, Part I. paragraph 43 (*Oeuvres de Fermat*, III. p. 352): To find three numbers ξ, η, ζ (ξ, ζ, η being in ascending order of magnitude) such that the following nine expressions may become squares:

(1) $\xi-\xi\eta\zeta$,	(4) $\eta-\xi-\xi\eta\zeta$,	(7) $\xi\eta-\xi\eta\zeta$,
(2) $\eta-\xi\eta\zeta$,	(5) $\zeta-\xi-\xi\eta\zeta$,	(8) $\eta\zeta-\xi\eta\zeta$,
(3) $\zeta-\xi\eta\zeta$,	(6) $\eta-\zeta-\xi\eta\zeta$,	(9) $\eta^2-\xi\eta\zeta$.

Take x, 1, $1-x$ as the values of ξ, η, ζ respectively. Then *six conditions*, namely, (1), (3), (4), (6), (7), (8), are all automatically satisfied.

By conditions (2) and (9) alike,

$$1-x+x^2=\text{a square}.$$

And, by (5), $$1-3x+x^2=\text{a square}.$$

Solving this double-equation in the usual way, we get $x=\frac{3}{8}$, and the numbers are $\frac{3}{8}$, 1, $\frac{5}{8}$.

Another solution can be obtained by putting $y+\frac{3}{8}$ in place of x in the two expressions, and so on. Cf. note on III. 18 above.

It would appear from a letter from Fermat to De Billy of 26 Aug. 1659 (*Oeuvres*, II. pp. 436–8) that this problem and the above single solution were De Billy's own. De Billy had supposed that this was the only solution, but Fermat observed that there were any number, as the above double-equation has any number of solutions. Fermat gave $\left(\frac{10416}{51865}, 1, \frac{41449}{51865}\right)$ as another solution.

Now, if the coefficient of x were the same on both sides, this would reduce to a simple equation, and x would be rational.

In order that this may be the case, we must put m for 2 in our assumption, where $3m - m = 6$ (the 6 being the given number in the original hypothesis). Thus $m = 3$.

We therefore assume

$$(3x-1)^3 - (3x-1) = 6x - x^2,$$

or $$27x^3 - 27x^2 + 6x = 6x - x^2,$$

and $$x = \tfrac{26}{27}.$$

The parts are $\frac{26}{27}$, $\frac{136}{27}$.

25. To divide a given number into three parts such that their continued product gives a cube the side of which is equal to the sum of the differences of the parts.

Given number 4.

Since the product is a cube, let it be $8x^3$, the side of which is $2x$.

Now (second part) − (first) + (third) − (second) + (third) − (first) = twice difference between third and first.

Therefore difference between third and first = half sum of differences $= x$.

Let the first be any multiple of x, say $2x$; therefore the third $= 3x$.

Hence second $= 8x^3/6x^2 = \frac{4}{3}x$; and, *if the second had lain between the first and third, the problem would have been solved.*

Now the second came from dividing 8 by 2 . 3, and the 2 and 3 are not two numbers at random but consecutive numbers.

Therefore we have to find two consecutive numbers such that, when 8 is divided by their product, the quotient lies between the numbers.

Assume m, $m+1$; therefore $8/(m^2+m)$ lies between m and $m+1$.

Therefore $$\frac{8}{m^2+m} + 1 > m + 1,$$

so that $$m^2 + m + 8 > m^3 + 2m^2 + m,$$

or $$8 > m^3 + m^2.$$

I form a cube such that it has m^3, m^2 as terms, that is, the cube $(m+\frac{1}{3})^3$, which is greater than m^3+m^2, and I put

$$(m+\tfrac{1}{3})^3 = 8;$$

therefore $m+\frac{1}{3}=2$, and $m=\frac{5}{3}$.

Assume now for first number $\frac{5}{3}x$; the third is $\frac{8}{3}x$, and the second is $\frac{9}{5}x$.

Multiplying throughout by 15, we take $25x$, $27x$, $40x$, and the product of these numbers is a cube the side of which is the sum of their differences.

The sum $= 92x = 4$, by hypothesis.

Therefore $x = \frac{1}{23}$,

and $\left(\frac{25}{23}, \frac{27}{23}, \frac{40}{23}\right)$ are the parts required.

[N.B. The condition $8/(m^2+m) < m+1$ is ignored in the work, and is *incidentally* satisfied.]

26. To find two numbers such that their product added to either gives a cube.

Let the first number be of the form m^3x, say $8x$.

Second x^2-1. Therefore one condition is satisfied, since

$$8x^3-8x+8x = \text{a cube}.$$

Also $8x^3-8x+x^2-1 = \text{a cube} = (2x-1)^3$, say.

Therefore $13x^2 = 14x$, and $x=\frac{14}{13}$.

The numbers are $\frac{112}{13}$, $\frac{27}{169}$.

27. To find two numbers such that their product *minus* either gives a cube.

Let the first be of the form m^3x, say $8x$, and the second x^2+1 (since $8x^3+8x-8x = \text{a cube}$).

Also $8x^3+8x-x^2-1$ must be a cube, "which is impossible[1]."

[1] Diophantus means that, if we are to get rid of the third power and the absolute term, we can only put the expression equal to $(2x-1)^3$, which gives a negative and therefore "impossible" value for x. But the equation is not really impossible, for we can get rid of the terms in x^3 and x^2 by putting

$$8x^3+8x-x^2-1=\left(2x-\frac{1}{12}\right)^3, \text{ whence } x=\frac{1727}{13752},$$

or we can make the term in x and the absolute term disappear by putting

$$8x^3+8x-x^2-1=\left(\frac{8}{3}x-1\right)^3, \text{ whence } x=\frac{549}{296}.$$

Diophantus has actually shown us how to do the former in IV. 25 just preceding.

Accordingly we assume for the first number an expression of the form m^3x+1, say $8x+1$, and for the second number x^2 (since $8x^3+x^2-x^2=$ a cube).

Also $8x^3+x^2-8x-1=$ a cube $=(2x-1)^3$, say.

Therefore $x=\frac{14}{13}$,

and the numbers are $\frac{125}{13}$, $\frac{196}{169}$.

28. To find two numbers such that their product $\pm$ their sum gives a cube.

Assume the first cube (product + sum) to be 64, and the second (product − sum) to be 8.

Therefore twice sum of numbers $=64-8=56$, and the sum $=28$, while the product + the sum $=64$; therefore the product $=36$.

Therefore we have to find two numbers such that their sum is 28 and their product 36. If $14+x$, $14-x$ are the numbers[1], we have $196-x^2=36$, or $x^2=160$; and, if 160 were a square, we should have a rational solution.

Now 160 arises from 14^2-36, and $14=\frac{1}{2}.28=\frac{1}{4}.56$ $=\frac{1}{4}$ (difference of two cubes); also $36=\frac{1}{2}$ (sum of the cubes).

Therefore we have to find two cubes such that

$(\frac{1}{4}$ of their difference$)^2-\frac{1}{2}$ their sum $=$ a square.

Let the sides of the cubes be $(z+1)$, $(z-1)$;

therefore $\frac{1}{4}$ of difference $=1\frac{1}{2}z^2+\frac{1}{2}$, and the square of this is $2\frac{1}{4}z^4+1\frac{1}{2}z^2+\frac{1}{4}$;

$\frac{1}{2}$ the sum of the cubes is z^3+3z;

therefore $2\frac{1}{4}z^4+1\frac{1}{2}z^2+\frac{1}{4}-z^3-3z=$ a square,

or $9z^4+6z^2+1-4z^3-12z=$ a square $=(3z^2+1-6z)^2$, say;

whence $32z^3=36z^2$, and $z=\frac{9}{8}$.

The sides of the cubes are therefore $\frac{17}{8}$, $\frac{1}{8}$, and the cubes $\frac{4913}{512}$, $\frac{1}{512}$.

Put now product of numbers + their sum $=\frac{4913}{512}$, and product − sum $=\frac{1}{512}$.

Therefore their sum $=\frac{2456}{512}$, and their product $=\frac{2457}{512}$.

Now let the first number $=x+$ half sum $=x+\frac{1228}{512}$,

and the second $=$ half sum $-x=\frac{1228}{512}-x$;

therefore $\frac{1507984}{262144}-x^2=\frac{2457}{512}$,

and $262144x^2=250000$.

[1] Cf. I. 27.

Therefore $x = \frac{500}{512}$,

and $\left(\frac{1728}{512}, \frac{728}{512}\right)$ is a solution.

Otherwise thus.

If any square number is divided into two parts one of which is its side, the product of the parts added to their sum gives a cube.

[That is, $x(x^2 - x) + x^2 - x + x =$ a cube.]

Let the square be x^2, and be divided into the parts $x, x^2 - x$.

Then, by the second condition of the problem,

$x^3 - x^2 - x^2 = x^3 - 2x^2 =$ a cube (less than x^3) $= (\frac{1}{2}x)^3$, say.

Therefore $8x^3 - 16x^2 = x^3$, so that $x = \frac{16}{7}$,

and $\left(\frac{16}{7}, \frac{144}{49}\right)$ is a solution.

29. To find four square numbers such that their sum added to the sum of their sides makes a given number[1].

Given number 12.

Now $x^2 + x + \frac{1}{4} =$ a square.

Therefore the sum of four squares + the sum of their sides $+ 1 =$ the sum of four other squares $= 13$, by hypothesis.

Therefore we have to divide 13 into four squares; then, if we subtract $\frac{1}{2}$ from each of their sides, we shall have the sides of the required squares.

[1] On this problem Bachet observes that Diophantus appears to assume, here and in some problems of Book v., that any number not itself a square is the sum of two or three or four squares. He adds that he has verified this statement for all numbers up to 325, but would like to see a scientific proof of the theorem. These remarks of Bachet's are the occasion for another of Fermat's famous notes: "I have been the first to discover a most beautiful theorem of the greatest generality, namely this: Every number is either a triangular number or the sum of two or three triangular numbers; every number is a square or the sum of two, three, or four squares; every number is a pentagonal number or the sum of two, three, four or five pentagonal numbers; and so on *ad infinitum*, for hexagons, heptagons and any polygons whatever, the enunciation of this general and wonderful theorem being varied according to the number of the angles. The proof of it, which depends on many various and abstruse mysteries of numbers, I cannot give here; for I have decided to devote a separate and complete work to this matter and thereby to advance arithmetic in this region of inquiry to an extraordinary extent beyond its ancient and known limits."

Unfortunately the promised separate work did not appear. The theorem so far as it relates to squares was first proved by Lagrange (*Nouv. Mémoires de l'Acad. de Berlin*, année 1770, Berlin 1772, pp. 123–133; *Oeuvres*, III. pp. 189–201), who followed up results obtained by Euler. Cf. also Legendre, *Zahlentheorie*, tr. Maser, I. pp. 212 sqq. Lagrange's proof is set out as shortly as possible in Wertheim's Diophantus, pp. 324–330. The theorem of Fermat in all its generality was proved by Cauchy (*Oeuvres*, II^e série, Vol. VI. pp. 320–353); cf. Legendre, *Zahlentheorie*, tr. Maser, II. pp. 332 sqq.

Now $13 = 4 + 9 = (\frac{64}{25} + \frac{36}{25}) + (\frac{144}{25} + \frac{81}{25})$,

and the sides of the required squares are $\frac{11}{10}, \frac{7}{10}, \frac{19}{10}, \frac{13}{10}$,

the squares themselves being $\frac{121}{100}, \frac{49}{100}, \frac{361}{100}, \frac{169}{100}$.

30. To find four squares such that their sum *minus* the sum of their sides is a given number.

Given number 4.

Now $x^2 - x + \frac{1}{4} =$ a square.

Therefore (the sum of four squares) − (sum of their sides) + 1 = the sum of four other squares = 5, by hypothesis.

Divide 5 into four squares, as $\frac{9}{25}, \frac{16}{25}, \frac{64}{25}, \frac{36}{25}$.

The sides of these squares *plus* $\frac{1}{2}$ in each case are the sides of the required squares.

Therefore sides of required squares are $\frac{11}{10}, \frac{13}{10}, \frac{21}{10}, \frac{17}{10}$,

and the squares themselves $\frac{121}{100}, \frac{169}{100}, \frac{441}{100}, \frac{289}{100}$.

31. To divide unity into two parts such that, if given numbers are added to them respectively, the product of the two sums gives a square.

Let 3, 5 be the numbers to be added; x, $1 - x$ the parts of 1.

Therefore $(x + 3)(6 - x) = 18 + 3x - x^2 =$ a square $= 4x^2$, say;

thus $18 + 3x = 5x^2$, *which does not give a rational result.*

Now 5 comes from a square + 1; and, in order that the equation may have a rational solution, we must substitute for the square taken (4) a square such that

$$(\text{the square} + 1) \cdot 18 + (\tfrac{3}{2})^2 = \text{a square}.$$

Put $\quad (m^2 + 1)18 + 2\frac{1}{4} =$ a square,

or $\quad 72m^2 + 81 =$ a square $= (8m + 9)^2$, say,

and $\quad m = 18, \ m^2 = 324$.

Hence we must put

$$(x + 3)(6 - x) = 18 + 3x - x^2 = 324x^2.$$

Therefore[1] $\quad 325x^2 - 3x - 18 = 0$,

$$x = \tfrac{78}{325} = \tfrac{6}{25},$$

and $\left(\frac{6}{25}, \frac{19}{25}\right)$ is a solution.

Otherwise thus.

The numbers to be added being 3, 5, assume the first of the two parts to be $x - 3$; the second is then $4 - x$.

Therefore $\quad x(9 - x) =$ a square $= 4x^2$, say,

and $\quad x = \frac{9}{5}$.

But I cannot take 3 *from* $\frac{9}{5}$, and x must be > 3 and < 4.

[1] Observe the solution of a mixed quadratic equation.

Now the value of x comes from $9/(\text{a square} + 1)$, and, since $x > 3$, this square $+ 1$ should be < 3, so that the square must be less than 2; but, since $x < 4$, the square $+ 1$ must be $> \frac{9}{4}$, so that the square must be $> \frac{5}{4}$.

Therefore I must find a square lying between $\frac{5}{4}$ and 2, or between $\frac{80}{64}$ and $\frac{128}{64}$.

$\frac{100}{64}$ or $\frac{25}{16}$ satisfies the condition.

Put now $$x(9-x) = \tfrac{25}{16}x^2;$$

therefore $$x = \tfrac{144}{41},$$

and $\left(\frac{21}{41}, \frac{20}{41}\right)$ is a solution.

32. To divide a given number into three parts such that the product of the first and second $\pm$ the third gives a square.

Given number 6.

Suppose third part $= x$, second $=$ any number less than 6, say 2; therefore first part $= 4 - x$.

The two remaining conditions require that $8 - 2x \pm x =$ a square,

or $$\left.\begin{matrix} 8 - x \\ 8 - 3x \end{matrix}\right\} \text{ are both squares. [Double-equation.]}$$

This *does not give a rational result* ("is not rational"), *since the ratio of the coefficients of x is not a ratio of a square to a square.*

But the coefficients of x are $2 - 1$ and $2 + 1$; therefore we must find a number y to replace 2 such that
$(y + 1)/(y - 1) =$ ratio of square to square $= \frac{4}{1}$, say.

Therefore $y + 1 = 4y - 4$, and $y = \frac{5}{3}$.

Put now second part $= \frac{5}{3}$; therefore first $= \frac{13}{3} - x$.

Therefore $$\tfrac{65}{9} - \tfrac{5}{3}x \pm x = \text{a square}.$$

That is, $$\left.\begin{matrix} 65 - 6x \\ 65 - 24x \end{matrix}\right\} \text{ are both squares,}$$

or $$\left.\begin{matrix} 260 - 24x \\ 65 - 24x \end{matrix}\right\} \text{ are both squares.}$$

The difference $= 195 = 15 \cdot 13$;

we put therefore $\frac{1}{4}(15 - 13)^2 = 65 - 24x$, and $x = \frac{8}{3}$.

Therefore the required parts are $\left(\frac{5}{3}, \frac{5}{3}, \frac{8}{3}\right)$[1].

[1] Fermat observes: "The following is an easier method of solution. Divide the number 6 into two in any manner, *e.g.* into 5 and 1. Divide their product less 1, that is 4, by 6, the given number: the result is $\frac{2}{3}$. Subtracting this first from 5 and then from 1,

33. To find two numbers such that the first with a fraction of the second is to the remainder of the second in a given ratio, and also the second with the same fraction of the first is to the remainder of the first in a given ratio.

Let the first with the fraction of the second = 3 times the remainder of the second, and the second with the same fraction of the first = 5 times the remainder of the first.

[The fraction may be either an aliquot part or not, τὸ αὐτὸ μέρος or τὰ αὐτὰ μέρη as Diophantus says, following the ordinary definition of those terms ("the same *part*" or "the same *parts*"): cf. Euclid VII. Deff. 3, 4.]

Let the second $= x + 1$, and let the part of it received by the first $= 1$;

therefore the first $= 3x - 1$ (since $3x - 1 + 1 = 3x$).

Since the second *plus* the fraction of the first = 5 times the remainder of the first,

the second + the first = 6 times the remainder of the first.

And first + second $= 4x$; therefore remainder of first $= \frac{2}{3}x$, and hence the second receives from the first $3x - 1 - \frac{2}{3}x$ or $\frac{7}{3}x - 1$.

We have therefore to secure that $\frac{7}{3}x - 1$ is the same fraction of $3x - 1$ that 1 is of $x + 1$.

This requires that $(\frac{7}{3}x - 1)(x + 1) = (3x - 1) . 1$;

therefore $\frac{7}{3}x^2 + \frac{4}{3}x - 1 = 3x - 1$, and $x = \frac{5}{7}$.

Accordingly the numbers are $\frac{8}{7}$, $\frac{12}{7}$; and 1 is $\frac{7}{12}$ of the second.

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we have as remainders $\frac{13}{5}$ and $\frac{1}{3}$, which are the first two parts of the number to be divided; the third is therefore $\frac{1}{2}$."

That is, if ξ, η, ζ be the required parts of the number a, Fermat divides a into two parts x, $a - x$ and then puts

$$\xi = x - \frac{x(a-x) - 1}{a} = \frac{x^2 + 1}{a},$$

$$\eta = a - x - \frac{x(a-x) - 1}{a} = \frac{(a-x)^2 + 1}{a},$$

whence $$\zeta = a - (\xi + \eta) = \frac{2(ax - x^2 - 1)}{a} = \frac{2\{x(a-x) - 1\}}{a}.$$

The three general expressions in x satisfy the conditions, and x may be given any value $< a$.

Multiply by 7 and the numbers are 8, 12, and the fraction is $\frac{7}{12}$; but 8 is not divisible by 12: so multiply by 3, and (24, 36) is a solution.

Lemma to the next problem.

To find two numbers indeterminately such that their product together with their sum is a given number.

Given number 8.

Assume the first number to be x, the second 3.

Therefore $3x + x + 3 =$ given number $= 8$; $x = \frac{5}{4}$, and the numbers are $\frac{5}{4}$, 3.

Now $\frac{5}{4}$ arises from $(8-3)/(3+1)$, where 3 is the assumed second number.

We may accordingly put for the second number (instead of 3) any (undetermined) number whatever[1]; then, substituting this for 3 in the above expression, we have the corresponding first number.

For example, we may take $x-1$ for the second number; the first is then $9-x$ divided by x, or $\frac{9}{x} - 1$.

34. To find three numbers such that the product of any two together with the sum of those two makes a given number[2].

[1] The Greek phrase is *ἐὰν ἄρα τάξωμεν τὸν βον ςοῦ οἱουδήποτε* (*οἱουδήποτε ς* in Lemma to IV. 36), "If we make the second" (literally "*put* the second *at*") "any ς whatever." But the ς is not here, as it is in the Lemma to IV. 36, the actual x of the problem, for Diophantus goes on to say "E.g. let the second be $x-1$." In the Lemma to IV. 34 the corresponding expression is "any quantity whatever" (*ὁσουδήποτε* without ς). The present Lemma amounts to saying that, if $xy+x+y=a$, then $x=(a-y)/(y+1)$.

[2] This determinate set of equations can of course be solved, with our notation, by a simple substitution.

The equations
$$\left.\begin{aligned} yz+y+z&=a\\ zx+z+x&=b\\ xy+x+y&=c\end{aligned}\right\}$$

are equivalent to
$$\left\{\begin{aligned}(y+1)(z+1)&=a+1,\\ (z+1)(x+1)&=b+1,\\ (x+1)(y+1)&=c+1,\end{aligned}\right. \quad \text{or} \quad \left\{\begin{aligned}\eta\zeta&=a+1,\\ \zeta\xi&=b+1,\\ \xi\eta&=c+1,\end{aligned}\right.$$

where
$$\xi=x+1,\quad \eta=y+1,\quad \zeta=z+1.$$

The solution is
$$\xi=x+1=\sqrt{\left\{\frac{(b+1)(c+1)}{a+1}\right\}} \text{ etc.}$$

In order that the result may be rational, it is only necessary that $(a+1)(b+1)(c+1)$ should be a square; it is not necessary that *each* of the expressions $a+1$, $b+1$, $c+1$ should be a square, as Diophantus says.

Necessary condition. Each number must be 1 less than some square[1].

Let (product + sum) of first and second $= 8$,
" " " second and third $= 15$,
" " " third and first $= 24$.

By the first equation, if we divide (8 − second) by (second + 1), we have the first number.

Let the second number be $x - 1$.

Therefore the first $= \frac{9-x}{x} = \frac{9}{x} - 1$.

Similarly the third number $= \frac{16}{x} - 1$.

The third equation remains, which gives

$$\frac{144}{x^2} - 1 = 24, \text{ and } x = \tfrac{12}{5}.$$

The numbers are $\frac{33}{12}, \frac{7}{5}, \frac{68}{12}$,

or, when reduced to a common denominator, $\frac{165}{60}, \frac{84}{60}, \frac{340}{60}$.

Lemma to the following problem.

To find two numbers indeterminately such that their product *minus* their sum is a given number.

Given number 8.

First number x, second 3, suppose; therefore

(product) − (sum) $= 3x - x - 3 = 2x - 3 = 8$, and $x = 5\tfrac{1}{2}$.

The first number is therefore $5\tfrac{1}{2}$, the second 3.

But $5\tfrac{1}{2}$ comes from $(8+3)/(3-1)$, and we may put for 3 any number whatever.

E.g. put the second number $= x + 1$; the first is then $x + 9$ divided by x, or $1 + \frac{9}{x}$.

35. To find three numbers such that the product of any two *minus* the sum of those two is a given number[2].

Necessary condition. Each of the given numbers must be 1 less than some square[2].

Let (product − sum) of first and second $= 8$,
" " " second and third $= 15$,
" " " third and first $= 24$.

[1] See last paragraph of preceding note.

[2] The notes to IV. 34 above apply, *mutatis mutandis*, to this problem as well.

By the first equation, if we divide $(8 + \text{second})$ by $(\text{second} - 1)$, we have the first number.

Assuming $x + 1$ for the second number, we have $1 + \frac{9}{x}$ for the first.

Similarly $1 + \frac{16}{x}$ is the third number, and two conditions are satisfied.

The third gives $\frac{144}{x^2} - 1 = 24$, and $x = \frac{12}{5}$.

The numbers are $\frac{57}{12}, \frac{17}{5}, \frac{92}{12}$,

or, with a common denominator, $\frac{285}{60}, \frac{204}{60}, \frac{460}{60}$.

Lemma to the following problem.

To find two numbers indeterminately such that their product has to their sum a given ratio.

Let the given ratio be $3:1$, the first number x, the second 5.

Therefore $5x = 3(5 + x)$, $x = 7\frac{1}{2}$; and the numbers are $7\frac{1}{2}$, 5.

But $7\frac{1}{2}$ arises from 15 divided by 2, while the 15 is the second number multiplied by the given ratio, and the 2 is the excess of the second number over the ratio.

Putting therefore x (instead of 5) for the second number, we have, for the first number, $3x$ divided by $x - 3$.

The numbers are therefore $3x/(x - 3)$, x.

36. To find three numbers such that the product of any two bears to the sum of those two a given ratio.

Let product of first and second be 3 times their sum.

" " second and third be 4 times their sum.

" " third and first be 5 times their sum.

Let second number be x; the first is therefore $3x/(x - 3)$, by the Lemma, and similarly the third is $4x/(x - 4)$.

Lastly $$\frac{3x}{x-3} \cdot \frac{4x}{x-4} = 5\left(\frac{3x}{x-3} + \frac{4x}{x-4}\right),$$

or $$12x^2 = 35x^2 - 120x.$$

Therefore $x = \frac{120}{23}$,

and the numbers are $\frac{360}{51}, \frac{120}{23}, \frac{480}{28}$.

37. To find three numbers such that the product of any two has to the sum of the three a given ratio[1].

Let product of first and second = 3 times sum of the three,
" " of second and third = 4 " "
" " of third and first = 5 " "

First seek three numbers such that the product of any two has to an *arbitrary* number (say 5) the given ratio.

Then product of first and second = 15; and, if x be the second, the first is $15/x$.

The product of second and third = 20; therefore third $= 20/x$.

It follows that $20 . 15/x^2 = 25$.

And, if the ratio of 20 . 15 to 25 were that of a square to a square, the problem would be solved.

Now $15 = 3 . 5$, and 20 is $4 . 5$, the 3 and 4 being fixed by the original hypothesis, but 5 being an *arbitrary* number.

We must therefore find a number m (to replace 5) such that $12m^2/5m$ = ratio of a square to a square.

Thus $12m^2 . 5m = 60m^3$ = a square $= 900m^2$, say; and $m = 15$.

Let then the sum of the three numbers be 15.

Product of first and second is therefore 45, and first $= 45/x$.

Similarly third $= 60/x$.

Therefore $\quad 45 . 60/x^2 = 75$, and $x = 6$.

Therefore the numbers are $7\frac{1}{2}$, 6, 10, and the sum of these $= 23\frac{1}{2}$.

Now, *if this sum were* 15 *instead, the problem would be solved.*

[1] Loria (*op. cit.* p. 130) quotes this problem as an instance of Diophantus' ingenious choice of unknowns. Here the equations are, with our notation,

$$yz = a(x+y+z),$$
$$zx = b(x+y+z),$$
$$xy = c(x+y+z),$$

and Diophantus chooses as his principal unknown the sum of the three numbers, $x+y+z=w$, say.

We may then write $x = cw/y$, $z = aw/y$, so that $zx = acw^2/y^2 = bw$, and $y^2 = \frac{ac}{b}w$.

Putting $w = \frac{ac}{b}\xi^2$, we have

$$x+y+z = \frac{ac}{b}\xi^2, \quad y = \frac{ac}{b}\xi, \quad z = a\xi, \quad x = c\xi,$$

from which, by eliminating x, y, z, we obtain $\xi = (bc+ca+ab)/ac$.

Hence $\quad x = (bc+ca+ab)/a, \quad y = (bc+ca+ab)/b, \quad z = (bc+ca+ab)/c.$

Put therefore for the sum of the three numbers $15x^2$, and for the numbers themselves $7\frac{1}{2}x$, $6x$, $10x$.

Therefore $23\frac{1}{2}x = 15x^2$, so that $x = \frac{47}{30}$,

and $\frac{352\frac{1}{2}}{30}, \frac{282}{30}, \frac{470}{30}$ is a solution.

38. To find three numbers such that their sum multiplied into the first gives a triangular number, their sum multiplied into the second a square, and their sum multiplied into the third a cube.

Let the sum be x^2, and let the numbers be m/x^2, n/x^2, p/x^2, where m, n, p are a triangular number, a square and a cube respectively;

say first number $= 6/x^2$, second $4/x^2$, third $8/x^2$.

But the sum is x^2; therefore $18/x^2 = x^2$, or $18 = x^4$.

Therefore *we must replace* 18 *by some fourth power.*

But 18 = sum of a triangular number, a square and a cube.

Let x^4 be the required fourth power, which must therefore be the sum of a triangular number, a square and a cube.

Let the square be $x^4 - 2x^2 + 1$;

therefore the triangular number + the cube $= 2x^2 - 1$.

Let the cube be 8; therefore the triangular number is $2x^2 - 9$.

But 8 *times a triangular number* + 1 = *a square*; therefore $16x^2 - 71$ = a square $= (4x - 1)^2$, say; thus $x = 9$, the triangular number is 153, the square 6400 and the cube 8.

Assume then as the numbers $153/x^2$, $6400/x^2$, $8/x^2$.

Therefore $6561/x^2 = x^2$, or $x^4 = 6561$, and $x = 9$.

Thus $\left(\frac{153}{81}, \frac{6400}{81}, \frac{8}{81}\right)$ is a solution[1].

[1] The procedure may be shown more generally thus.

Let ξ, η, ζ be the required numbers; suppose

$$\xi + \eta + \zeta = x^2,$$

and

$$\xi = \frac{\alpha(\alpha+1)}{2x^2}, \quad \eta = \frac{\beta^2}{x^2}, \quad \zeta = \frac{\gamma^3}{x^2}.$$

It follows that

$$x^4 = \frac{\alpha(\alpha+1)}{2} + \beta^2 + \gamma^3.$$

Suppose now that $\beta = x^2 - z^2$ [Diophantus and Bachet assume $z = 1$].

Then

$$\frac{\alpha(\alpha+1)}{2} = 2z^2x^2 - z^4 - \gamma^3.$$

Eight times the left hand side plus 1 gives a square (by the property of triangular numbers); that is,

$$(2\alpha+1)^2 = 16z^2x^2 - 8z^4 - 8\gamma^3 + 1 = \text{a square}$$
$$= (4zx - k)^2, \text{ say,}$$

39. To find three numbers such that the difference of the greatest and the middle has to the difference of the middle and the least a given ratio, and also the sum of any two is a square.

Ratio $3:1$. Since middle number + least = square, let the square be 4.

Therefore middle > 2; let it be $x+2$, so that least $= 2-x$.

Therefore difference of greatest and middle $= 6x$, whence the greatest $= 7x+2$.

Therefore $\left.\begin{matrix} 8x+4 \\ 6x+4 \end{matrix}\right\}$ are both squares. [*Double-equation.*]

Take the difference $2x$, split it into factors, say $\frac{1}{2}x$, 4, and proceed by the rule; therefore $x = 112$.

But I cannot take 112 *from* 2; therefore x must be found to be < 2, so that $6x+4 < 16$.

Thus there are to be three squares $8x+4$, $6x+4$ and 4 (the 4 arising from 2.2), and the difference of the greatest and middle is $\frac{1}{3}$ of the difference of the middle and least.

We have therefore to find three squares having this property and such that the least $= 4$ and the middle < 16.

Let side of middle square be $z+2$; therefore excess of middle over least $= z^2+4z$, whence excess of greatest over middle $= \frac{1}{3}z^2 + 1\frac{1}{3}z$, and therefore the greatest

$$= 1\tfrac{1}{3}z^2 + 5\tfrac{1}{3}z + 4.$$

This must be a square; therefore, multiplying by 9, we have

$$12z^2 + 48z + 36 = \text{a square},$$

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whence $$x = \frac{8z^4 + 8y^3 + k^2 - 1}{8kz}.$$

But $\frac{a(a+1)}{2}$ must be integral, and therefore a integral, so that $\frac{1}{2}(4zx - k - 1)$ must be integral; that is, $\frac{8z^4 + 8y^3 - (k+1)^2}{4k}$ must be integral.

Bachet assumes that it is necessary, with Diophantus, to take $k=1$, observing that trial will show that the problem can hardly be solved otherwise. On this Fermat remarks that Bachet's trial had not been carried far enough. We may, he says, put for y^3 any cube, for instance, with side of the form $3n+1$. Suppose, for example, we take 7^3. Then [z being 1] we have to make

$$2x^2 - 344 \text{ a triangle},$$

and therefore $16x^2 - 2751$ a square, and we may take, if we please, $4x-3$ as the side of this square [so that k is in this case 3].

By varying the cubes we may use an unlimited variety of odd numbers, besides 3, as values for k which will satisfy the required condition.

Loria (*op. cit.* p. 138) points out that the problem could have been more simply solved by substituting x for x^2 and z for z^2 in the above assumptions. The ultimate expression to be made a square would then have been $16zx - 8z^2 - 8y^3 + 1$, and we could have equated this to λ^2, thus finding x.

or $\quad 3z^2 + 12z + 9 = \text{a square} = (mz - 3)^2$, say.

It follows that $z = (6m + 12)/(m^2 - 3)$, which must be < 2.

Therefore $\quad 6m + 12 < 2m^2 - 6$, or $2m^2 > 6m + 18$.

"When we solve such an equation[1], we multiply half the coefficient of x into itself—this gives 9—then multiply the coefficient of x^2 into the units—$2 . 18 = 36$—add this last number to the 9, making 45, and take the side [square root] of 45, which is not less than 7; add half the coefficient of x—making a number not less than 10—and divide the result by the coefficient of x^2; the result is not less than 5."

[$3^2 + 18 . 2 = 45$, and $\frac{1}{2}\sqrt{45} + \frac{3}{2}$ is not less than $\frac{3}{2} + \frac{7}{2}$.]

We may therefore put $m = \frac{3}{2} + \frac{7}{2}$, or 5, and we thus have

$$3z^2 + 12z + 9 = (3 - 5z)^2.$$

Therefore $z = \frac{21}{11}$, and the side of the middle square is $\frac{43}{11}$, the square itself being $\frac{1849}{121}$.

Turning to the original problem, we put $6x + 4 = \frac{1849}{121}$, and $x = \frac{1365}{726}$, which *is* less than 2.

The greatest of the required numbers $= 7x + 2 = \dfrac{11007}{726}$,

the middle $= x + 2 = \dfrac{2817}{726}$,

and the least $= 2 - x = \dfrac{87}{726}$.

The denominator not being a square, we can make it a square by dividing out by 6; the result is

$$\frac{1834\frac{1}{2}}{121},\ \frac{469\frac{1}{2}}{121},\ \frac{14\frac{1}{2}}{121},$$

or again, to avoid the $\frac{1}{2}$ in the numerators, we may multiply numerators, and denominators, by 4; thus

$$\frac{7338}{484},\ \frac{1878}{484},\ \frac{58}{484} \text{ is a solution}^2.$$

[1] I have quoted Diophantus' exact words here, with the few added by Tannery, "making a number not less than 10...coefficient of x^2," in order to show the precise rule by which Diophantus solved a complete quadratic.

When he says $\sqrt{45}$ is not less than 7, Diophantus is not seeking exact limits. Since $\sqrt{45}$ is between 6 and 7 we cannot take a smaller integral value than 7 in order to satisfy the conditions of the problem (cf. p. 65 above).

[2] A note in the *Inventum Novum* (Part II, paragraph 26) remarks upon the prolix and involved character of Diophantus' solution and gives a shorter alternative. The problem is to solve

$$\xi - \eta = m(\eta - \zeta), \qquad (\xi > \eta > \zeta, \text{ and } m = 3, \text{ say})$$
$$\eta + \zeta = u^2,$$
$$\zeta + \xi = v^2,$$
$$\xi + \eta = w^2.$$

40. To find three numbers such that the difference of the squares of the greatest and the middle numbers has to the difference of the middle and the least a given ratio, and also the sum of each pair is a square.

Ratio $3 : 1$.

Let greatest + middle number = the square $16x^2$; therefore greatest $> 8x^2$: let it be $8x^2 + 2$; hence middle $= 8x^2 - 2$.

And, since greatest + middle > greatest + least,

$$16x^2 > (\text{greatest} + \text{least}) > 8x^2;$$

let greatest + least $= 9x^2$, say; therefore least $= x^2 - 2$.

Now difference of squares of greatest and middle $= 64x^2$, and difference of middle and least $= 7x^2$.

But 64 is not equal to $3 \cdot 7$ or 21.

Now 64 comes from $32 \cdot 2$; therefore we must find a number m (in place of 2) such that $32m = 21$.

Therefore $m = \frac{21}{32}$.

Assume now greatest number $= 8x^2 + \frac{21}{32}$, middle $= 8x^2 - \frac{21}{32}$, least $= x^2 - \frac{21}{32}$.

[And difference of squares of greatest and middle

$$= 21x^2 = 3 \cdot 7x^2.]$$

The only condition left is: middle + least = square; that is,

$$9x^2 - \tfrac{42}{32} = \text{a square} = (3x - 6)^2, \text{ say}.$$

Therefore $x = \frac{597}{576}$,

and $\left(\dfrac{3069000}{331776}, \dfrac{2633544}{331776}, \dfrac{138681}{331776}\right)$ is a solution.

Take an arbitrary square number, say 4, for the sum of η, ζ; suppose $2 + x = \eta$, $2 - x = \zeta$, so that $\eta - \zeta = 2x$; therefore $\xi - \eta = 3(\eta - \zeta) = 6x$, whence $\xi = 2 + 7x$.

The last two conditions require that

$$\left.\begin{array}{l}4 + 8x \\ 4 + 6x\end{array}\right\} \text{ shall be squares.}$$

Replace x by $\dfrac{1}{6}y^2 + \dfrac{2}{3}y$. This will make $4 + 6x$ a square. It remains that

$$4 + \frac{16}{3}y + \frac{4}{3}y^2 = \text{a square}$$

$$= \left(2 + \frac{5}{4}y\right)^2, \text{ say}.$$

Thus
$$y\left(\frac{25}{16} - \frac{4}{3}\right) = \left(\frac{16}{3} - 5\right),$$

and $y = \dfrac{16}{11}$, so that $x = \dfrac{1}{6}y^2 + \dfrac{2}{3}y = \dfrac{160}{121}$.

The numbers are therefore $\dfrac{1362}{121}, \dfrac{402}{121}, \dfrac{82}{121}$.

BOOK V

1. To find three numbers in geometrical progression such that each of them *minus* a given number gives a square.

Given number 12.

Find a *square* which exceeds 12 by a square. "This is easy [II. 10]; $42\frac{1}{4}$ is such a number."

Let the first number be $42\frac{1}{4}$, the third x^2; therefore the middle number $= 6\frac{1}{2}x$.

Therefore $\left.\begin{array}{r} x^2 - 12 \\ 6\frac{1}{2}x - 12 \end{array}\right\}$ are both squares;

their difference $= x^2 - 6\frac{1}{2}x = x(x - 6\frac{1}{2})$; half the difference of the factors multiplied into itself $= \frac{169}{16}$; therefore, putting $6\frac{1}{2}x - 12 = \frac{169}{16}$, we have $x = \frac{361}{104}$,

and $\left(42\frac{1}{4}, \frac{2346\frac{1}{2}}{104}, \frac{130321}{10816}\right)$ is a solution.

2. To find three numbers in geometrical progression such that each of them when added to a given number gives a square.

Given number 20.

Take a square which when added to 20 gives a square, say 16.

Put for one of the extremes 16, and for the other x^2, so that the middle term $= 4x$.

Therefore $\left.\begin{array}{r} x^2 + 20 \\ 4x + 20 \end{array}\right\}$ are both squares.

Their difference is $x^2 - 4x = x(x - 4)$, and the usual method gives $4x + 20 = 4$, *which is absurd*, because the 4 ought to be some number greater than 20.

But the $4 = \frac{1}{4}(16)$, while the 16 is a square which when added to 20 makes a square; therefore, to replace 16, we must find some square greater than $4 . 20$ and such that when increased by 20 it makes a square.

Now $81 > 80$; therefore, putting $(m + 9)^2$ for the required square, we have

$$(m + 9)^2 + 20 = \text{square} = (m - 11)^2, \text{ say};$$

therefore $m = \frac{1}{2}$, and the square $= (9\frac{1}{2})^2 = 90\frac{1}{4}$.

Assume now for the numbers $90\frac{1}{4}$, $9\frac{1}{2}x$, x^2, and we have

$$\left.\begin{array}{r} x^2+20 \\ 9\frac{1}{2}x+20 \end{array}\right\} \text{both squares.}$$

The difference $=x(x-9\frac{1}{2})$, and we put $9\frac{1}{2}x+20=\frac{361}{16}$.
Therefore $x=\frac{41}{152}$, and

$$\left(90\frac{1}{4},\ \frac{389\frac{1}{2}}{152},\ \frac{1681}{23104}\right) \text{ is a solution.}$$

3. Given one number, to find three others such that any one of them, or the product of any two of them, when added to the given number, gives a square.

Given number 5.

"*We have it in the Porisms* that if, of two numbers, each, as well as their product, when added to one and the same given number, severally make squares, the two numbers are obtained from the squares of consecutive numbers[1]."

Take then the squares $(x+3)^2$, $(x+4)^2$, and, subtracting the given number 5 from each, put for the first number x^2+6x+4, and for the second $x^2+8x+11$, and let the third[2] be twice their sum *minus* 1, or

$$4x^2+28x+29.$$

[1] On this Porism, see pp. 99, 100 *ante*.

[2] The Porism states that, if a be the given number, the numbers x^2-a, $(x+1)^2-a$ satisfy the conditions.

In fact, their product $+a=\{x(x+1)\}^2-a(2x^2+2x+1)+a^2+a$

$$=\{x(x+1)\}^2-2ax(x+1)+a^2=\{x(x+1)-a\}^2.$$

Diophantus here adds, without explanation, that, if X, Y denote the above two numbers, we should assume for the third required number $Z=2(X+Y)-1$. We want *three* numbers such that *any two* satisfy the same conditions as X, Y. Diophantus takes for the third $Z=2(X+Y)-1$ because, as is easily seen, with this assumption two out of the three additional conditions are thereby satisfied.

For
$$\begin{aligned} Z=2(X+Y)-1&=2(2x^2+2x+1)-4a-1 \\ &=(2x+1)^2-4a; \end{aligned}$$

therefore
$$\begin{aligned} XZ+a&=x^2(2x+1)^2-a\{(2x+1)^2+4x^2\}+4a^2+a \\ &=x^2(2x+1)^2-a\cdot 4x(2x+1)+4a^2 \\ &=\{x(2x+1)-2a\}^2, \end{aligned}$$

while
$$\begin{aligned} YZ+a&=(x+1)^2(2x+1)^2-a\{(2x+1)^2+4(x+1)^2\}+4a^2+a \\ &=(x+1)^2(2x+1)^2-a(8x^2+12x+4)+4a^2 \\ &=\{(x+1)(2x+1)-2a\}^2. \end{aligned}$$

The only condition remaining is then

$$Z+a=\text{a square},$$

or
$$(2x+1)^2-3a=\text{a square}=(2x-k)^2, \text{ say,}$$

and x is found.

Cf. pp. 100, 104 above.

Therefore $4x^2+28x+34 = \text{a square} = (2x-6)^2$, say.

Hence $x = \frac{1}{26}$, and $\left(\frac{2861}{676}, \frac{7645}{676}, \frac{20336}{676}\right)$ is a solution[1].

4. Given one number, to find three others such that any one of them, or the product of any two, *minus* the given number gives a square.

Given number 6.

Take two consecutive squares x^2, x^2+2x+1.

Adding 6 to each, we assume for the first number x^2+6, and for the second x^2+2x+7.

For the third[2] we take twice the sum of the first and second *minus* 1, or $4x^2+4x+25$.

Therefore third *minus* $6 = 4x^2+4x+19 = \text{square} = (2x-6)^2$, say.

Therefore $x = \frac{17}{28}$,

and $\left(\frac{4993}{784}, \frac{6729}{784}, \frac{22660}{784}\right)$ is a solution.

[The same Porism is assumed as in the preceding problem but with a *minus* instead of a *plus*. Cf. p. 99 above.]

5. To find three squares such that the product of any two added to the sum of those two, or to the remaining square, gives a square.

"*We have it in the Porisms*" that, if the squares on any two consecutive numbers be taken, and a third number be also taken which exceeds twice the sum of the squares by 2, we have three numbers such that the product of any two added to those two or to the remaining number gives a square[3].

[1] Diophantus having solved the problem of finding three numbers ξ, η, ζ satisfying the six equations

$$\xi+a=r^2, \qquad \eta\zeta+a=u^2,$$
$$\eta+a=s^2, \qquad \zeta\xi+a=v^2,$$
$$\zeta+a=t^2, \qquad \xi\eta+a=w^2,$$

Fermat observes that we can deduce the solution of the problem

To find four numbers such that the product of any pair added to a given number produces a square.

Taking three numbers, as found by Diophantus, satisfying the above six conditions, we take $x+1$ as the fourth number. We then have three conditions which remain to be satisfied. These give a "triple-equation" to be solved by Fermat's method.

[2] Diophantus makes this assumption for the same reason as in the last problem, v. 3. The second note on p. 201 covers this case if we substitute $-a$ for a throughout.

[3] On this Porism, see pp. 100–1 *ante*.

Assume as the first square x^2+2x+1, and as the second x^2+4x+4, so that third number $=4x^2+12x+12$.

Therefore $x^2+3x+3=\text{a square}=(x-3)^2$, say, and $x=\frac{2}{3}$.

Therefore $\left(\frac{25}{9}, \frac{64}{9}, \frac{196}{9}\right)$ is a solution.

6. To find three numbers such that each *minus* 2 gives a square, and the product of any two *minus* the sum of those two, or *minus* the remaining number, gives a square.

Add 2 to each of three numbers found as in the Porism quoted in the preceding problem.

Let the numbers so obtained be x^2+2, x^2+2x+3, $4x^2+4x+6$.

All the conditions are now satisfied[1], except one, which gives

$$4x^2+4x+6-2=\text{a square}.$$

Divide by 4, and $x^2+x+1=\text{a square}=(x-2)^2$, say.

Therefore $x=\frac{3}{5}$,

and $\left(\frac{59}{25}, \frac{114}{25}, \frac{246}{25}\right)$ is a solution.

Lemma I to the following problem.

To find two numbers such that their product added to the squares of both gives a square.

Suppose first number x, second any number (m), say 1.

Therefore $x \cdot 1+x^2+1=x^2+x+1=\text{a square}=(x-2)^2$, say.

Thus $x=\frac{3}{5}$, and

$\left(\frac{3}{5}, 1\right)$ is a solution, or $(3, 5)$.

Lemma II to the following problem.

To find three right-angled triangles (*i.e.* three right-angled triangles *in rational numbers*[2]) which have equal areas.

We must first find two numbers such that their product + the sum of their squares = a square, *e.g.* 3, 5, as in the preceding problem.

[1] The numbers are x^2+2, $(x+1)^2+2$, $2\{x^2+(x+1)^2+1\}+2$; and if X, Y, Z denote these numbers respectively, it is easily verified that

$$XY-(X+Y)=(x^2+x+1)^2, \qquad XY-Z=(x^2+x)^2,$$
$$XZ-(X+Z)=(2x^2+x+2)^2, \qquad XZ-Y=(2x^2+x+3)^2,$$

and
$$YZ-(Y+Z)=(2x^2+3x+3)^2, \qquad YZ-X=(2x^2+3x+4)^2.$$

[2] All Diophantus' right-angled triangles must be understood to be right-angled triangles with sides expressible in rational numbers. In future I shall say "right-angled triangle" simply, for brevity.

Now form right-angled triangles from the pairs of numbers[1]

$$(7, 3),\ (7, 5),\ (7, 3+5)$$

[*i.e.* the right-angled triangles $(7^2+3^2, 7^2-3^2, 2.7.3)$, etc.].

The triangles are (40, 42, 58), (24, 70, 74), (15, 112, 113), the area of each being 840.

[1] Diophantus here tacitly assumes that, if $ab+a^2+b^2=c^2$, and right-angled triangles be formed from (c, a), (c, b) and $(c, a+b)$ respectively, their areas are equal. The areas are of course $(c^2-a^2)ca$, $(c^2-b^2)cb$ and $\{(a+b)^2-c^2\}(a+b)c$, and it is easy to see that each $=abc(a+b)$.

Nesselmann suggests that Diophantus discovered the property as follows. Let the triangles formed from (n, m), (q, m), (r, m) have their areas equal; therefore

$$n(m^2-n^2)=q(m^2-q^2)=r(r^2-m^2).$$

It follows, first, since $\quad m^2n-n^3=m^2q-q^3,$

that $\quad m^2=(n^3-q^3)/(n-q)=n^2+nq+q^2.$

Again, given (q, m, n), to find r.

We have $\quad q(m^2-q^2)=r(r^2-m^2),$

and $m^2-q^2=n^2+nq$, from above;

therefore $\quad q(n^2+nq)=r(r^2-n^2-nq-q^2),$

or $\quad q(n^2+nr)+q^2(n+r)=r(r^2-n^2).$

Dividing by $r+n$, we have $\quad qn+q^2=r^2-rn;$

therefore $\quad (q+r)n=r^2-q^2,$

and $\quad r=q+n.$

Fermat observes that, given any rational right-angled triangle, say z, b, d, where z is the hypotenuse, it is possible to find an infinite number of other rational right-angled triangles having the same area. Form a right-angled triangle from $z^2, 2bd$; this gives the triangle $z^4+4b^2d^2$, $z^4-4b^2d^2$, $4z^2bd$. Divide each of these sides by $2z(b^2-d^2)$, b being $>d$; and we have a triangle with the same area $\left(\frac{1}{2}bd\right)$ as the original triangle. Trying this method with Diophantus' first triangle (40, 42, 58), we obtain as the new triangle

$$\frac{1412881}{1189},\ \frac{1412880}{1189},\ \frac{1681}{1189}.$$

The method gives $\left(\frac{7}{10}, \frac{120}{7}, \frac{1201}{70}\right)$ as a right-angled triangle with area equal to that of (3, 4, 5).

Another method of finding other rational right-angled triangles having the same area as a given right-angled triangle is explained in the *Inventum Novum*, Part I, paragraph 38 (*Œuvres de Fermat*, III. p. 348).

Let the given triangle be 3, 4, 5, so that it is required to find a new rational right-angled triangle with area 6.

Let $3, x+4$ be the perpendicular sides; therefore

the square of the hypotenuse $=x^2+8x+25=$ a square.

Again, the area is $\frac{3}{2}x+6$; and, as this is to be 6, it must be six times a certain square; that is, $\frac{3}{2}x+6$ divided by 6 must be a square, and this again multiplied by 36 must be a square; therefore

$$9x+36=\text{a square}.$$

Accordingly we have to solve the double-equation

$$\left.\begin{array}{r} x^2+8x+25=u^2 \\ 9x+36=v^2 \end{array}\right\}.$$

7. To find three numbers such that the square of any one $\pm$ the sum of the three gives a square.

Since, in a right-angled triangle,

$$(\text{hypotenuse})^2 \pm \text{twice product of perps.} = \text{a square},$$

we make the three required numbers hypotenuses and the sum of the three four times the area.

Therefore we must find three right-angled triangles having the same area, *e.g.*, as in the preceding problem,

$$(40, 42, 58),\ (24, 70, 74),\ (15, 112, 113).$$

Reverting to the substantive problem, we put for the numbers $58x, 74x, 113x$; their sum $245x$ = four times the area of any one of the triangles $= 3360x^2$.

Therefore $x = \frac{7}{96}$,

and $\left(\frac{406}{96}, \frac{518}{96}, \frac{791}{96}\right)$ is a solution.

Lemma to the following problem.

Given three squares, it is possible to find three numbers such that the products of the three pairs shall be respectively equal to those squares.

This gives $$x = -\frac{6725600}{2405601},$$

whence $$x+4 = \frac{2896804}{2405601}.$$

The triangle is thus found to be

$$3, \quad \frac{2896804}{2405601}, \quad \frac{7776485}{2405601}.$$

The area is 6 times a certain square, namely $\frac{724201}{2405601}$, the root of which is $\frac{851}{1551}$.

Dividing each of the above sides by $\frac{851}{1551}$, we obtain a triangle with area 6, namely

$$\frac{4653}{851}, \quad \frac{3404}{1551}, \quad \frac{7776485}{1319901}.$$

Another solution of the double-equation, $x = -\frac{13959}{3600}$, giving $x+4 = \frac{49}{400}$, leads to the same triangle $\left(\frac{7}{10}, \frac{120}{7}, \frac{1201}{70}\right)$ as that obtained by Fermat's rule (see above).

The method of the *Inventum Novum* has a feature in common with the procedure in the ancient Greek problem reproduced and commented on by Heiberg and Zeuthen (*Bibliotheca Mathematica*, VIII₃, 1907/8, p. 122), where it is required to find a rational right-angled triangle, having given the area, 5 feet, and where the 5 is multiplied by a square number containing 6 as a factor and such that the product "can form the area of a right-angled triangle." 36 is taken and the area becomes 180, which is the area of (9, 40, 41). The sides of the latter triangle are then divided by 6, and we have the required triangle (cf. p. 119, *ante*).

Squares 4, 9, 16.

First number x, so that the others are $4/x$, $9/x$; and $36/x^2=16$.

Therefore $x=\frac{6}{4}$, and the numbers are $(1\frac{1}{2}, 2\frac{2}{3}, 6)$.

We observe that $x=\frac{6}{4}$, where 6 is the product of 2 and 3, and 4 is the side of 16.

Hence the following *rule*. Take the product of two sides (2, 3), divide by the side of the third square 4 [the result is the first number]; divide 4, 9 respectively by the result, and we have the second and third numbers.

8. To find three numbers such that the product of any two ± the sum of the three gives a square.

As in Lemma II to the 7th problem, we find three right-angled triangles with equal areas; the squares of their hypotenuses are 3364, 5476, 12769.

Now find, as in the last Lemma, three numbers such that the products of the three pairs are equal to these squares respectively, which we take because each ± 4.(area) or 3360 gives a square; the three numbers then are

$\frac{4292}{113}x$, $\frac{3277}{37}x$ [$\frac{380132}{4292}x$ Tannery],
$\frac{4181}{29}x$ [$\frac{618788}{4292}x$ Tannery].

It remains that the sum of the three $= 3360x^2$.

Therefore $\frac{32824806}{121249}x$ [$\frac{131299224}{484996}x$ Tannery] $= 3360x^2$.

Therefore $x=\frac{32824806}{407396640}$ [$\frac{131299224}{1629586560}$ or $\frac{781543}{9699920}$ Tannery],

[and the numbers are $\frac{781543}{255380}$, $\frac{781543}{109520}$, $\frac{781543}{67280}$].

9. To divide unity into two parts such that, if the same given number be added to either part, the result will be a square.

Necessary condition. The given number must not be odd and the double of it + 1 must not be divisible by any prime number which, when increased by 1, is divisible by 4 [*i.e.* any prime number of the form $4n-1$][1].

Given number 6. Therefore 13 must be divided into two squares each of which > 6. If then we divide 13 into two squares the difference of which < 1, we solve the problem.

[1] For a discussion of the text of this condition see pp. 107–8, *ante*.

Take half of 13 or $6\frac{1}{2}$, and we have to add to $6\frac{1}{2}$ a small fraction which will make it a square,

or, multiplying by 4, we have to make $\frac{1}{x^2}+26$ a square,

i.e. $26x^2+1=$ a square $=(5x+1)^2$, say, whence $x=10$.

That is, in order to make 26 a square, we must add $\frac{1}{100}$, or, to make $6\frac{1}{2}$ a square, we must add $\frac{1}{400}$, and

$$\frac{1}{400}+6\frac{1}{2}=\left(\frac{51}{20}\right)^2.$$

Therefore we must divide 13 *into two squares such that their sides may be as nearly as possible equal to* $\frac{51}{20}$. [This is the παρισότητος ἀγωγή described above, pp. 95–8.]

Now $13=2^2+3^2$. Therefore we seek two numbers such that 3 *minus* the first $=\frac{51}{20}$, so that the first $=\frac{9}{20}$, and 2 *plus* the second $=\frac{51}{20}$, so that the second $=\frac{11}{20}$.

We write accordingly $(11x+2)^2$, $(3-9x)^2$ for the required squares [substituting x for $\frac{1}{20}$].

The sum $=202x^2-10x+13=13$.

Therefore $x=\frac{5}{101}$, and the sides are $\frac{257}{101}$, $\frac{258}{101}$.

Subtracting 6 from the squares of each, we have, as the parts of unity,

$$\frac{4843}{10201},\ \frac{5358}{10201}.$$

10. To divide unity into two parts such that, if we add different given numbers to each, the results will be squares.

Let the numbers[1] be 2, 6 and let them and the unit be represented in the figure, where $DA=2$, $AB=1$, $BE=6$, and G is a point in AB so chosen that DG, GE are both squares.

Now $DE=9$. *Therefore we have to divide* 9 *into two squares such that one of them lies between* 2 *and* 3.

Let the latter square be x^2, so that the other is $9-x^2$, where $3>x^2>2$.

Take two squares, one >2, the other <3 [the former being the smaller], say $\frac{289}{144}$, $\frac{361}{144}$.

[1] Loria (*op. cit.* p. 150 *n.*), as well as Nesselmann, observes that Diophantus omits to state the necessary condition, namely that the sum of the two given numbers *plus* 1 must be the sum of two squares.

Therefore, if we can make x^2 lie between these, we shall solve the problem.

We must have $x > \frac{17}{12}$ and $< \frac{19}{12}$.

Hence, in making $9 - x^2$ a square, we must find

$$x > \tfrac{17}{12} \text{ and } < \tfrac{19}{12}.$$

Put $9 - x^2 = (3 - mx)^2$, say, whence $x = 6m/(m^2 + 1)$.

Therefore $$\frac{17}{12} < \frac{6m}{m^2+1} < \frac{19}{12}.$$

The first inequality gives $72m > 17m^2 + 17$; and

$$36^2 - 17 \cdot 17 = 1007,$$

the square root of which[1] is not greater than 31;

therefore $m \ngtr \dfrac{31+36}{17}$, *i.e.* $m \ngtr \dfrac{67}{17}$.

Similarly from the inequality $19m^2 + 19 > 72m$ we find[2]

$$m \nless \tfrac{66}{19}.$$

Let $m = 3\frac{1}{2}$. Therefore $9 - x^2 = (3 - 3\frac{1}{2}x)^2$, and $x = \frac{84}{53}$.

Therefore $x^2 = \frac{7056}{2809}$,

and the segments of 1 are $\left(\frac{1438}{2809}, \frac{1371}{2809}\right)$.

11. To divide unity into three parts such that, if we add the same number to each of the parts, the results are all squares.

Necessary condition[3]. The given number must not be 2 or any multiple of 8 increased by 2.

Given number 3. Thus 10 is to be divided into three squares such that each > 3.

Take $\frac{1}{3}$ of 10, or $3\frac{1}{3}$, and find x so that $\dfrac{1}{9x^2} + 3\frac{1}{3}$ may be a square, or $30x^2 + 1 =$ a square $= (5x + 1)^2$, say.

Therefore $x = 2$, $x^2 = 4$, $1/x^2 = \frac{1}{4}$, and

$$\tfrac{1}{36} + 3\tfrac{1}{3} = \tfrac{121}{36} = \text{a square.}$$

Therefore we have to divide 10 into three squares each of which is as near as possible to $\frac{121}{36}$. [παρισότητος ἀγωγή.]

Now $10 = 3^2 + 1^2 =$ the sum of the three squares $9, \frac{16}{25}, \frac{9}{25}$.

Comparing the sides $3, \frac{4}{5}, \frac{3}{5}$ with $\frac{11}{6}$,

or (multiplying by 30) 90, 24, 18 with 55, we must make each side approach 55.

[1] *I.e.* the *integral* part of the root is $\ngtr 31$. The limits taken in each case are *a fortiori* limits as explained above, pp. 61–3.

[2] See p. 61, *ante*.

[3] See pp. 108–9, *ante*.

[Since then $\frac{55}{30} = 3 - \frac{35}{30} = \frac{4}{5} + \frac{31}{30} = \frac{3}{5} + \frac{37}{30}$], we put for the sides of the required numbers

$$3 - 35x, \quad 31x + \tfrac{4}{5}, \quad 37x + \tfrac{3}{5}.$$

The sum of the squares $= 3555x^2 - 116x + 10 = 10$.

Therefore $x = \frac{116}{3555}$,

and this solves the problem.

12. To divide unity into three parts such that, if three different given numbers be added to the parts respectively, the results are all squares.

Given numbers 2, 3, 4. Then I have to divide 10 into three squares such that the first > 2, the second > 3, and the third > 4.

Let us add $\frac{1}{2}$ of unity to each, and we have to find three squares such that their sum is 10, while the first lies between 2, $2\frac{1}{2}$, the second between 3, $3\frac{1}{2}$, and the third between 4, $4\frac{1}{2}$.

It is necessary, first, to divide 10 (the sum of two squares) into two squares one of which lies between 2, $2\frac{1}{2}$; then, if we subtract 2 from the latter square, we have one of the required parts of unity.

Next divide the other square into two squares, one of which lies between 3, $3\frac{1}{2}$;

subtracting 3 from the latter square, we have the second of the required parts of unity.

Similarly we can find the third part[1].

[1] Diophantus only thus briefly indicates the course of the solution. Wertheim solves the problem in detail after Diophantus' manner; and, as this is by no means too easy, I think it well to reproduce his solution.

I. It is first necessary to divide 10 into two squares one of which lies between 2 and 3. We use the παρισότητος ἀγωγή.

The first square must be in the neighbourhood of $2\frac{1}{2}$; and we seek a small fraction $\frac{1}{x^2}$ which when added to $2\frac{1}{2}$ gives a square: in other words, we must make $4\left(2\frac{1}{2} + \frac{1}{x^2}\right)$ a square. This expression may be written $10 + \left(\frac{1}{y}\right)^2$, and, to make this a square, we put

$$10y^2 + 1 = (3y + 1)^2, \text{ say,}$$

whence $y = 6$, $y^2 = 36$, $x^2 = 144$, so that $2\frac{1}{2} + \frac{1}{x^2} = \frac{361}{144} = \left(\frac{19}{12}\right)^2$, which is an approximation to the first of the two squares the sum of which is 10.

The second of these squares approximates to $7\frac{1}{2}$, and we seek a small fraction $\frac{1}{x^2}$ such that $7\frac{1}{2} + \frac{1}{x^2}$ is a square, or $30 + \left(\frac{2}{x}\right)^2 = 30 + \left(\frac{1}{y}\right)^2$, say $=$ a square.

13. To divide a given number into three parts such that the sum of any two of the parts gives a square.

Given number 10.

Put $$30y^2+1=(5y+1)^2, \text{ say};$$

therefore $y=2$, $y^2=4$, $x^2=16$, so that $7\frac{1}{2}+\frac{1}{x^2}=\frac{121}{16}=\left(\frac{11}{4}\right)^2=\left(\frac{33}{12}\right)^2$.

Now, since $10=1^2+3^2$, and $\frac{19}{12}=1+\frac{7}{12}$, while $\frac{33}{12}=3-\frac{3}{12}$,

we put $$(1+7x)^2+(3-3x)^2=10, \qquad \text{[Cf. V. 9]}$$

so that $$x=\frac{2}{29},$$

$$(1+7x)^2=\left(1+\frac{14}{29}\right)^2=\left(\frac{43}{29}\right)^2=\frac{1849}{841},$$

$$(3-3x)^2=\left(3-\frac{6}{29}\right)^2=\left(\frac{81}{29}\right)^2=\frac{6561}{841}.$$

Therefore the two squares into which 10 is divided are $\frac{1849}{841}$ and $\frac{6561}{841}$, and the first of these lies in fact between 2 and $2\frac{1}{2}$.

II. We have next to divide the square $\frac{6561}{841}$ into two squares, one of which, which we will call x^2, lies between 3 and 4. [The method of V. 10 is here applicable.]

Instead of 3, 4 take $\frac{49}{16}, \frac{64}{16}$ as the limits.

Therefore $$\frac{49}{16}<x^2<\frac{64}{16},$$

or $$\frac{7}{4}<x<\frac{8}{4}.$$

And $\frac{6561}{841}-x^2$ must be a square $=\left(\frac{81}{29}-kx\right)^2$, say,

which gives $$x=\frac{162k}{29(1+k^2)};$$

k has now to be chosen such that

(1) $$\frac{162k}{29(1+k^2)}>\frac{7}{4},$$

from which it follows that $$k<2{\cdot}8\ldots,$$

and (2) $$\frac{162k}{29(1+k^2)}<\frac{8}{4},$$

whence $$k>2{\cdot}3\ldots.$$

We may therefore put $$k=2{\cdot}5.$$

Therefore $$x=\frac{1620}{841},\quad x^2=\frac{2624400}{707281},$$

and $$\frac{6561}{841}-x^2=\frac{2893401}{707281}.$$

The three required squares into which 10 is divided are therefore

$$\frac{1849}{841},\quad \frac{2624400}{707281},\quad \frac{2893401}{707281}.$$

And if we subtract 2 from the first, 3 from the second and 4 from the third, we obtain as the required parts of unity

$$\frac{140447}{707281},\quad \frac{502557}{707281},\quad \frac{64277}{707281}.$$

Since the sum of each pair of parts is a square less than 10, while the sum of the three pairs is twice the sum of the three parts or 20,

we have to divide 20 *into three squares each of which is* < 10.

But 20 is the sum of two squares, 16 and 4;

and, if we put 4 for one of the required squares, we have to divide 16 into two squares, each of which is < 10, or, in other words, into two squares, one of which lies between 6 and 10. This we learnt how to do[1] [V. 10].

We have, when this is done, three squares such that each is < 10, while their sum is 20;

and by subtracting each of these squares from 10 we obtain the parts of 10 required.

14. To divide a given number into four parts such that the sum of any three gives a square.

Given number 10.

Three times the sum of the parts = the sum of four squares.

Therefore 30 has to be divided into four squares, each of which is < 10.

(1) If we use the method of approximation (παρισότης), we have to make each square approximate to $7\frac{1}{2}$;

[1] Wertheim gives a solution in full, thus.

Let the squares be x^2, $16-x^2$, of which one, x^2, lies between 6 and 10.

Put instead of 6 and 10 the limits $\frac{25}{4}$ and 9, so that

$$\frac{5}{2} < x < 3.$$

To make $16 - x^2$ a square, we put

$$16 - x^2 = (4 - kx)^2,$$

whence
$$x = \frac{8k}{1+k^2}.$$

Now (1) $\frac{8k}{1+k^2} > \frac{5}{2}$, and (2) $\frac{8k}{1+k^2} < 3$.

These conditions give, as limits for k, $2\cdot84\ldots$ and $2\cdot21\ldots$.

We may therefore *e.g.* put $k = 2\frac{1}{2}$.

Then
$$x = \frac{80}{29}, \quad x^2 = \frac{6400}{841}, \quad 16 - x^2 = \frac{7056}{841}.$$

The required three squares making up 20 are $4, \frac{6400}{841}, \frac{7056}{841}$.

Subtracting these respectively from 10, we have the required parts of the given number 10, namely
$$6, \quad \frac{2010}{841}, \quad \frac{1354}{841}.$$

then, when the squares are found, we subtract each from 10, and so find the required parts.

(2) *Or*, observing that $30 = 16 + 9 + 4 + 1$, we take 4, 9 for two of the squares, and then divide 17 into two squares, each of which < 10.

If then we divide 17 into two squares, one of which lies between $8\frac{1}{2}$ and 10, as we have learnt how to do[1] [cf. V. 10], the squares will satisfy the conditions.

We shall then have divided 30 into four squares, each of which is less than 10, two of them being 4, 9 and the other two the parts of 17 just found.

Subtracting each of the four squares from 10, we have the required parts of 10, two of which are 1 and 6.

15. To find three numbers such that the cube of their sum added to any one of them gives a cube.

Let the sum be x and the cubes $7x^3$, $26x^3$, $63x^3$.

Therefore $96x^3 = x$, or $96x^2 = 1$.

But 96 is not a square; we must therefore replace it by a square in order to solve the problem.

[1] Wertheim gives a solution of this part of the problem.

As usual, we make $8\frac{1}{2} + \frac{1}{x^2}$, or $34 + \left(\frac{2}{x}\right)^2$, a square.

Putting $\frac{2}{x} = \frac{1}{y}$, we must make $34 + \left(\frac{1}{y}\right)^2$ a square.

Let $$34y^2 + 1 = \text{a square} = (6y - 1)^2,$$
and we obtain $$y = 6,\ y^2 = 36,\ x^2 = 144.$$
Thus $$8\tfrac{1}{2} + \frac{1}{144} = \frac{1225}{144} = \left(\frac{35}{12}\right)^2,$$
and $\frac{35}{12}$ is an approximation to the side of each of the required squares.

Next, since $17 = 1^2 + 4^2$, and $\frac{35}{12} = 1 + \frac{23}{12} = 4 - \frac{13}{12}$,
we put $$17 = (1 + 23x)^2 + (4 - 13x)^2,$$
and we obtain $$x = \frac{29}{349}.$$
The squares are then $$(1 + 23x)^2 = \left(\frac{1016}{349}\right)^2 = \frac{1032256}{121801},$$
and $$(4 - 13x)^2 = \left(\frac{1019}{349}\right)^2 = \frac{1038361}{121801}.$$

Subtracting each of these from 10, we have the third and fourth of the required parts of 10, namely
$$\frac{185754}{121801},\ \frac{179649}{121801}.$$

Now 96 is the sum of three numbers, each of which is 1 less than a cube;

therefore we have to find three numbers such that each of them is a cube less 1, and the sum of the three is a square.

Let the sides of the cubes[1] be $m+1$, $2-m$, 2, whence the numbers are m^3+3m^2+3m, $7-12m+6m^2-m^3$, 7; their sum $=9m^2-9m+14=$ a square $=(3m-4)^2$, say;

therefore $m=\frac{2}{15}$,

and the numbers are $\frac{1538}{3375}$, $\frac{18577}{3375}$, 7.

Reverting to the original problem, we put x for the sum of the numbers, and for the numbers respectively

$$\tfrac{1538}{3375}x^3,\ \tfrac{18577}{3375}x^3,\ 7x^3,$$

whence $\frac{43740}{3375}x^3=x$,

that is (if we divide out by 15 and by x),

$$2916x^2=225, \text{ and } x=\tfrac{15}{54}.$$

The numbers are therefore found.

16. To find three numbers such that the cube of their sum *minus* any one of them gives a cube.

Let the sum be x, and the numbers $\frac{7}{8}x^3$, $\frac{26}{27}x^3$, $\frac{63}{64}x^3$.

Therefore $\frac{4877}{1728}x^3=x$,

and, if $\frac{4877}{1728}$ were the ratio of a square to a square, the problem would be solved.

But $\frac{4877}{1728}=3-$(the sum of three cubes).

Therefore we must find three cubes, each of which <1, and such that $(3-\text{their sum})=$ a square.

If, *a fortiori*, the sum of the three cubes is made <1, the square will be >2. Let[2] it be $2\frac{1}{4}$.

[1] If a^3, b^3, c^3 are the three cubes, so that $a^3+b^3+c^3-3$ has to be a square, Diophantus chooses c^3 arbitrarily (8) and then makes such assumptions for the sides of a^3, b^3, being linear expressions in m, that, in the expression to be turned into a square, the coefficient of m^3 vanishes, and that of m^2 is a square. If $a=m$, the condition is satisfied by putting $b=3k^2-m$, where k is any number.

[2] Bachet, finding no way of hitting upon $2\frac{1}{4}$ as the particular square to be taken in order that the difference between it and 3 may be separable into three cubes, and observing that he could not solve the problem if he took another arbitrary square between 2 and 3, *e.g.* $2\frac{7}{9}$, instead of $2\frac{1}{4}$, concluded that Diophantus must have hit upon $2\frac{1}{4}$, which does enable the problem to be solved, by accident.

Fermat would not admit this and considered that the method used by Diophantus for finding $2\frac{1}{4}$ as the square to be taken should not be difficult to discover. Fermat accordingly suggested a method as follows.

Let $x-1$ be the side of the required square lying between 2 and 3. Then $3-(x-1)^2 =2+2x-x^2$, and this has to be separated into three cubes. Fermat assumes for the sides

We have therefore to find three cubes the sum of which $= \frac{3}{4}$ or $\frac{162}{216}$;

that is, we have to divide 162 into three cubes.

But $162 = 125 + 64 - 27$;

and "we have it in the Porisms" that *the difference of two cubes can be transformed into the sum of two cubes*[1].

Having thus found the three cubes[2], we start again, and $x = 2\frac{1}{4}x^3$, so that $x = \frac{2}{3}$.

The three numbers are thus determined.

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of two of the required cubes two linear expressions in x such that, when the sum of their cubes is subtracted from $2 + 2x - x^2$, the result only contains terms in x^2 and x^3 or in x and units.

The first alternative is secured if the sides of the first and second cubes are $1 - \frac{1}{3}x$ and $1 + x$ respectively; for

$$2 + 2x - x^2 - \left(1 - \frac{1}{3}x\right)^3 - (1 + x)^3 = -4\tfrac{1}{3}x^2 - \frac{26}{27}x^3.$$

This latter expression has to be made a cube, for which purpose we put

$$-4\tfrac{1}{3}x^2 - \frac{26}{27}x^3 = -\frac{m^3x^3}{27}, \text{ say,}$$

which gives a value for x. We have only to see that this value makes $\frac{1}{3}x$ less than 1, and we can easily choose m so as to fulfil this condition.

[*E.g.* suppose $m = 5$, and we find $x = \frac{13}{11}$, so that

$$\frac{1}{3}x = \frac{13}{33}, \quad 1 - \frac{1}{3}x = \frac{20}{33}, \quad 1 + x = \frac{72}{33},$$

and the side of the third cube is $-\frac{65}{33}$.

The square $(x-1)^2 = \left(\frac{2}{11}\right)^2$, and in fact $3 - \left\{\left(\frac{20}{33}\right)^3 + \left(\frac{72}{33}\right)^3 - \left(\frac{65}{33}\right)^3\right\} = \left(\frac{2}{11}\right)^2$.]

We then have three cubes which make up the excess of 3 over a certain square; but, while the first of these cubes is < 1, the second is > 1 and the third is negative. Hence we must, like Diophantus, proceed to transform the difference between the two latter cubes into the sum of two other cubes.

It will, however, be seen by trial that even Fermat's method is not quite general, for it will not, as a matter of fact, give the particular solution obtained by Diophantus in which the square is $2\frac{1}{4}$.

[1] On the transformation of the difference of two cubes into the sum of two cubes, see pp. 101–3, *ante*.

[2] Vieta's rule gives $4^3 - 3^3 = \left(\frac{303}{91}\right)^3 + \left(\frac{40}{91}\right)^3$. It follows that

$$\frac{3}{4} = \frac{162}{216} = \left(\frac{5}{6}\right)^3 + \left(\frac{101}{182}\right)^3 + \left(\frac{20}{273}\right)^3;$$

and, since $x^3 = \frac{8}{27}$, the required numbers are

$$\frac{91}{216} \cdot \frac{8}{27}, \quad \frac{4998267}{6028568} \cdot \frac{8}{27}, \quad \frac{20338417}{20346417} \cdot \frac{8}{27}.$$

17. To find three numbers such that each of them *minus* the cube of their sum gives a cube.

Let the sum be x and the numbers $2x^3$, $9x^3$, $28x^3$.

Therefore $39x^2 = 1$;

and we must replace 39 by a square which is the sum of three cubes $+ 3$;

therefore we must find three cubes such that their sum $+ 3$ is a square.

Let their sides[1] be m, $3 - m$, and any number, say 1.

Therefore $9m^2 + 31 - 27m =$ a square $= (3m - 7)^2$, say, so that $m = \frac{6}{5}$, and the sides of the cubes are $\frac{6}{5}$, $\frac{9}{5}$, 1.

Starting again, we put x for the sum, and for the numbers

$$\tfrac{341}{125}x^3,\ \tfrac{854}{125}x^3,\ \tfrac{250}{125}x^3,$$

whence $1445x^2 = 125$, $x^2 = \frac{25}{289}$, and $x = \frac{5}{17}$.

The required numbers are thus found.

18. To find three numbers such that their sum is a square and the cube of their sum added to any one of them gives a square.

Let the sum be x^2 and the numbers $3x^6$, $8x^6$, $15x^6$.

It follows that $26x^4 = 1$; and, if 26 were a fourth power, the problem would be solved.

To replace it by a fourth power, we have to find three numbers such that each increased by 1 gives a square, while the sum of the three gives a fourth power.

Let these numbers be $m^4 - 2m^2$, $m^2 + 2m$, $m^2 - 2m$ [the sum being m^4]; these are indeterminate numbers satisfying the conditions.

Putting any number, say 3, for m, we have as the required auxiliary numbers 63, 15, 3.

Starting again, we put x^2 for the sum and $3x^6$, $15x^6$, $63x^6$ for the required numbers,

and we have $81x^6 = x^2$, so that $x = \frac{1}{3}$.

The numbers are thus found $\left(\frac{3}{729}, \frac{15}{729}, \frac{63}{729}\right)$.

19. To find three numbers such that their sum is a square and the cube of their sum *minus* any one of them gives a square.

[There is obviously a lacuna in the text after this enunciation; for the next words are "And we have *again* to divide 2 *as before*,"

[1] Cf. note on V. 15. In this case, if one of the cubes is chosen arbitrarily and m^3 is another, we have only to put $(3k^2 - m)$ for the side of the third cube in order that, in the expression to be made a square, the term in m^3 may vanish, and the term in m^2 may be a square.

whereas there is nothing in our text to which they can refer, and the lines which follow are clearly no part of the solution of V. 19.

Bachet first noticed the probability that three problems intervened between V. 19 and V. 20, and he gave solutions of them. But he seems to have failed to observe that the eight lines or so in the text between the enunciation of V. 19 and the enunciation of V. 20 belonged to the solution of the last of the three missing problems. The first of the missing problems is connected with V. 18 and 19, making a natural trio with them, while the second and third similarly make with V. 20 a set of three. The enunciations were doubtless somewhat as follows.

19*a*. To find three numbers such that their sum is a square and any one of them *minus* the cube of their sum gives a square.

19*b*. To find three numbers such that their sum is a given number and the cube of their sum *plus* any one of them gives a square.

19*c*. To find three numbers such that their sum is a given number and the cube of their sum *minus* any one of them gives a square.

The words then in the text after the enunciation of V. 19 evidently belong to this last problem.]

The given sum is 2, the cube of which is 8.

We have to subtract each of the numbers from 8 and thereby make a square.

Therefore we have to divide 22 into three squares, each of which is greater than 6;

after which, by subtracting each of the squares from 8, we find the required numbers.

But we have already shown [cf. V. 11] how to divide 22 into three squares, each of which is greater than 6—and less than 8, Diophantus should have added.

[The above is explained by the fact that, by addition, three times the cube of the sum *minus* the sum itself is the sum of three squares, and three times the cube of the sum *minus* the sum $= 3 \cdot 8 - 2 = 22$.][1]

[1] Wertheim adds a solution in Diophantus' manner. We have to find what small fraction of the form $\frac{1}{x^2}$ we have to add to $\frac{22}{3}$ or $\frac{66}{9}$, and therefore to 66, in order to make a square. In order that $66 + \frac{1}{x^2}$ may be a square, we put

$$66x^2 + 1 = \text{square} = (1 + 8x)^2, \text{ say,}$$

which gives $$x = 8 \text{ and } x^2 = 64.$$

20. To divide a given fraction into three parts such that any one of them *minus* the cube of their sum gives a square.

Given fraction $\frac{1}{4}$.

Therefore each part $= \frac{1}{64} +$ a square.

Therefore the sum of the three $= \frac{1}{4} =$ the sum of three squares $+ \frac{3}{64}$.

Hence we have to divide $\frac{13}{64}$ into three squares, "which is easy[1]."

21. To find three squares such that their continued product added to any one of them gives a square.

Let the "solid content" $= x^2$.

We want now three squares, each of which increased by 1 gives a square.

They can be got from right-angled triangles[2] by dividing the square of one of the sides about the right angle by the square of the other.

Let the squares then be

$$\frac{9}{16}x^2,\ \frac{25}{144}x^2,\ \frac{64}{225}x^2.$$

The continued product $= \frac{14400}{518400}x^6 = x^2$, by hypothesis.

Therefore $\frac{120}{720}x^2 = 1$; and, if $\frac{120}{720}$ were a square, the problem would be solved.

We have therefore to increase 66 by $\frac{1}{64}$, and therefore $7\frac{1}{3}$ by $\frac{1}{576}$, in order to make a square. And in fact $7\frac{1}{3} + \frac{1}{576} = \left(\frac{65}{24}\right)^2$.

Next, since $22 = 3^2 + 3^2 + 2^2$, and $65 - 48 = 17$, while $72 - 65 = 7$, we put

$$22 = (3 - 7x)^2 + (3 - 7x)^2 + (2 + 17x)^2,$$

and $$x = \frac{16}{387}.$$

Therefore the sides of the squares are $\frac{1049}{387}, \frac{1049}{387}, \frac{1046}{387}$,

the squares themselves $\frac{1100401}{149769}, \frac{1100401}{149769}, \frac{1094116}{149769}$,

and the required parts of 2 are $\frac{97751}{149769}, \frac{97751}{149769}, \frac{104036}{149769}$.

[1] As Wertheim observes, $\frac{13}{64} = \frac{9}{64} + \frac{1}{25} + \frac{9}{400}$, and the required fractions into which $\frac{1}{4}$ is divided are $\frac{250}{1600}, \frac{89}{1600}, \frac{61}{1600}$.

[2] If a, b be the perpendiculars, c the hypotenuse in a right-angled triangle,

$$\frac{a^2}{b^2} + 1 = \frac{c^2}{b^2} = \text{a square}.$$

Diophantus uses the triangles (3, 4, 5), (5, 12, 13), (8, 15, 17).

As it is not, we must find three right-angled triangles such that, if b's are their bases, and p's are their perpendiculars, $p_1p_2p_3b_1b_2b_3 =$ a square;

and, if we assume one triangle arbitrarily (3, 4, 5), we have to make $12p_1p_2b_1b_2$ a square, or $3p_1b_1/p_2b_2$ a square.

"This is easy[1]," and the three triangles are (3, 4, 5), (9, 40, 41), (8, 15, 17) or similar to them.

[1] Diophantus does not give the work here, but only the result. Bachet obtains it in this way. Suppose it required to find three rational right-angled triangles (h_1, p_1, b_1), (h_2, p_2, b_2) and (h_3, p_3, b_3) such that $p_1p_2p_3/b_1b_2b_3$ is the ratio of a square to a square. One triangle (h_1, p_1, b_1) being chosen arbitrarily, form two others by putting

$$h_2 = h_1^2 + p_1^2, \quad p_2 = h_1^2 - p_1^2 = b_1^2, \quad b_2 = 2h_1p_1,$$
$$h_3 = h_1^2 + b_1^2, \quad p_3 = h_1^2 - b_1^2 = p_1^2, \quad b_3 = 2h_1b_1,$$

and we have

$$\frac{p_1p_2p_3}{b_1b_2b_3} = \left(\frac{p_1}{2h_1}\right)^2 = \text{a square}.$$

If now $h_1 = 5$, $p_1 = 4$, $b_1 = 3$, the triangles (h_2, p_2, b_2) and (h_3, p_3, b_3) are (41, 9, 40) and (34, 16, 30) respectively. Dividing the sides of the latter throughout by 2 (which does not alter the ratio), we have Diophantus' second and third triangles (9, 40, 41) and (8, 15, 17).

Fermat, in his note on the problem, gives the following general rule for finding two right-angled triangles the areas of which are in the ratio $m : n$ $(m > n)$.

Form (1) the greater triangle from $2m + n$, $m - n$, and the lesser from $m + 2n$, $m - n$,
or (2) the greater from $2m - n$, $m + n$, the lesser from $2n - m$, $m + n$,
or (3) the greater from $6m$, $2m - n$, the lesser from $4m + n$, $4m - 2n$,
or (4) the greater from $m + 4n$, $2m - 4n$, the lesser from $6n$, $m - 2n$.

The alternative (2) gives Diophantus' solution if we put $m = 3$, $n = 1$ and substitute $m - 2n$ for $2n - m$.

Fermat continues as follows: We can deduce a method of finding *three* right-angled triangles the areas of which are in the ratio of three given numbers, provided that the sum of two of these numbers is equal to four times the third. Suppose *e.g.* that m, n, q are three numbers such that $m + q = 4n$ $(m > q)$. Now form the following triangles:

(1) from $m + 4n$, $2m - 4n$,
(2) from $6n$, $m - 2n$,
(3) from $4n + q$, $4n - 2q$.

[If A_1, A_2, A_3 be the areas, we have, as a matter of fact,

$$A_1/m = A_2/n = A_3/q = -6m^3 + 36m^2n + 144mn^2 - 384n^3.]$$

We can derive, says Fermat, a method of *finding three right-angled triangles the areas of which themselves form a right-angled triangle.* For we have only to find a triangle such that the sum of the base and hypotenuse is four times the perpendicular. This is easy, and the triangle will be similar to (17, 15, 8); the three triangles will then be formed

(1) from $17 + 4 \cdot 8$, $2 \cdot 17 - 4 \cdot 8$ or 49, 2,
(2) from $6 \cdot 8$, $17 - 2 \cdot 8$ or 48, 1,
(3) from $4 \cdot 8 + 15$, $4 \cdot 8 - 2 \cdot 15$ or 47, 2.

[The areas of the three right-angled triangles are in fact 234906, 110544 and 207270, and these numbers form the sides of a right-angled triangle.]

Hence also we can derive a method of *finding three right-angled triangles the areas of which are in the ratio of three given squares such that the sum of two of them is equal to*

Starting again, we put for the squares

$$\tfrac{9}{16}x^2,\ \tfrac{225}{64}x^2,\ \tfrac{81}{1600}x^2.$$

Equating the product of these to x^2, we find x to be rational [$x=\frac{16}{9}$, and the squares are $\frac{16}{9}$, $\frac{100}{9}$, $\frac{4}{25}$].

22. To find three squares such that their continued product *minus* any one of them gives a square.

Let the solid content be x^2, and let the numbers be obtained from right-angled triangles, being

$$\tfrac{16}{25}x^2,\ \tfrac{25}{169}x^2,\ \tfrac{64}{289}x^2.$$

Therefore the continued product $\left(\dfrac{4 \cdot 5 \cdot 8}{5 \cdot 13 \cdot 17}\right)^2 x^6 = x^2$,

or $$\left(\frac{4 \cdot 5 \cdot 8}{5 \cdot 13 \cdot 17}\right)^2 x^4 = \frac{25600}{1221025} x^4 = 1.$$

If then $\dfrac{25600}{1221025}$ were a fourth power, *i.e.* if $\dfrac{4 \cdot 5 \cdot 8}{5 \cdot 13 \cdot 17}$ were a square, the problem would be solved.

We have therefore to find three right-angled triangles with hypotenuses h_1, h_2, h_3 respectively, and with p_1, p_2, p_3 as one of the perpendiculars in each respectively, such that

$$h_1 h_2 h_3 p_1 p_2 p_3 = \text{a square}.$$

Assuming one of the triangles to be (3, 4, 5), so that *e.g.* $h_3 p_3 = 5 \cdot 4 = 20$, we must have

$$5 h_1 p_1 h_2 p_2 = \text{a square}.$$

This is satisfied if $h_1 p_1 = 5 h_2 p_2$.

With a view to this we have first (cf. the last proposition) to find two right-angled triangles such that, if x_1, y_1 are the two *perpendiculars* in one and x_2, y_2 the two *perpendiculars* in the other, $x_1 y_1 = 5 x_2 y_2$. From such a pair of triangles we can form two more right-angled triangles such that the product of the *hypotenuse* and *one perpendicular* in one is five times the product of the *hypotenuse* and *one perpendicular* in the other[1].

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four times the third, and we can also in the same way *find three right-angled triangles of the same area;* we can also *construct*, in an infinite number of ways, *two right-angled triangles the areas of which are in a given ratio*, by multiplying one of the terms of the ratio or the two terms by given squares, etc.

[1] Diophantus' procedure is only obscurely indicated in the Greek text. It was explained by Schulz in his edition (cf. Tannery in *Oeuvres de Fermat*, I. p. 323, note).

Since the triangles found satisfying the relation $x_1y_1 = 5x_2y_2$ are (5, 12, 13) and (3, 4, 5) respectively[1], we have in fact to find two new right-angled triangles from them, namely the triangles (h_1, p_1, b_1) and (h_2, p_2, b_2), such that

$$h_1p_1 = 30 \text{ and } h_2p_2 = 6,$$

the numbers 30 and 6 being the areas of the two triangles mentioned.

These triangles are $(6\frac{1}{2}, \frac{60}{13}, [\frac{119}{26}])$ and $(2\frac{1}{2}, \frac{12}{5}, [\frac{7}{10}])$ respectively.

Starting again, we take for the numbers

$$\tfrac{16}{25}x^2,\ \tfrac{576}{625}x^2,\ \tfrac{14400}{28561}x^2.$$

[$\frac{12}{5}$ divided by $2\frac{1}{2}$ gives $\frac{24}{25}$, and $\frac{60}{13}$ divided by $6\frac{1}{2}$ gives $\frac{120}{169}$.]

The product $= x^2$:

therefore, taking the square root, we have

$$\frac{4 \cdot 24 \cdot 120}{5 \cdot 25 \cdot 169}x^2 = 1,$$

so that $x = \frac{65}{48}$, and the required squares are found.

23. To find three squares such that each *minus* the product of the three gives a square.

Having given a rational right-angled triangle (z, x, y), Diophantus knows how to find a rational right-angled triangle (h, p, b) such that $hp = \frac{1}{2}xy$. We have in fact to put

$$h = \frac{1}{2}z,\ p = \frac{xy}{z},\ \text{whence } b^2 = h^2 - p^2 = \frac{1}{4}\left(\frac{z^4 - 4x^2y^2}{z^2}\right) = \left(\frac{x^2 - y^2}{2z}\right)^2.$$

Thus, having found two triangles (5, 12, 13) and (3, 4, 5) with areas in the ratio of 5 to 1 (see next paragraph of text with note thereon), Diophantus takes

$$h_1 = \frac{1}{2} \cdot 13 = 6\tfrac{1}{2},\ p_1 = \frac{5 \cdot 12}{13} = \frac{60}{13};\ \text{and similarly } h_2 = \frac{1}{2} \cdot 5 = 2\tfrac{1}{2},\ p_2 = \frac{3 \cdot 4}{5} = \frac{12}{5}.$$

Cossali (after Bachet) gives a formula for three right-angled triangles such that the solid content of the three hypotenuses has to the solid content of three perpendiculars (one in each triangle) the ratio of a square to a square; his triangles are

$$(1)\ i, b, p\ [i = \text{ipotenusa}], \qquad (2)\ \frac{4p^2 + b^2}{b},\ \frac{4p^2 - b^2}{b},\ \frac{4bp}{b} = 4p,$$

$$(3)\ \frac{ib^2 + 4ip^2}{b},\ \frac{b \cdot 4bp + p(4p^2 - b^2)}{b},\ \frac{p \cdot 4bp - b(4p^2 - b^2)}{b} = b^2,$$

and, in fact,

$$\frac{i(b^2 + 4p^2)(ib^2 + 4ip^2)}{b^2} : p \cdot 4p \cdot b^2 = \frac{i^2(b^2 + 4p^2)^2}{b^2} : 4p^2b^2.$$

If $i = 5$, $b = 4$, $p = 3$, we can get from this triangle the triangles (13, 5, 12) and (65, 63, 16), and our equation is $\frac{3 \cdot 12 \cdot 16}{5 \cdot 13 \cdot 65}x^2 = 1$.

[1] These triangles can be obtained by putting $m = 5$, $n = 1$ in Fermat's *fourth* formula (note on last proposition). By that formula the triangles are formed from (9, 6) and (6, 3) respectively; and, dividing out by 3, we form the triangles from (3, 2) and (2, 1) respectively.

Let the "solid content" be x^2, and let the squares be formed from right-angled triangles, as before.

If we take the same triangles as those found in the last problem and put for the three squares

$$[illegible]x^2,\quad [illegible]x^2,\quad [illegible]x^2,$$

each of these *minus* the continued product (x^2) gives a square.

It remains that their product $= x^2$;

this gives $x =$ [illegible], and the problem is solved.

24. To find three squares such that the product of any two increased by 1 gives a square.

Product of first and second $+ 1 =$ a square, and the third is a square; therefore "solid content" + each = a square.

The problem therefore reduces to V. 21 above[1].

[1] De Billy in the *Inventum Novum*, Part II. paragraph 18 (*Oeuvres de Fermat*, III. pp. 373–4), extends this problem, showing how to find *four* numbers, three of which (only) are squares, having the given property, i.e. to solve the equations

$$x_2^2x_3^2 + 1 = r^2,\quad x_1^2x_4 + 1 = u^2,$$
$$x_3^2x_1^2 + 1 = s^2,\quad x_2^2x_4 + 1 = v^2,$$
$$x_1^2x_2^2 + 1 = t^2,\quad x_3^2x_4 + 1 = w^2.$$

First seek three square numbers satisfying the conditions of Diophantus' problem V. 24, say $\left(\frac{9}{16}, \frac{25}{4}, \frac{256}{81}\right)$, the solution of V. 21 given in Bachet's edition. We have then to find a fourth number (x, say) such that

$$\left.\begin{array}{l}\frac{9}{16}x+1\\ \frac{25}{4}x+1\\ \frac{256}{81}x+1\end{array}\right\}\text{ are all squares.}$$

Substitute $\frac{16}{9}y^2 + \frac{32}{9}y$ for x, so as to make the first expression a square. We have then to solve the double-equation

$$\left.\begin{array}{l}\frac{100}{9}y^2 + \frac{200}{9}y + 1 = u^2\\ \frac{4096}{729}y^2 + \frac{8192}{729}y + 1 = v^2\end{array}\right\},$$

which can be solved by the ordinary method.

De Billy does not give the solution, but it may be easily supplied thus.

The difference

$$= \left(\frac{10}{3} + \frac{64}{27}\right)\left(\frac{10}{3} - \frac{64}{27}\right)(y^2 + 2y)$$
$$= \frac{154}{27}y\left(\frac{26}{27}y + \frac{2 \cdot 26}{27}\right),$$

25. To find three squares such that the product of any two *minus* 1 gives a square.

This reduces, similarly, to V. 22 above.

26. To find three squares such that, if we subtract the product of any two of them from unity, the result is a square.

This again reduces to an earlier problem, V. 23.

27. Given a number, to find three squares such that the sum of any two added to the given number makes a square.

Given number 15.

Let one of the required squares be 9;

I have then to find two other squares such that each $+ 24 =$ a square, and their sum $+ 15 =$ a square.

To find two squares, each of which $+ 24 =$ a square, take two pairs of numbers which have 24 for their product[1].

Let one pair of factors be $4/x$, $6x$, and let the side of one square be half their difference or $\frac{2}{x} - 3x$.

Let the other pair of factors be $3/x$, $8x$, and let the side of the other square be half their difference or $\frac{1\frac{1}{2}}{x} - 4x$.

Therefore each of the squares $+ 24$ gives a square.

It remains that their sum $+ 15 =$ a square;

therefore $\left(\frac{1\frac{1}{2}}{x} - 4x\right)^2 + \left(\frac{2}{x} - 3x\right)^2 + 15 =$ a square,

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Equating the square of half the sum of the factors to the larger expression, we have

$$\left(\frac{10}{3}y + \frac{26}{27}\right)^2 = \frac{100}{9}y^2 + \frac{200}{9}y + 1,$$

whence $$y = -\frac{53}{11520}, \text{ and } y^2 + 2y = -\frac{1218311}{(11520)^2}.$$

Therefore $x = \frac{16}{9}(y^2 + 2y) = -\frac{1218311}{74649600}$, which satisfies the equations. In fact $\frac{9}{16}x + 1 = \left(\frac{11467}{11520}\right)^2$, $\frac{25}{4}x + 1 = \left(\frac{3275}{3456}\right)^2$, and $\frac{256}{81}x + 1 = \left(\frac{4733}{4860}\right)^2$.

But even here, as the value of x which we have found is negative, we ought, strictly speaking, to deduce a further value by substituting $y - \frac{1218311}{74649600}$ for x in the equations and solving again, which would of course lead to very large numbers.

[1] The text adds the words "and [let us take] sides about the right angle in a right-angled triangle." I think these words must be a careless interpolation: they are not wanted and give no sense; nor do they occur in the corresponding place in the next problem.

or $$\frac{6\frac{1}{4}}{x^2} + 25x^2 - 9 = \text{a square} = 25x^2, \text{ say.}$$

Therefore $x = \frac{5}{6}$, and the problem is solved[1].

28. Given a number, to find three squares such that the sum of any two *minus* the given number makes a square.

Given number 13.

Let one of the squares be 25;

I have then to find two other squares such that each $+ 12 =$ a square, and (sum of both) $- 13 =$ a square.

Divide 12 into factors in two ways, and let the factors be $(3x, 4/x)$ and $(4x, 3/x)$.

Take as the sides of the squares half the differences of the factors, *i.e.* let the squares be

$$\left(1\tfrac{1}{2}x - \frac{2}{x}\right)^2, \quad \left(2x - \frac{1\frac{1}{2}}{x}\right)^2.$$

Each of these $+ 12$ gives a square.

It remains that the sum of the squares $- 13 =$ a square,

or $$\frac{6\frac{1}{4}}{x^2} + 6\tfrac{1}{4}x^2 - 25 = \text{a square} = \frac{6\frac{1}{4}}{x^2}, \text{ say.}$$

Therefore $x = 2$, and the problem is solved[2].

[1] Diophantus has found values of ξ, η, ζ satisfying the equations

$$\left.\begin{aligned} \eta^2 + \zeta^2 + a &= u^2 \\ \zeta^2 + \xi^2 + a &= v^2 \\ \xi^2 + \eta^2 + a &= w^2 \end{aligned}\right\}.$$

Fermat shows how to find *four* numbers (not squares) satisfying the corresponding conditions, namely that the sum of any two added to a shall give a square. Suppose $a = 15$.

Take three numbers satisfying the conditions of Diophantus' problem, say $9, \frac{1}{100}, \frac{529}{225}$.

Assume $x^2 - 15$ as the first of the four required numbers; and let the second be $6x + 9$ (because 9 is one of the square numbers taken and 6 is twice its side); for the same reason let the third number be $\frac{1}{5}x + \frac{1}{100}$ and the fourth $\frac{46}{15}x + \frac{529}{225}$.

Three of the conditions are now fulfilled since each of the last three numbers added to the first $(x^2 - 15)$ *plus* 15 gives a square. The three remaining conditions give the triple-equation

$$\begin{aligned} 6\tfrac{1}{5}x + 9 + \frac{1}{100} + 15 &= 6\tfrac{1}{5}x + \left(\frac{49}{10}\right)^2 = u^2, \\ \frac{136}{15}x + 9 + \frac{529}{225} + 15 &= \frac{136}{15}x + \left(\frac{77}{15}\right)^2 = v^2, \\ \frac{49}{15}x + \frac{1}{100} + \frac{529}{225} + 15 &= \frac{49}{15}x + \left(\frac{25}{6}\right)^2 = w^2. \end{aligned}$$

[2] Fermat observes that *four* numbers (not squares) with the property indicated can be found by the same procedure as that shown in the note to the preceding problem.

If a is the given number, put $x^2 + a$ for the first of the four required numbers.

29. To find three squares such that the sum of their squares is a square.

Let the squares be x^2, 4, 9 respectively[1].

Therefore $x^4 + 97 =$ a square $= (x^2 - 10)^2$, say;

whence $x^2 = \frac{3}{20}$.

If the ratio of 3 to 20 were the ratio of a square to a square, the problem would be solved; but it is not.

Therefore *I have to find two squares* (p^2, q^2, *say*) *and a number* (m, *say*) *such that* $m^2 - p^4 - q^4$ *has to* $2m$ *the ratio of a square to a square.*

Let $p^2 = z^2$, $q^2 = 4$ and $m = z^2 + 4$.

Therefore $m^2 - p^4 - q^4 = (z^2 + 4)^2 - z^4 - 16 = 8z^2$.

Hence $8z^2/(2z^2 + 8)$, or $4z^2/(z^2 + 4)$, must be the ratio of a square to a square.

Put $$z^2 + 4 = (z + 1)^2, \text{ say;}$$

therefore $z = 1\frac{1}{2}$, and the squares are $p^2 = 2\frac{1}{4}$, $q^2 = 4$, while $m = 6\frac{1}{4}$;

or, if we take 4 times each, $p^2 = 9$, $q^2 = 16$, $m = 25$.

Starting again, we put for the squares x^2, 9, 16;

then the sum of the squares $= x^4 + 337 = (x^2 - 25)^2$, and $x = \frac{12}{5}$.

The required squares are $\frac{144}{25}$, 9, 16.

30. [The enunciation of this problem is in the form of an epigram, the meaning of which is as follows.]

A man buys a certain number of measures (χόες) of wine, some at 8 drachmas, some at 5 drachmas each. He pays for them a *square* number of drachmas; and if we add 60 to this number, the result is a square, the side of which is equal to the whole number of measures. Find how many he bought at each price.

Let $x =$ the whole number of measures; therefore $x^2 - 60$ was the price paid, which is a square $= (x - m)^2$, say.

If now k^2, l^2, m^2 represent three numbers satisfying the conditions of the present problem of Diophantus, put for the second of the required numbers $2kx + k^2$, for the third $2lx + l^2$, and for the fourth $2mx + m^2$. These satisfy three conditions, since each of the last three numbers added to the first $(x^2 + a)$ less the number a gives a square. The remaining three conditions give a triple-equation.

[1] "Why," says Fermat, "does not Diophantus seek *two* fourth powers such that their sum is a square? This problem is in fact impossible, as by my method I am in a position to prove with all rigour." It is probable that Diophantus knew the fact without being able to prove it generally. That neither the sum nor the difference of two fourth powers can be a square was proved by Euler (*Commentationes arithmeticae*, I. pp. 24 sqq., and *Algebra*, Part II. c. XIII.).

Now $\frac{1}{5}$ of the price of the five-drachma measures + $\frac{1}{8}$ of the price of the eight-drachma measures $= x$;

so that $x^2 - 60$, the total price, has to be divided into two parts such that $\frac{1}{5}$ of one + $\frac{1}{8}$ of the other $= x$.

We cannot have a real solution of this unless

$$x > \tfrac{1}{8}(x^2 - 60) \text{ and } < \tfrac{1}{5}(x^2 - 60).$$

Therefore $\quad 5x < x^2 - 60 < 8x.$

(1) Since $\quad x^2 > 5x + 60,$

$x^2 = 5x +$ a number greater than 60,

whence x is[1] *not less than* 11.

(2) $\quad x^2 < 8x + 60$

or $\quad x^2 = 8x +$ some number less than 60,

whence x is[1] *not greater than* 12.

Therefore $\quad 11 < x < 12.$

Now (from above) $x = (m^2 + 60)/2m$;

therefore $22m < m^2 + 60 < 24m$.

Thus (1) $\quad 22m = m^2 +$ (some number less than 60),

and therefore m is[2] *not less than* 19.

(2) $\quad 24m = m^2 +$ (some number greater than 60),

and therefore m is[2] *less than* 21.

Hence we put $m = 20$, and

$$x^2 - 60 = (x - 20)^2,$$

so that $x = 11\frac{1}{2}$, $x^2 = 132\frac{1}{4}$, and $x^2 - 60 = 72\frac{1}{4}$.

Thus we have to divide $72\frac{1}{4}$ into two parts such that $\frac{1}{5}$ of one part *plus* $\frac{1}{8}$ of the other $= 11\frac{1}{2}$.

Let the first part be $5x$.

Therefore $\frac{1}{8}$ (second part) $= 11\frac{1}{2} - x$,

or second part $= 92 - 8x$;

therefore $5x + 92 - 8x = 72\frac{1}{4}$,

and $x = \frac{79}{12}$.

Therefore the number of five-drachma χόες $= \frac{79}{12}$.

" " " " eight-drachma " $= \frac{59}{12}$.

[1] For an explanation of these limits see p. 60, *ante*.
[2] See p. 62, *ante*.

BOOK VI

1. To find a (rational) right-angled triangle such that the hypotenuse *minus* each of the sides gives a cube[1].

Let the required triangle be formed from x, 3.

Therefore hypotenuse $= x^2 + 9$, perpendicular $= 6x$, base $= x^2 - 9$.

Thus $x^2 + 9 - (x^2 - 9) = 18$ should be a cube, but it is not.

Now $18 = 2 \cdot 3^2$; therefore we must replace 3 by m, where $2 \cdot m^2$ is a cube; and $m = 2$.

We form, therefore, a right-angled triangle from x, 2, namely $(x^2 + 4,\ 4x,\ x^2 - 4)$; and one condition is satisfied.

The other gives $x^2 - 4x + 4 =$ a cube;

therefore $(x-2)^2$ is a cube, or $x - 2$ is a cube $= 8$, say.

Thus $x = 10$,

and the triangle is $(40, 96, 104)$.

2. To find a right-angled triangle such that the hypotenuse added to each side gives a cube.

Form a triangle, as before, from two numbers; and, as before, one of them must be such that twice its square is a cube, *i.e.* must be 2.

We form a triangle from x, 2, namely $x^2 + 4$, $4x$, $4 - x^2$; therefore $x^2 + 4x + 4$ must be a cube, while x^2 must be less than 4, or $x < 2$.

Thus $x + 2 =$ a cube which must be < 4 and $> 2 = \frac{27}{8}$, say.

Therefore $x = \frac{11}{8}$,

and the triangle is $\left(\frac{135}{64},\ 5\frac{1}{2},\ \frac{377}{64}\right)$,

or, if we multiply by the common denominator, $(135, 352, 377)$.

3. To find a right-angled triangle such that its area added to a given number makes a square.

Let 5 be the given number, $(3x, 4x, 5x)$ the required triangle.

[1] Diophantus' expressions are ὁ ἐν τῇ ὑποτεινούσῃ, "the (*number*) in (or representing) the hypotenuse," ὁ ἐν ἑκατέρᾳ τῶν ὀρθῶν, "the (*number*) in (or representing) each of the perpendicular sides," ὁ ἐν τῷ ἐμβαδῷ, "the (*number*) in (or representing) the area," etc. It will be convenient to say "the hypotenuse," etc. simply. It will be observed that, as between the numbers representing sides and area, all idea of dimension is ignored.

Therefore $6x^2 + 5 = \text{a square} = 9x^2$, say,

or $3x^2 = 5$.

But 5 should have to 5 the ratio of a square to a square.

Therefore we must find a right-angled triangle and a number such that the difference between the square of the number and the area of the triangle has to 5 the ratio of a square to a square, *i.e.* $= \frac{1}{5}$ of a square.

Form a right-angled triangle from $\left(m, \dfrac{1}{m}\right)$;

thus the area is $m^2 - \dfrac{1}{m^2}$.

Let the number be $m + \dfrac{2 \cdot 5}{m}$, so that we must have

$$4 \cdot 5 + \frac{101}{m^2} = \tfrac{1}{5} \text{ of a square};$$

therefore $4 \cdot 25 + \dfrac{505}{m^2} = \text{a square}$,

or $100m^2 + 505 = \text{a square} = (10m + 5)^2$, say,

and $m = \frac{24}{5}$.

The auxiliary triangle must therefore be formed from $\frac{24}{5}$, $\frac{5}{24}$, and the auxiliary number sought is $\frac{413}{60}$.

Put now for the original triangle (hx, px, bx), where (h, p, b) is the right-angled triangle formed from $\frac{24}{5}$, $\frac{5}{24}$;

this gives $\frac{1}{2}pbx^2 + 5 = \frac{170569}{3600}x^2$,

and we have the solution.

[The perpendicular sides of the right-angled triangle are

$$\left(\frac{24^2}{5^2} - \frac{5^2}{24^2}\right)x = \tfrac{331151}{14400}x \text{ and } 2x;$$

whence $\frac{331151}{14400}x^2 + 5 = \frac{170569}{3600}x^2$,

$x = \frac{24}{53}$,

and the triangle is

$$\left(\tfrac{331151}{31800},\ \tfrac{48}{53},\ \tfrac{332401}{31800}\right).]$$

4. To find a right-angled triangle such that its area *minus* a given number makes a square.

Given number 6, triangle $(3x, 4x, 5x)$, say.

Therefore $6x^2 - 6 = \text{square} = 4x^2$, say.

Thus, in this case, we must find a right-angled triangle and a number such that

$$(\text{area of triangle}) - (\text{number})^2 = \tfrac{1}{6} \text{ of a square}.$$

Form a triangle from $m, \dfrac{1}{m}$.

Its area is $m^2 - \frac{1}{m^2}$, and let the number be $m - \frac{6}{2} \cdot \frac{1}{m}$.

Therefore $6 - \frac{10}{m^2} = \frac{1}{6}$ (a square),

or $36m^2 - 60 =$ a square $= (6m - 2)^2$, say.

Therefore $m = \frac{8}{3}$, and the auxiliary triangle is formed from $(\frac{8}{3}, \frac{3}{8})$, the auxiliary number being $\frac{37}{24}$.

We start again, substituting for 3, 4, 5 in the original hypothesis the sides of the auxiliary triangle just found, and putting $(\frac{37}{24})^2 x^2$ in place of $4x^2$; and the solution is obvious.

[The auxiliary triangle is $(\frac{4015}{576}, 2, \frac{4177}{576})$, whence

$$\frac{4015}{576}x^2 - 6 = (\tfrac{37}{24})^2 x^2, \text{ and } x = \tfrac{8}{7},$$

so that the required triangle is $(\frac{4015}{504}, \frac{16}{7}, \frac{4177}{504})$.]

5. To find a right-angled triangle such that, if its area be subtracted from a given number, the remainder is a square.

Given number 10, triangle $(3x, 4x, 5x)$, say.

Thus $10 - 6x^2 =$ a square; and we have to find a right-angled triangle and a number such that

(area of triangle) + (number)$^2 = \frac{1}{10}$ of a square.

Form a triangle from $m, \frac{1}{m}$, the area being $m^2 - \frac{1}{m^2}$,

and let the number be $\frac{1}{m} + 5m$.

Therefore $26m^2 + 10 = \frac{1}{10}$ of a square,

or $\quad 260m^2 + 100 =$ a square,

or again $\quad 65m^2 + 25 =$ a square $= (8m + 5)^2$, say,

whence $\quad m = 80$.

The rest is obvious.

The required triangle is $\frac{40959999}{825600}, \frac{2}{129}, \frac{40960001}{825600}$.]

6. To find a right-angled triangle such that the area added to one of the perpendiculars makes a given number.

Given number 7, triangle $(3x, 4x, 5x)$.

Therefore $6x^2 + 3x = 7$.

In order that this might be solved, it would be necessary that (half coefficient of x)^2 + product of coefficient of x^2 and absolute term should be a square;

but $(1\frac{1}{2})^2 + 6 \cdot 7$ is not a square.

Hence we must find, to replace (3, 4, 5), a right-angled triangle such that

($\frac{1}{2}$ one perpendicular)^2 + 7 times area = a square.

Let one perpendicular be m, the other 1.

$$\left.\begin{array}{l}\text{Therefore } 3\tfrac{1}{2}m + \tfrac{1}{4} = \text{a square, or } 14m + 1 = \text{a square.}\\ \text{Also, since the triangle is rational, } m^2 + 1 = \text{a square.}\end{array}\right\}$$

The difference $m^2 - 14m = m(m-14)$;

and putting, as usual, $7^2 = 14m + 1$,

we have $m = \frac{24}{7}$.

The auxiliary triangle is therefore $(\frac{24}{7}, 1, \frac{25}{7})$ or $(24, 7, 25)$.

Starting afresh, we take as the triangle $(24x, 7x, 25x)$.

Therefore $84x^2 + 7x = 7$,

and $x = \frac{1}{4}$.

We have then $\left(6, \frac{7}{4}, \frac{25}{4}\right)$ as the solution[1].

7. To find a right-angled triangle such that its area *minus* one of the perpendiculars is a given number.

Given number 7.

As before, we have to find a right-angled triangle such that

$(\frac{1}{2}$ one perpendicular$)^2 + 7$ times area $=$ a square;

this triangle is $(7, 24, 25)$.

Let then the triangle of the problem be $(7x, 24x, 25x)$.

Therefore $84x^2 - 7x = 7$,

$$x = \tfrac{1}{3},$$

and the problem is solved[2].

[1] Fermat observes that this problem and the next can be solved by another method. "Form in this case," he says, "a triangle from the given number and 1, and divide the sides by the sum of the given number and 1; the quotients will give the required triangle."

In fact, if we take as the sides of the required triangle

$$(a^2+1)x, \ (a^2-1)x, \ 2ax,$$

where a is the given number, we have

$$(a^2-1)ax^2 + 2ax = a,$$

one root of which is $$x = -\frac{1}{a^2-1} + \frac{a}{a^2-1} = \frac{1}{a+1},$$

and the sides of the required triangle are therefore

$$\frac{a^2+1}{a+1}, \ \frac{a^2-1}{a+1}, \ \frac{2a}{a+1}.$$

The solution is really the same as that of Diophantus.

[2] Similarly in this case we may, with Fermat, form the triangle from the given number and 1, and divide the sides by the difference between the given number and 1, and we shall have the required triangle.

In VI. 6, 7, Diophantus has found triangles ζ, ξ, η (ζ being the hypotenuse), such that

$$(1) \quad \frac{1}{2}\xi\eta + \xi = a,$$

and $$(2) \quad \frac{1}{2}\xi\eta - \xi = a.$$

Fermat enunciates the third case

$$(3) \quad \xi - \frac{1}{2}\xi\eta = a,$$

8. To find a right-angled triangle such that the area added to the sum of the perpendiculars makes a given number.

Given number 6.

Again I have to find a right-angled triangle such that

$$(\tfrac{1}{2}\text{ sum of perpendiculars})^2 + 6 \text{ times area} = \text{a square}.$$

Let m, 1 be the perpendicular sides of this triangle;

therefore $\frac{1}{4}(m+1)^2 + 3m = \frac{1}{4}m^2 + 3\frac{1}{2}m + \frac{1}{4} =$ a square,

while $m^2 + 1$ must also be a square.

Therefore $\left.\begin{matrix} m^2 + 14m + 1 \\ m^2 + 1 \end{matrix}\right\}$ are both squares.

The difference is $2m \cdot 7$, and we put

$$m^2 - 7m + 12\tfrac{1}{4} = m^2 + 1,$$

whence $m = \frac{45}{28}$,

and the auxiliary triangle is $(\frac{45}{28}, 1, \frac{53}{28})$, or $(45, 28, 53)$.

Assume now for the triangle of the problem

$$(45x, 28x, 53x).$$

Therefore $630x^2 + 73x = 6$;

x is rational $[= \frac{1}{18}]$, and the solution follows.

9. To find a right-angled triangle such that the area *minus* the sum of the perpendiculars is a given number.

Given number 6.

As before, we find a subsidiary right-angled triangle such that $(\frac{1}{2}$sum of perpendiculars$)^2 + 6$ times area $=$ a square.

This is found to be $(28, 45, 53)$ as before.

Taking $(28x, 45x, 53x)$ for the required triangle,

$$630x^2 - 73x = 6;$$

$x = \frac{6}{35}$, and the problem is solved[1].

10. To find a right-angled triangle such that the sum of its area, the hypotenuse, and one of the perpendiculars is a given number.

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observing that Diophantus and Bachet appear not to have known the solution, but that it can be solved "by our method." He does not actually give the solution; but we may compare his solutions of similar problems in the *Inventum Novum*, *e.g.* those given in the notes to VI. 11 and VI. 15 below and in the Supplement. The essence of the method is that, if the first value of x found in the ordinary course is such as to give a negative value for one of the sides, we can derive from it a fresh value which will make all the sides positive.

[1] Here likewise, Diophantus having solved the problem

$$\frac{1}{2}\xi\eta - (\xi + \eta) = a,$$

Fermat enunciates, as to be solved by his method, the corresponding problem

$$\xi + \eta - \frac{1}{2}\xi\eta = a.$$

Given number 4.

If we assumed as the triangle (hx, px, bx), we should have

$$\tfrac{1}{2}pbx^2 + hx + bx = 4;$$

and, in order that the solution may be rational, we must find a right-angled triangle such that

$$\tfrac{1}{4}(\text{hyp.} + \text{one perp.})^2 + 4 \text{ times area} = \text{a square}.$$

Form a right-angled triangle from $1, m+1$.

Then $\tfrac{1}{4}(\text{hyp.} + \text{one perp.})^2 = \tfrac{1}{4}(m^2 + 2m + 2 + m^2 + 2m)^2$
$$= m^4 + 4m^3 + 6m^2 + 4m + 1,$$
and 4 times area $= 4(m+1)(m^2+2m)$
$$= 4m^3 + 12m^2 + 8m.$$

Therefore

$m^4 + 8m^3 + 18m^2 + 12m + 1 = \text{a square} = (6m + 1 - m^2)^2$, say,

whence $m = \tfrac{4}{5}$, and the auxiliary triangle is formed from $(1, \tfrac{9}{5})$ or $(5, 9)$. This triangle is $(56, 90, 106)$ or $(28, 45, 53)$.

We assume therefore $28x, 45x, 53x$ for the original triangle, and we have $630x^2 + 81x = 4$.

Therefore $x = \tfrac{4}{105}$, and the problem is solved.

11. To find a right-angled triangle such that its area *minus* the sum of the hypotenuse and one of the perpendiculars is a given number.

Given number 4.

We have then to find an auxiliary triangle with the same property as in the last problem;

therefore $(28, 45, 53)$ will serve the purpose.

We put for the triangle of the problem $(28x, 45x, 53x)$, and we have $630x^2 - 81x = 4$;

$x = \tfrac{1}{6}$, and the problem is solved[1].

[1] Diophantus has in VI. 10, 11 shown us how to find a rational right-angled triangle ζ, ξ, η (ζ being the hypotenuse) such that

$$(1)\quad \frac{1}{2}\xi\eta + \zeta + \xi = a,$$

$$(2)\quad \frac{1}{2}\xi\eta - (\zeta + \xi) = a.$$

Fermat, in the *Inventum Novum*, Part III. paragraph 33 (*Oeuvres de Fermat*, III. p. 389), propounds and solves the corresponding problem

$$(3)\quad \zeta + \xi - \frac{1}{2}\xi\eta = a.$$

In the particular case taken by Fermat $a = 4$. He proceeds thus:

First find a rational right-angled triangle in which (since $a = 4$)

$$\left\{\frac{1}{2}(\zeta + \xi)\right\}^2 - 4\cdot\frac{1}{2}\xi\eta = \text{a square}.$$

Lemma I to the following problem.

To find a right-angled triangle such that the difference of the perpendiculars is a square, the greater alone is a square, and further the area added to the lesser perpendicular gives a square.

Let the triangle be formed from two numbers, the greater perpendicular being twice their product.

Hence I must find two numbers such that (1) twice their product is a square and (2) twice their product exceeds the difference of their squares by a square.

This is true of any two numbers the greater of which = twice the lesser.

Form then the triangle from x, $2x$, and two conditions are satisfied.

The third gives $6x^4 + 3x^2 =$ a square, or $6x^2 + 3 =$ a square.

I have therefore to find a number such that 6 times its square + 3 = a square;

one such number is 1, and there are an infinite number of others[1].

If $x = 1$, the triangle is formed from 1, 2.

Suppose it formed from $x+1$, x; the sides then are

$$\zeta = 2x^2 + 2x + 1, \quad \xi = 2x + 1, \quad \eta = 2x^2 + 2x.$$

Thus
$$\left\{\frac{1}{2}(\zeta + \xi)\right\}^2 - 4 \cdot \frac{1}{2}\xi\eta = x^4 + 4x^3 + 6x^2 + 4x + 1 - 4(2x^3 + 3x^2 + x)$$
$$= x^4 - 4x^3 - 6x^2 + 1$$
$$= \text{a square}$$
$$= (x^2 - 2x + 1)^2, \text{ say.}$$

Therefore $-6x^2 = 6x^2 - 4x$, $x = \frac{1}{3}$, and $x + 1 = \frac{4}{3}$.

The triangle formed from $\frac{4}{3}$, $\frac{1}{3}$ is $\left(\frac{17}{9}, \frac{15}{9}, \frac{8}{9}\right)$. Thus we may take as the auxiliary triangle (17, 15, 8).

Take now $17x$, $15x$, $8x$ for the sides of the triangle originally required to be found. We have then

$$\zeta + \xi - \frac{1}{2}\xi\eta = 32x - 60x^2 = 4;$$

whence $x = \frac{1}{3}$, and the required triangle is $\left(\frac{17}{3}, \frac{15}{3}, \frac{8}{3}\right)$.

[The auxiliary right-angled triangle was of course necessary to be found in order to make the final quadratic give a rational result.]

Bachet adds after VI. 11 a solution of the problem represented by

$$\frac{1}{2}\xi\eta - \zeta = a,$$

to which Fermat adds the enunciation of the corresponding problem

$$\zeta - \frac{1}{2}\xi\eta = a.$$

[1] Though there are an infinite number of values of x for which $6x^2 + 3$ becomes a square, the resulting triangles are all similar. For, if x be any one of the values, the triangle is

Lemma II to the following problem.

Given two numbers the sum of which is a square, an infinite number of squares can be found such that, when the square is multiplied by one of the given numbers and the product is added to the other, the result is a square.

Given numbers 3, 6.

Let x^2+2x+1 be the required square which, say, when multiplied by 3 and then increased by 6, gives a square.

We have $3x^2+6x+9 =$ a square;

and, since the absolute term is a square, an infinite number of solutions can be found.

Suppose, *e.g.* $3x^2+6x+9=(3-3x)^2$,

and $x=4$.

The side of the required square is 5, and an infinite number of other solutions can be found.

12. To find a right-angled triangle such that the area added to either of the perpendiculars gives a square.

Let the triangle be $(5x, 12x, 13x)$.

Therefore (1) $30x^2+12x =$ a square $=36x^2$, say,

and $x=2$.

But (2) we must also have

$30x^2+5x =$ a square.

This is however not a square when $x=2$.

Therefore I must find a square m^2x^2, to replace $36x^2$, such that $12/(m^2-30)$, the value of x obtained from the first equation, is real and satisfies the condition

$30x^2+5x =$ a square.

This gives, by substitution,

$(60m^2+2520)/(m^4-60m+900) =$ a square,

or $60m^2+2520 =$ a square.

This could be solved [by the preceding Lemma II] *if* $60+2520$ *were equal to a square.*

Now 60 arises from $5 \cdot 12$, *i.e.* from the product of the perpendicular sides of (5, 12, 13);

2520 is $30 \cdot 12 \cdot (12-5)$, *i.e.* the continued product of the area, the greater perpendicular, and the difference between the perpendiculars.

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formed from x, $2x$, and its sides are therefore $3x^2$, $4x^2$, $5x^2$; that is, the triangles are all similar to (3, 4, 5). Fermat shows in his note on the following problem, VI. 12, how to find any number of triangles satisfying the conditions of this Lemma and *not* similar to (3, 4, 5). See p. 235, note.

Hence we must find an auxiliary triangle such that (product of perps.) + (continued product of area, greater perp. and difference of perps.) = a square.

Or, *if we make the greater perpendicular a square* and divide out by it, we must have

(lesser perp.) + (product of area and diff. of perps.) = a square.

Then, assuming that we have found two numbers, (1) the product of the area and the difference of the perpendiculars and (2) the lesser perpendicular, satisfying these conditions, we have to find a square (m^2) such that the product of this square into the second of the numbers, when added to the first number, gives a square[1].

[1] The text of this sentence is unsatisfactory. Bachet altered the reading of the MSS. So did Tannery, but more by way of filling out. The version above follows Tannery's text, which is as follows: *ἀπάγεται εἰς τὸ δύο ἀριθμοὺς εὑρόντας* [for *ὄντας* of MSS.] <*τόν τε ὑπὸ*> *τοῦ ἐμβαδοῦ καὶ τῆς ὑπεροχῆς τῶν ὀρθῶν*, <*καὶ τὸν ἐν τῇ ἐλάσσονι τῶν ὀρθῶν*>, *αὖθις* [for *αὐτῆς* of MSS.] *ζητεῖν* □*όν τινα, ὃς πολλαπλασιασθεὶς ἐπὶ ἕνα τὸν δοθέντα*, <*καὶ προσλαβὼν τὸν ἕτερον*>, *ποιεῖ τετράγωνον.*

The argument would then be this. If (h, p, b) be the triangle $(b > p)$, we have to make

$$bp + \frac{1}{2}bp(b-p)b \text{ a square,}$$

or, if b is a square, $\quad p + \frac{1}{2}bp(b-p)$ must be a square.

The ultimate equation to be solved (corresponding to $60m^2 + 2520 =$ a square) is

$$bpm^2 + \frac{1}{2}bp(b-p)b = \text{a square,}$$

or, if b is a square, $\quad pm^2 + \frac{1}{2}bp(b-p) =$ a square;

and therefore, according to Tannery's text, "the problem is reduced to this: Having found two numbers $\frac{1}{2}bp(b-p)$ and p [satisfying the conditions, namely that their sum is a square, while b is also a square], to find after that a square such that the product of it and the latter number added to the former number gives a square."

The difficulty is that, with the above readings, there is nothing to correspond exactly to the phraseology of the enunciation of Lemma I, which speaks, not of making $p + \frac{1}{2}bp(b-p)$ a square when b is a square, but of making $b-p$, b and $p + \frac{1}{2}bp$ all simultaneously squares. But the *particular solution* of the Lemma is really equivalent to making b and $p + \frac{1}{2}bp(b-p)$ simultaneously squares. For the triangle is formed from a, $2a$; this method of making b a square $(=4a^2)$ incidentally makes $b-p$ a square $(=a^2)$, and $p + \frac{1}{2}bp$ becomes $3a^2 + 6a^4$, while $p + \frac{1}{2}bp(b-p)$ becomes $3a^2 + 6a^6$. Since the solution actually used is $a = 1$, the effect is the same whichever way the problem is stated. And in any case, whether the expression to be made a square is $3a^2m^2 + 6a^4$ or $3a^2m^2 + 6a^6$, the problem equally reduces to that of making $3m'^2 + 6$ a square.

How to solve these problems is shown in the Lemmas.

The auxiliary triangle is (3, 4, 5). [Lemma I.]

Accordingly, putting for the original triangle $(3x, 4x, 5x)$,

we have $\left.\begin{matrix} 6x^2+4x \\ 6x^2+3x \end{matrix}\right\}$ both squares.

Let $x=\dfrac{4}{m^2-6}$ be the solution of the first equation;

then $$x^2=\frac{16}{m^4-12m^2+36}.$$

The second equation therefore gives

$$\frac{96}{m^4-12m^2+36}+\frac{12}{m^2-6}=\text{a square},$$

whence $$12m^2+24=\text{a square},$$

and we have therefore to find a square (m^2) such that twelve times it + 24 is a square; this is possible, since 12 + 24 is a square [Lemma II].

A solution is $m^2=25$,

whence $x=\frac{4}{19}$,

and $\left(\frac{12}{19}, \frac{16}{19}, \frac{20}{19}\right)$ is the required triangle[1].

13. To find a right-angled triangle such that its area *minus* either perpendicular gives a square.

We have to find an auxiliary triangle exactly as in the last problem;

Bachet's reading is *ἀπάγεται εἰς τὸ δύο ἀριθμῶν δοθέντων τοῦ τε ἐμβαδοῦ, καὶ τῆς ἐλάσσονος τῶν περὶ τὴν ὀρθήν, αὐτοῖς ζητεῖν τετράγωνόν τινα, ὃς πολλαπλασιασθεὶς ἐπὶ ἕνα τῶν δοθέντων, καὶ προσλαβὼν τὸν ἕτερον, ποιῇ τετράγωνον.*

[1] Fermat observes that Diophantus gives only one species of triangle satisfying the condition, namely triangles similar to (3, 4, 5), but that by his (Fermat's) method an infinite number of triangles of different species can be found to satisfy the conditions, the first being derived from Diophantus' triangle, the second from the new triangle, and so on.

Suppose that the triangle (3, 4, 5) has been found satisfying the condition that

$$\xi\eta+\xi(\xi-\eta)\cdot\frac{1}{2}\xi\eta=\text{a square},$$

where ξ, η are the perpendicular sides and $\xi>\eta$.

To derive a second such triangle from the first (3, 4, 5), assume the greater of the two perpendicular sides to be 4 and the lesser $3+x$.

Then $$\xi\eta+\xi(\xi-\eta)\cdot\frac{1}{2}\xi\eta=36-12x-8x^2=\text{a square}.$$

Also $$\zeta^2=\xi^2+\eta^2=25+6x+x^2=\text{a square}.$$

We have therefore simply to solve the double-equation

$$\left.\begin{aligned} 36-12x-8x^2&=u^2 \\ 25+6x+x^2&=v^2 \end{aligned}\right\},$$

which is a matter of no difficulty. As a matter of fact, the usual method gives

$$x+3=\frac{20667}{5932289}, \text{ and the triangle is } \left(\frac{20667}{5932289}, 4, \frac{23729165}{5932289}\right).$$

this triangle is (3, 4, 5), and accordingly we assume for the triangle of the problem $(3x, 4x, 5x)$.

One condition then gives $6x^2 - 4x = \text{a square} = m^2x^2$, say $(m^2 < 6)$,

and $$x = \frac{4}{6 - m^2}.$$

The second condition gives $6x^2 - 3x = \text{a square}$; and, by substitution,

$$\frac{96}{m^4 - 12m^2 + 36} - \frac{12}{6 - m^2} = \text{a square},$$

or $$24 + 12m^2 = \text{a square}.$$

This is satisfied by $m = 1$,

whence $x = \frac{4}{5}$, and the required triangle is $\left(\frac{12}{5}, \frac{16}{5}, 4\right)$.

Or, if we do not wish to use the value 1 for m,

let $m = z + 1$, and (dividing by 4) we have

$$3m^2 + 6 = 3z^2 + 6z + 9 = \text{a square};$$

z must be found to be not greater than $\frac{13}{9}$ (in order that m^2 may be less than 6), and m will not be greater than $\frac{22}{9}$. The solution is then rational[1].

14. To find a right-angled triangle such that its area *minus* the hypotenuse or *minus* one of the perpendiculars gives a square.

Let the triangle be $(3x, 4x, 5x)$.

Therefore $$\left.\begin{array}{l} 6x^2 - 5x \\ 6x^2 - 3x \end{array}\right\} \text{are both squares.}$$

Making the latter a square $(= m^2x^2)$, we have

$$x = \frac{3}{6 - m^2} \quad (m^2 < 6).$$

[1] Diophantus having solved the problem of finding a right-angled triangle ζ, η, ξ (ζ being the hypotenuse) such that

$$\left.\begin{array}{l} \frac{1}{2}\xi\eta - \xi \\ \frac{1}{2}\xi\eta - \eta \end{array}\right\} \text{are both squares,}$$

Fermat enunciates, as susceptible of solution by his method, but otherwise very difficult, the corresponding problem of making

$$\left.\begin{array}{l} \xi - \frac{1}{2}\xi\eta \\ \eta - \frac{1}{2}\xi\eta \end{array}\right\} \text{both squares.}$$

This problem was solved by Euler (*Novi Commentarii Acad. Petropol.* 1749, II. (1751), pp. 49 sqq. = *Commentationes arithmeticae*, I. pp. 62–72).

The first equation then gives

$$\frac{54}{m^4 - 12m^2 + 36} - \frac{15}{6 - m^2} = \text{a square},$$

or $$15m^2 - 36 = \text{a square}.$$

This equation we cannot solve because 15 *is not the sum of two squares*[1]. Therefore we must change the assumed triangle.

Now (with reference to the triangle 3, 4, 5) $15m^2$ = the continued product of a square less than the area, the hypotenuse, and one perpendicular;

while 36 = the continued product of the area, the perpendicular, and the difference between the hypotenuse and the perpendicular.

Therefore we have to find a right-angled triangle (h, p, b, say) and a square (m^2) less than 6 such that

$$m^2hp - \tfrac{1}{2}pb \,.\, p(h-p) \text{ is a square.}$$

If we form the triangle from two numbers X_1, X_2 and suppose that $p = 2X_1X_2$, and if we then divide throughout by $(X_1 - X_2)^2$ which is equal to $h - p$, we must find a square $z^2 [= m^2/(X_1 - X_2)^2]$ such that

$$z^2hp - \tfrac{1}{2}pb \,.\, p \text{ is a square.}$$

The problem can be solved if X_1, X_2 *are "similar plane numbers*[2]*."*

Form the auxiliary triangle from similar plane numbers accordingly, say 4, 1. [The conditions are then satisfied[3].]

[The equation for m then becomes

$$8 \,.\, 17m^2 - 4 \,.\, 15 \,.\, 8 \,.\, 9 = \text{a square},$$

or $$136m^2 - 4320 = \text{a square}.]$$

Let[4] $m^2 = 36$. [This satisfies the equation, and 36 < area of triangle.]

[1] See p. 70 above.

[2] Diophantus states this without proof. [A "plane number" being of the form $a \,.\, b$, a plane number similar to it is of the form $\frac{m}{n} a \,.\, \frac{m}{n} b$ or $\frac{m^2}{n^2} ab$.]

The fact stated may be verified thus. We have

$$z^2 (X_1^2 + X_2^2)\, 2X_1X_2 - X_1X_2 (X_1^2 - X_2^2)\, 2X_1X_2 = \text{a square}.$$

The condition is satisfied if $z^2 = X_1X_2$, for the expression then reduces to $4X_1^2X_2^2 \,.\, X_2^2$. In that case X_1X_2 is a square, or X_1/X_2 is a square.

[3] Since $X_1 = 4$, $X_2 = 1$, we have $h = 17$, $p = 8$, $b = 15$, $z^2 = X_1X_2 = 4$, and

$$z^2hp - \frac{1}{2}pb \,.\, p = 4 \,.\, 17 \,.\, 8 - 4 \,.\, 15 \,.\, 8 = 2 \,.\, 32 = 64, \text{ a square.}$$

[4] The reason for this assumption is that, by hypothesis, $z^2 = m^2/(X_1 - X_2)^2$, or $4 = m^2/3^2$, and $m^2 = 36$.

The triangle formed from 4, 1 being (8, 15, 17), we assume $8x$, $15x$, $17x$ for the original triangle.

We now put $60x^2 - 8x = 36x^2$,

and $x = \frac{1}{3}$.

The required triangle is therefore $\left(\frac{8}{3}, 5, \frac{17}{3}\right)$.

Lemma to the following problem.

Given two numbers, if, when some square is multiplied into one of the numbers and the other number is subtracted from the product, the result is a square, another square larger than the aforesaid square can always be found which has the same property.

Given numbers 3, 11, side of square 5, say, so that $3 \cdot 25 - 11 = 64$, a square.

Let the required square be $(x+5)^2$.

Therefore

$$3(x+5)^2 - 11 = 3x^2 + 30x + 64 = \text{a square}$$
$$= (8-2x)^2, \text{ say},$$

and $x = 62$.

The side of the new square is 67, and the square itself 4489.

15. To find a right-angled triangle such that the area added to either the hypotenuse or one of the perpendiculars gives a square.

In order to guide us to a proper assumption for the required triangle, we have, in this case, to seek a triangle (h, p, b, say) and a square (m^2) such that $m^2 > \frac{1}{2}pb$, the area, and

$$m^2hp - \tfrac{1}{2}pb \cdot p(h-p) \text{ is a square.}$$

Let the triangle be formed from 4, 1, the square (m^2) being 36, as before;

but, the triangle being (8, 15, 17), *the square is not greater than the area.*

We must therefore, as in the preceding Lemma, replace 36 by a greater square.

Now $hp = 136$, and $\frac{1}{2}pb \cdot p(h-p) = 60 \cdot 8 \cdot 9 = 4320$,

so that $136m^2 - 4320 = \text{a square}$,

which is satisfied by $m^2 = 36$; and we have to find a larger square (z^2) such that

$$136z^2 - 4320 = \text{a square.}$$

Put $z = m + 6$, and we have

$$(m^2 + 12m + 36)\,136 - 4320 = \text{a square},$$

or $136m^2 + 1632m + 576 = \text{a square} = (km - 24)^2$, say.

This equation has any number of solutions; *e.g.*, putting $k = 16$, we have

$$m = 20,\quad z = 26,\quad \text{and } z^2 = 676.$$

We therefore put $(8x, 15x, 17x)$ for the original triangle, and then assume

$$60x^2 + 8x = 676x^2,$$

whence $x = \frac{1}{77}$, and the problem is solved[1].

[1] In VI. 14, 15 Diophantus has shown how to find a rational right-angled triangle ξ, η, ζ (where ζ is the hypotenuse) such that

$$(1)\quad \left.\begin{matrix} \frac{1}{2}\xi\eta - \zeta \\ \frac{1}{2}\xi\eta - \xi \end{matrix}\right\}\text{ are both squares,}$$

or

$$(2)\quad \left.\begin{matrix} \frac{1}{2}\xi\eta + \zeta \\ \frac{1}{2}\xi\eta + \xi \end{matrix}\right\}\text{ are both squares.}$$

In the *Inventum Novum*, Part 1, paragraphs 26, 40 (*Oeuvres de Fermat*, III. pp. 341–2, 349–50) is given Fermat's solution of a third case in which

$$\left.\begin{matrix} \zeta - \frac{1}{2}\xi\eta \\ \xi - \frac{1}{2}\xi\eta \end{matrix}\right\}\text{ are both squares.}$$

This depends on the *Lemma*: To find a rational right-angled triangle in which

$$\zeta\left(\xi + \frac{1}{2}\eta\right) - \frac{1}{2}\xi\eta = \text{a square}.$$

Form a right-angled triangle from $x + 1$, 1; the sides are then

$$x^2 + 2x + 2,\quad x^2 + 2x,\quad 2x + 2.$$

We must therefore have

$$(x^2 + 2x + 2)(x^2 + 3x + 1) - (x + 1)(x^2 + 2x) = \text{a square},$$

or

$$x^4 + 4x^3 + 6x^2 + 6x + 2 = \text{a square}$$
$$= (x^2 + 2x + 1)^2, \text{ say}.$$

Therefore $x = -\frac{1}{2}$, and the triangle has one of its sides $x^2 + 2x$ negative. Instead therefore of forming the triangle from $\frac{1}{2}$, 1 or from 1, 2, we form it from $x + 1$, 2 and repeat the operation. The sides are then

$$x^2 + 2x + 5,\quad x^2 + 2x - 3,\quad 4x + 4,$$

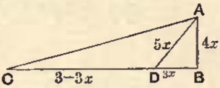
and we have

$$(x^2 + 2x + 5)(x^2 + 4x - 1) - (2x + 2)(x^2 + 2x - 3) = \text{a square},$$

or

$$x^4 + 4x^3 + 6x^2 + 20x + 1 = \text{a square}$$
$$= (1 + 10x - x^2)^2, \text{ say},$$

16. To find a right-angled triangle such that the number representing the (portion intercepted within the triangle of the) bisector of an acute angle is rational[1].



Suppose the bisector $AD = 5x$, and one segment of the base $(DB) = 3x$; therefore the perpendicular $= 4x$.

Let the whole base CB be some multiple of 3, say 3; then

$$CD = 3 - 3x.$$

But, since AD bisects the angle CAB,

$$AC : CD = AB : BD;$$

therefore the hypotenuse $AC = \frac{5}{3}(3 - 3x) = 5 - 5x$.

---

whence $x = \frac{23}{6}$, and the required auxiliary triangle is formed from $\frac{29}{6}$, 2 or from 29, 12, the sides being accordingly 985, 697, 696.

(Fermat observes that the same result is obtained by putting $y - \frac{1}{2}$ for x in the expression $x^4 + 4x^3 + 6x^2 + 6x + 2$; for we must have

$$y^4 + 2y^3 + \frac{3}{2}y^2 + \frac{5}{2}y + \frac{1}{16} = \text{a square} = \left(\frac{1}{4} + 5y - y^2\right)^2, \text{ say,}$$

whence $y = \frac{23}{12}$, so that $x = y - \frac{1}{2} = \frac{17}{12}$, and the triangle is formed from $\frac{29}{12}$, 1 or from 29, 12, as before.)

We now return to the original problem of solving

$$\left.\begin{aligned} \zeta - \frac{1}{2}\xi\eta &= u^2 \\ \xi - \frac{1}{2}\xi\eta &= v^2 \end{aligned}\right\}.$$

We assume for the required triangle $(985x, 697x, 696x)$ and we have $\frac{1}{2}\xi\eta = 242556x^2$, so that

$$\left.\begin{aligned} 985x - 242556x^2 \\ 697x - 242556x^2 \end{aligned}\right\} \text{ must both be squares.}$$

Assume that $$697x - 242556x^2 = (697x)^2,$$

and we have $$x - 348x^2 = 697x^2,$$

whence $x = \frac{1}{1045}$, and the required triangle is $\left(\frac{985}{1045}, \frac{697}{1045}, \frac{696}{1045}\right)$.

[The $985x - 242556x^2$ is a square by virtue of the sides 985, 697, 696 satisfying the conditions of the Lemma; for $985x - 242556x^2 = \frac{985}{1045} - \frac{1}{2}\cdot\frac{697 \cdot 696}{(1045)^2}$, which is a square if $985 \cdot 1045 - \frac{1}{2} \cdot 697 \cdot 696$ is a square, and $1045 = 697 + \frac{1}{2} \cdot 696$.]

[1] Why did not Diophantus propound the analogous problem "To find a right-angled triangle such that the sides are rational and the bisector of the right angle is also rational"? Evidently because he knew it to be impossible, as is clear when (a, c being the perpendiculars) the bisector is expressed as $\frac{ac}{a+c}\sqrt{2}$. (Loria, *op. cit.* p. 148 *n.*)

Therefore [by Eucl. I. 47]

$$16x^2 - 32x + 16 = 16x^2 + 9,$$

and $$x = \tfrac{7}{32}.$$

If we multiply throughout by 32, the perpendicular = **28**, the base = **96**, the hypotenuse = **100**, and the bisector = **35**.

17. To find a right-angled triangle such that the area added to the hypotenuse gives a square, while the perimeter is a cube.

Let the area be x and the hypotenuse some square *minus* x, say $16 - x$.

The product of the perpendiculars $= 2x$;

therefore, if one of them be 2, the other is x, and the perimeter = 18, *which is not a cube.*

Therefore we must find some square which, when 2 is added to it, becomes a cube[1].

[1] "Did Diophantus know that the equation $u^2 + 2 = v^3$ only admits of *one* solution $u = 5$, $v = 3$? Probably not" (Loria, *op. cit.* p. 155). The fact was noted by Fermat (on the present proposition) and proved by Euler.

Euler's proof (*Algebra*, Part II. Arts. 188, 193) is, I think, not too long to be given here. Art. 188 shows how to find x, y such that $ax^2 + cy^2$ may be a cube. Separate $ax^2 + cy^2$ into its factors $x\sqrt{a} + y\sqrt{(-c)}$, $x\sqrt{a} - y\sqrt{(-c)}$, and assume

$$x\sqrt{a} + y\sqrt{(-c)} = \{p\sqrt{a} + q\sqrt{(-c)}\}^3,$$
$$x\sqrt{a} - y\sqrt{(-c)} = \{p\sqrt{a} - q\sqrt{(-c)}\}^3,$$

the product $(ap^2 + cq^2)^3$ being a cube and equal to $ax^2 + cy^2$.

To find values for x and y, we write out the expansions of the cubes in full, and

$$x\sqrt{a} + y\sqrt{(-c)} = ap^3\sqrt{a} + 3ap^2q\sqrt{(-c)} - 3cpq^2\sqrt{a} - cq^3\sqrt{(-c)},$$
$$x\sqrt{a} - y\sqrt{(-c)} = ap^3\sqrt{a} - 3ap^2q\sqrt{(-c)} - 3cpq^2\sqrt{a} + cq^3\sqrt{(-c)},$$

whence $$x = ap^3 - 3cpq^2,$$
$$y = 3ap^2q - cq^3.$$

For example, suppose it is required to make $x^2 + y^2$ a cube. Here $a = 1$ and $c = 1$, so that $$x = p^3 - 3pq^2,$$
$$y = 3p^2q - q^3,$$

while $x^2 + y^2 = (p^2 + q^2)^3$. If now $p = 2$ and $q = 1$, we find $x = 2$ and $y = 11$, whence

$$x^2 + y^2 = 125 = 5^3.$$

Now (Art. 193) let it be required to find, if possible, in integral numbers, other squares besides 25 which, when added to 2, give cubes.

Since $x^2 + 2$ has to be made a cube, and 2 is double of a square, let us first determine the cases in which $x^2 + 2y^2$ becomes a cube. Here $a = 1$, $c = 2$, so that

$$x = p^3 - 6pq^2, \quad y = 3p^2q - 2q^3;$$

therefore, since $y = \pm 1$, we must have

$$3p^2q - 2q^3 \text{ or } q(3p^2 - 2q^2) = \pm 1;$$

consequently q must be a divisor of 1.

Let, then, $q = 1$, and we shall have $3p^2 - 2 = \pm 1$.

With the upper sign we have $3p^2 = 3$ and, taking $p = -1$, we find $x = 5$; with the lower sign we get an irrational value of p which is of no use.

Let the side of the square be $m+1$, and that of the cube $m-1$.

Therefore $m^3-3m^2+3m-1=m^2+2m+3$,

from which m is found[1] to be 4.

Hence the side of the square $=5$, and that of the cube $=3$.

Assuming now x for the area of the original triangle, $25-x$ for its hypotenuse, and 2, x for the perpendiculars, we find that the perimeter is a cube.

But (hypotenuse)$^2=$ sum of squares of perpendiculars;

therefore $x^2-50x+625=x^2+4$;

$x=\frac{621}{50}$, and the problem is solved.

18. To find a right-angled triangle such that the area added to the hypotenuse gives a cube, while the perimeter is a square.

Area x, hypotenuse some cube *minus* x, perpendiculars x, 2.

Therefore we have to find a cube which, when 2 is added to it, becomes a square.

Let the side of the cube be $m-1$.

Therefore $m^3-3m^2+3m+1=$ a square $=(1\frac{1}{2}m+1)^2$, say.

Thus $m=\frac{21}{4}$, and the cube $=(\frac{17}{4})^3=\frac{4913}{64}$.

Put now x for the area, x, 2 for the perpendiculars, and $\frac{4913}{64}-x$ for the hypotenuse;

and x is found from the equation $(\frac{4913}{64}-x)^2=x^2+4$.

[$x=\frac{24121185}{628864}$, and the triangle is $(2, \frac{24121185}{628864}, \frac{24153953}{628864})$.]

19. To find a right-angled triangle such that its area added to one of the perpendiculars gives a square, while the perimeter is a cube.

Make a right-angled triangle from some indeterminate odd number[2], say $2x+1$;

then the altitude $=2x+1$, the base $=2x^2+2x$, and the hypotenuse $=2x^2+2x+1$.

It follows that there is no square except 25 which has the required property.

Fermat says ("Relation des nouvelles découvertes en la science des nombres," *Oeuvres*, II. pp. 433–4) that it was by a special application of his method of *descente*, such as that by which he proved that *a cube cannot be the sum of two cubes*, that he proved (1) that *there is only one integral square which when increased by 2 gives a cube*, and (2) that *there are only two squares in integers which, when added to 4, give cubes.* The latter squares are 4, 121 (as proved by Euler, *Algebra*, Part II. Art. 192).

[1] See pp. 66, 67 above.

[2] This is the method of formation of right-angled triangles attributed to Pythagoras. If m is any odd number, the sides of the right-angled triangle formed therefrom are m, $\frac{1}{2}(m^2-1)$, $\frac{1}{2}(m^2+1)$, for $m^2+\left\{\frac{1}{2}(m^2-1)\right\}^2=\left\{\frac{1}{2}(m^2+1)\right\}^2$. Cf. Proclus, *Comment. on Eucl.* I. (ed. Friedlein), p. 428, 7 sqq., etc. etc.

Since the perimeter = a cube,

$$4x^2 + 6x + 2 = (4x + 2)(x + 1) = \text{a cube};$$

and, if we divide all the sides by $x + 1$, we have to make $4x + 2$ a cube.

Again, the area + one perpendicular = a square.

Therefore $\dfrac{2x^3 + 3x^2 + x}{(x+1)^2} + \dfrac{2x+1}{x+1} = \text{a square}$;

that is, $\dfrac{2x^3 + 5x^2 + 4x + 1}{x^2 + 2x + 1} = 2x + 1 = \text{a square}$.

But $4x + 2 = \text{a cube}$;

therefore we must find a cube which is double of a square; this is of course 8.

Therefore $4x + 2 = 8$, and $x = 1\frac{1}{2}$.

The required triangle is $\left(\frac{8}{5}, \frac{15}{5}, \frac{17}{5}\right)$.

20. To find a right-angled triangle such that the sum of its area and one perpendicular is a cube, while its perimeter is a square.

Proceeding as in the last problem, we have to make

$$\left.\begin{array}{l} 4x + 2 \text{ a square} \\ 2x + 1 \text{ a cube} \end{array}\right\}.$$

We have therefore to seek a square which is double of a cube; this is 16, which is double of 8.

Therefore $4x + 2 = 16$, and $x = 3\frac{1}{2}$.

The triangle is $\left(\frac{16}{9}, \frac{63}{9}, \frac{65}{9}\right)$.

21. To find a right-angled triangle such that its perimeter is a square, while its perimeter added to its area gives a cube.

Form a right-angled triangle from x, 1.

The perpendiculars are then $2x$, $x^2 - 1$, and the hypotenuse $x^2 + 1$.

Hence $2x^2 + 2x$ should be a square,

and $x^3 + 2x^2 + x$ a cube.

It is easy to make $2x^2 + 2x$ a square; let $2x^2 + 2x = m^2x^2$; therefore $x = 2/(m^2 - 2)$.

By the second condition,

$$\frac{8}{(m^2-2)^3} + \frac{8}{(m^2-2)^2} + \frac{2}{m^2-2} \text{ must be a cube},$$

i.e. $$\frac{2m^4}{(m^2-2)^3} = \text{a cube}.$$

Therefore $2m^4 =$ a cube, or $2m =$ a cube $= 8$, say.

Thus $m = 4$, $x = \frac{2}{14} = \frac{1}{7}$, $x^2 = \frac{1}{49}$.

But one of the perpendiculars of the triangle is $x^2 - 1$, and we cannot subtract 1 from $\frac{1}{49}$.

Therefore we must find another value for x greater than 1;

hence $$2 < m^2 < 4.$$

And we have therefore to find a cube such that $\frac{1}{4}$ of the square of it is greater than 2, but less than 4.

If z^3 be this cube,

$$2 < \tfrac{1}{4}z^6 < 4,$$

or $$8 < z^6 < 16.$$

This is satisfied by $z^6 = \frac{729}{64}$, or $z^3 = \frac{27}{8}$.

Therefore $m = \frac{27}{16}$, $m^2 = \frac{729}{256}$, and $x = \frac{512}{217}$, the square of which is > 1.

Thus the triangle is known $\left[\frac{1024}{217}, \frac{215055}{47089}, \frac{309233}{47089}\right]$.

22. To find a right-angled triangle such that its perimeter is a cube, while the perimeter added to the area gives a square.

(1) We must first see how, given two numbers, a triangle may be formed such that its perimeter = one of the numbers and its area = the other.

Let 12, 7 be the numbers, 12 being the perimeter, 7 the area.

Therefore the product of the two perpendiculars

$$= 14 = \frac{1}{x} \cdot 14x.$$

If then $\frac{1}{x}$, $14x$ are the perpendiculars,

$$\text{hypotenuse} = \text{perimeter} - \text{sum of perps.} = 12 - \frac{1}{x} - 14x.$$

Therefore [by Eucl. I. 47]

$$\frac{1}{x^2} + 196x^2 + 172 - \frac{24}{x} - 336x = \frac{1}{x^2} + 196x^2;$$

that is, $$172 = 336x + \frac{24}{x},$$

or $$172x = 336x^2 + 24.$$

This equation gives no rational solution, because $86^2 - 24 \cdot 336$ *is not a square.*

Now $172 = (\text{perimeter})^2 + 4$ times area,

$24 \cdot 336 = 8$ times area multiplied by $(\text{perimeter})^2$.

(2) Let now the area $= m$, and the perimeter = any number which is both a square and a cube, say 64.

Therefore $\{\frac{1}{2}(64^2+4m)\}^2-8.64^2.m$ must be a square,

or $\quad 4m^2-24576m+4194304=$ a square.

$$\left.\begin{array}{ll}\text{Therefore } m^2-6144m+1048576= \text{a square} \\ \text{Also} \qquad\qquad\qquad m+64= \text{a square}\end{array}\right\}.$$

To solve this double-equation, multiply the second by such a number as will make the absolute term the same as the absolute term in the first.

Then, if we take the difference and the factors as usual, the equations are solved.

[After the second equation is multiplied by 16384, the double-equation becomes

$$\left.\begin{array}{r}m^2-6144m+1048576= \text{a square} \\ 16384m+1048576= \text{a square}\end{array}\right\}.$$

The difference is $m^2-22528m$.

If m, $m-22528$ are taken as the factors, we find $m=7680$, which is an impossible value for the area of a right-angled triangle of perimeter 64.

We therefore take as the factors $11m$, $\frac{1}{11}m-2048$; then, when the square of half the difference is equated to the smaller of the two expressions to be made squares, we have

$$(\tfrac{60}{11}m+1024)^2=16384m+1048576,$$

and $m=\frac{39424}{225}$.

Returning now to the original problem, we put $\frac{1}{x}$, $2mx$ for the perpendicular sides of the required triangle, and we have

$$\left(64-\frac{1}{x}-2mx\right)^2=\frac{1}{x^2}+4m^2x^2,$$

which leads, when the value of m is substituted, to the equation

$$78848x^2-8432x+225=0.$$

The solution of this equation is rational, namely

$$x=\frac{527\pm 23}{9856}=\frac{25}{448} \text{ or } \frac{9}{176}.$$

Diophantus would of course use the first value, which would give $(\frac{448}{25}, \frac{176}{9}, \frac{5968}{225})$ as the required right-angled triangle. The second value of x clearly gives the same triangle.]

23. To find a right-angled triangle such that the square of its hypotenuse is also the sum of a different square and the side of that square, while the quotient obtained by dividing the square of the hypotenuse by one of the perpendiculars of the triangle is the sum of a cube and the side of the cube.

Let one of the perpendiculars be x, the other x^2.

Therefore $(\text{hypotenuse})^2 =$ the sum of a square and its side; also $\dfrac{x^4+x^2}{x} = x^3 + x =$ the sum of a cube and its side.

It remains that $x^4 + x^2$ must be a square.

Therefore $x^2 + 1 =$ a square $= (x-2)^2$, say.

Therefore $x = \frac{3}{4}$, and the triangle is found $[\frac{3}{4}, \frac{9}{16}, \frac{15}{16}]$.

24. To find a right-angled triangle such that one perpendicular is a cube, the other is the difference between a cube and its side, and the hypotenuse is the sum of a cube and its side.

Let the hypotenuse be $x^3 + x$, and one perpendicular $x^3 - x$.

Therefore the other perpendicular $= 2x^2 =$ a cube $= x^3$, say.

Thus $x = 2$, and the triangle is (6, 8, 10).

It is on Bachet's note to VI. 22 that Fermat explains his method of solving *triple-equations*, as to which see the Supplement, Section V.

[No. 20 of the problems on right-angled triangles which Bachet appended to Book VI. ("To find a right-angled triangle such that its area is equal to a given number") is the occasion of Fermat's remarkable note upon the theorem discovered by him to the effect that *The area of a right-angled triangle the sides of which are rational numbers cannot be a square number.*

This note will be given in full, with other information on the same subject, in the Supplement.]

ON POLYGONAL NUMBERS

All numbers from 3 upwards in order are polygonal, containing as many angles as they have units, *e.g.* 3, 4, 5, etc.

> "As with regard to squares it is obvious that they are such because they arise from the multiplication of a number into itself, so it was found that any polygonal multiplied into a certain number depending on the number of its angles, with the addition to the product of a certain square also depending on the number of the angles, turned out to be a square. This I shall prove, first showing how any assigned polygonal number may be found from a given side, and the side from a given polygonal number. I shall begin by proving the preliminary propositions which are required for the purpose."

1. If there are three numbers with a common difference, then 8 times the product of the greatest and middle + the square of the least = a square, the side of which is the sum of the greatest and twice the middle number.

Let the numbers be AB, BC, BD in the figure, and we have to prove $8AB \cdot BC + BD^2 = (AB + 2BC)^2$.

By hypothesis $AC = CD$, $AB = BC + CD$, $BD = BC - CD$.

Now $8AB \cdot BC = 4AB \cdot BC + (4BC^2 + 4BC \cdot CD)$.

Therefore $8AB \cdot BC + BD^2$

$$= 4AB \cdot BC + 4BC^2 + (4BC \cdot CD + BD^2)$$

$$= 4AB \cdot BC + 4BC^2 + AB^2, \qquad \text{[Eucl. II. 8]}$$

and we have to see how $AB^2 + 4AB \cdot BC + 4BC^2$ can be made a square.

[Diophantus does this by producing BA to E, so that $AE = BC$, and then proving that

$$AB^2 + 4AB \cdot BC + 4BC^2 = (BE + EA)^2.]$$

It is indeed obvious that

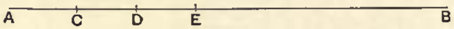
$$AB^2 + 4AB \cdot BC + 4BC^2 = (AB + 2BC)^2.$$

2. If there are any numbers, as many as we please, in A.P., the difference between the greatest and the least is equal to the common difference multiplied by the number of terms less one.

[That is, if in an A.P. the first term is a, the common difference b and the greatest term l, n being the number of terms, then

$$l-a=(n-1)\,b.]$$

Let AB, BC, BD, BE have a common difference.



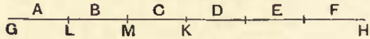
Now AC, CD, DE are all equal.

Therefore $EA = AC \times$ (number of terms AC, CD, DE)
$= AC \times$ (number of terms in series -1).

3. If there are as many numbers as we please in A.P., then (greatest + least) × number of terms = double the sum of the terms.

[That is, with the usual notation, $2s=n\,(l+a)$.]

(1) Let the numbers be A, B, C, D, E, F, the number of them being *even*.



Let GH contain as many units as there are numbers, and let GH, being even, be bisected at K. Divide GK into units at L, M.

Since $\qquad F-D=C-A,$

$\qquad\qquad F+A=C+D.$

But $\qquad F+A=(F+A)\,.\,GL$;

therefore $\qquad C+D=(F+A)\,.\,LM.$

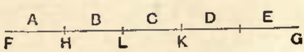
Similarly $\qquad E+B=(F+A)\,.\,MK.$

Therefore, by addition,

$$A+B+C+D+E+F=(F+A)\,.\,GK.$$

Therefore $\qquad 2\,(A+B+\ldots)=2\,(F+A)\,.\,GK$
$\qquad\qquad\qquad =(F+A)\,.\,GH.$

(2) Let the number of terms be *odd*, the terms being A, B, C, D, E.



Let there be as many units in FG as there are terms, so that there is an odd number of units.

Let FH be one of them; bisect HG at K, and divide HK into units, at L.

Since $E - C = C - A,$

$$E + A = 2C = 2C \,.\, LK.$$

Similarly $B + D = 2C \,.\, LH.$

Therefore $A + E + B + D = 2C \,.\, HK$

$$= C \,.\, HG.$$

Also $C = C \,.\, HF;$

therefore, by addition,

$$A + B + C + D + E = C \,.\, FG;$$

and, since $2C = A + E,$

$$2(A + B + C + D + E) = (A + E) \,.\, FG.$$

4. If there are as many numbers as we please beginning with 1 and increasing by a common difference, then the sum of all × 8 times the common difference + the square of (common difference − 2) = a square, the side of which diminished by 2 = the common difference multiplied by a number which when increased by 1 is double of the number of terms.

[The A.P. being $1, 1 + b, \ldots 1 + (n-1)b$, and s the sum, we have to prove that

$$s \,.\, 8b + (b-2)^2 = \{b(2n-1) + 2\}^2,$$

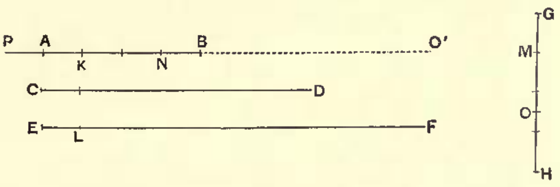
i.e. $8bs = 4b^2n^2 - 4(b-2)nb,$

or $2s = bn^2 - (b-2)n$

$$= n\{2 + (n-1)b\}.$$

The proof being cumbrous, I shall add the generalised algebraic equivalent in a column parallel to the text.]

Let AB, CD, EF be the terms in A.P. after 1.	$1 + b, 1 + 2b, 1 + 3b, \ldots.$



Let GH contain as many units as there are terms including 1.	n
Difference between EF and 1 = (diff. between AB and 1) × (GH − 1). [Prop. 2]	$l - 1 = (n-1)b$

Put AK, EL, GM each equal to unity.

Therefore $LF = KB \cdot MH$.

Make $KN = 2$, and we have to inquire whether

$$(\text{sum of terms}) \times 8KB + NB^2 = \{2 + KB(GH + HM)\}^2.$$

Call the expression on the left-hand side X.

Now sum of terms

$$= \tfrac{1}{2}(FE + EL) \cdot GH \quad \text{[Prop. 3]}$$
$$= \tfrac{1}{2}(LF + 2EL) \cdot GH$$
$$= \tfrac{1}{2}(KB \cdot MH \cdot GH + 2GH),$$

since $LF = KB \cdot MH$ [above].

$$s = \tfrac{1}{2}(l + 1)n$$
$$= \tfrac{1}{2}(l - 1 + 2)n$$
$$= \tfrac{1}{2}\{(n-1)bn + 2n\}$$

Bisecting MH at O, we have

(sum of terms)

$$= KB \cdot GH \cdot HO + GH.$$

We have therefore to inquire whether

$$(KB \cdot GH \cdot HO + GH) \cdot 8KB + NB^2$$

is a square.

$$X = bn \cdot \frac{n-1}{2} \cdot 8b + 8bn + (b-2)^2$$

Now $KB \cdot GH \cdot HO \cdot 8KB$

$$= 8GH \cdot HO \cdot KB^2$$
$$= 4GH \cdot HM \cdot KB^2.$$

Is then

$$4GH \cdot HM \cdot KB^2 + 8KB \cdot GH + NB^2$$

a square?

$$= 4n(n-1)b^2 + 8bn + (b-2)^2$$

Now $8GH \cdot KB$

$$= 4GM \cdot KB + 4(GH + HM)KB.$$

Also $4GM \cdot KB = 2NK \cdot KB$;

and, adding NB^2, the right-hand side becomes $KB^2 + KN^2$. [Eucl. II. 7]

Is then $4GH \cdot HM \cdot KB^2$

$$+ 4(GH + HM)KB + KB^2 + KN^2$$

a square?

$$= 4n(n-1)b^2 + 4\{n + (n-1)\}b + b^2 + 2^2$$

Again, $KB^2 + 4GH \cdot HM \cdot KB^2$

$$= GM^2 \cdot KB^2 + 4GH \cdot HM \cdot KB^2$$
$$= (GH + HM)^2 \cdot KB^2. \quad \text{[Eucl. II. 8]}$$

Is then $(GH + HM)^2 \cdot KB^2$

$$+ 4(GH + HM)KB + KN^2$$

a square?

$$= (n + n - 1)^2 b^2 + 4\{n + (n-1)\}b + 2^2$$

Make the number NO' equal to

$$(GH + HM) \cdot KB;$$

thus $(GH+HM)^2 \,.\, KB^2 = NO'^2$, as will be proved later[1].

Also $4NO' = 2 \,.\, NO' \,.\, NK$, since $NK = 2$.

Is then $NO'^2 + NK^2 + 2NO' \,.\, NK$ a square?

Yes; it is the square on KO'.

And

$O'K - 2 = NO' = KB(GH+HM)$, while $GH + HM + 1 =$ (twice number of terms).

Thus the proposition is proved.

Therefore

$$X = [(n+n-1)\delta + 2]^2$$
$$= [(2n-1)\delta + 2]^2.$$

"The above being premised, I say that,

[5] If there be as many terms as we please in A.P. beginning from 1, the sum of the terms is polygonal; for it has as many angles as the common difference increased by 2 contains units, and its side is the number of the terms set out including 1."

The numbers being as set out in the figure of Prop. 4, we have, by that proposition,

$$(\text{sum of terms}) \,.\, 8KB + NB^2 = KO'^2.$$

Taking another unit AP, we have $KP = 2$, while $KN = 2$; therefore PB, BK, BN are in A.P., so that

$$8PB \,.\, BK + NB^2 = (PB + 2KB)^2; \quad \text{[Prop. 1]}$$

and $\quad PB + 2KB - 2 = PB + 2KB - PK = 3KB$,

while $3 + 1 = 2 \,.\, 2$, or 3 is one less than the double of 2.

Now, since the sum of the terms of the progression

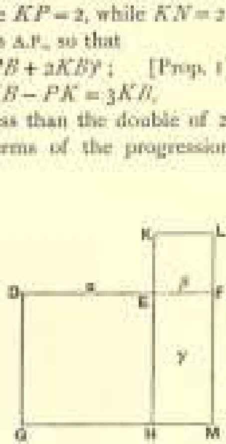
[1] *Deferred lemma.*

To prove that $(GH+HM)^2 \,.\, KB^2$
$= [(GH+HM) \,.\, KB]^2$.

Let $\alpha = GH + HM$,
$\beta = KB$,
$\gamma = (GH+HM) \,.\, KB$.

Place DE (equal to α) and EF (equal to β) in a straight line.

Describe squares DH, EL on DE, EF and complete the figure.



Then $\quad DE : EF = DH : HF$,
and $\quad HE : EK = HF : EL$.

Therefore HF is a mean proportional between the two squares,

that is $\quad DH \,.\, FK = HF^2$,
or $\quad \alpha^2 \,.\, \beta^2 = (\alpha\beta)^2$.

including unity satisfies the same formula[1] [literally "does the same problem"] as *PB* does,

while *PB* is *any* number and is also always a polygonal, the first after unity (for *AP* is a unit and *AB* is the term next after it), and has 2 for its side,

it follows that the sum of all the terms of the progression is a polygonal with the same number of angles as *PB*, the number of its angles being the same as the number of units in the number which is greater by 2, or *PK*, than the common difference *KB*, and that its side is *GH* which is equal to the number of terms including 1.

And thus is demonstrated what is stated by Hypsicles in his definition, namely, that,

"If there are as many numbers as we please beginning from 1 and increasing by the same common difference, then, when the common difference is 1, the sum of all the terms is a triangular number; when 2, a square; when 3, a pentagonal number [and so on]. And the number of the angles is called after the number exceeding the common difference by 2, and the side after the number of terms including 1."

[In other words, if there be an arithmetical progression

$$1,\ 1+b,\ 1+2b, \dots 1+(n-1)b,$$

the sum of the n terms, or $\frac{1}{2}n\{2+(n-1)b\}$, is the nth polygonal number which has $(b+2)$ angles.]

Hence, since we have triangles when the common difference is 1, the sides of the triangles will be the greatest term in each case, and the product of the greatest term and the greatest term increased by 1 is double the triangle.

[1] Nesselmann (pp. 475–6), exhibits this result thus.

Take the A.P. $\quad 1,\ 1+b,\ 1+2b, \dots 1+(n-1)b.$

If s is the sum, $\quad 8sb+(b-2)^2=\{b(2n-1)+2\}^2.$

If now we take the three terms $b-2$, b, $b+2$, also in A.P.,

$$8b(b+2)+(b-2)^2=\{(b+2)+2b\}^2$$
$$=(3b+2)^2.$$

Now $b+2$ is the sum of the first two terms of the first series, and corresponds therefore to s when $n=2$; and $3=2\cdot2-1$, so that 3 corresponds to $2n-1$.

Hence s and $b+2$ are subject to the same law; and therefore, as $b+2$ is a polygonal number with $b+2$ angles, s is also a polygonal number (the nth) with $b+2$ angles.

And, since PB is a polygonal with as many angles as there are units in it,

and $8PB.(PB-2)+(PB-4)^2=$ a square (from above, BK being equal to $PB-2$, and NB to $PB-4$),

the definition of polygonal numbers will be as follows:

Every polygonal multiplied by 8 times (number of angles -2) + square of (number of angles -4) = a square[1].

The Hypsiclean definition and the new one being thus simultaneously proved, it remains to show how, when the side is given, the prescribed polygonal is found.

For, having given the side GH and the number of angles, we know KB.

Therefore $(GH+HM)KB$, which is equal to NO', is also given; therefore $KO'(=NO'+NK$ or $NO'+2)$ is given.

Therefore KO'^2 is given;

and, subtracting from it the given square on NB, we obtain the remaining term which is equal to the required polygonal multiplied by $8KB$. Thus the required polygonal can be found.

Similarly, given the polygonal number, we can find its side GH. Q. E. D.

Rules for practical use.

(1) *To find the number from the side.*

Take the side, double it, subtract 1, and multiply the remainder by (number of angles -2). Add 2 to the product; and from the square of the sum subtract the square of (number of angles -4). Dividing the remainder by 8 times (number of angles -2), we have the required number.

[1] Hultsch points out (art. Diophantos in Pauly-Wissowa's *Real-Encyclopädie der classischen Altertumswissenschaften*) that this formula

$$8P(a-2)+(a-4)^2=\text{a square}$$

shows that Diophantus intended it to be applied not only to cases where a is greater than 4 but also where $a=4$ or less. For 36, as Diophantus must have known, besides being the second 36-gon, is also a triangle, a square, and a 13-gon, inasmuch as

$$8.36(3-2)+(3-4)^2=289=17^2,$$
$$8.36(4-2)+(4-4)^2=576=24^2,$$
$$8.36(13-2)+(13-4)^2=3249=57^2.$$

And indeed it is evident from Def. 9 of the *Arithmetica* that $(3-4)^2=1$, while it is equally obvious that $(4-4)^2=0$.

[If P be the nth a-gonal number,

$$P \,.\, 8(a-2) + (a-4)^2 = \{2 + (2n-1)(a-2)\}^2,$$

or
$$P = \frac{\{2+(2n-1)(a-2)\}^2 - (a-4)^2}{8(a-2)}.]$$

(2) *To find the side from the number.*

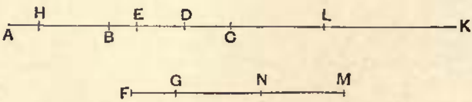
Multiply the number by 8 times (number of angles -2); add to the product the square of (number of angles -4). We thus get a square. Subtract 2 from the side of this square and divide the remainder by (number of angles -2). Add 1 to the quotient, and half the result gives the side required[1].

$$\left[n = \frac{1}{2}\left(\frac{\sqrt{\{P\,.\,8(a-2)+(a-4)^2\}}-2}{a-2}+1\right).\right]$$

Given a number, to find in how many ways it can be polygonal.

Let AB be the given number, BC the number of angles, and in BC take $CD = 2$, $CE = 4$.

[*Algebraical equivalent.*]
Number $AB = P$.
Number of angles $BC = a$.



Since the polygonal AB has BC angles,

(1) $8AB\,.\,BD + BE^2 =$ a square $= FG^2$, say.

Cut off AH equal to 1;

therefore $8AB\,.\,BD$
$= 4AH\,.\,BD + 4(AB+BH)BD$.

Make DK equal to $4(AB+BH)$, and for $4AH\,.\,BD$ put $2BD\,.\,DE$.

$8P(a-2) + (a-4)^2$
$= \{2+(2n-1)(a-2)\}^2$
$= X^2$, say.

But $8P(a-2)$
$= 4(a-2) + 4(2P-1)(a-2)$
$= 2(a-2)\,.\,2 + 4(2P-1)(a-2)$.
$DK = 4(2P-1)$

[1] Fermat has the following note. "A very beautiful and wonderful proposition which I have discovered shall be set down here without proof. *If, in the series of natural numbers beginning with* 1, *any number n be multiplied into the next following,* $n+1$, *the product is twice the nth triangular number; if n be multiplied into the* $(n+1)$*th triangular number, the product is three times the nth tetrahedral number; if n be multiplied into the* $(n+1)$*th tetrahedral number, the product is four times the nth triangulo-triangular number* [*figured number of 4th order*]; *and so on, ad infinitum.* I do not think there can be, in the theory of numbers, any theorem more beautiful or more general. The margin is too small, and I am not at liberty, to give the proof." (Cf. Letter to Roberval of 4 November 1636, *Oeuvres de Fermat*, II. pp. 84, 85.) For a proof, see Wertheim's Diophantus, pp. 318–20.

Therefore

(2) $FG^2 = KD.DB + 2BD.DE + BE^2$,

$$X^2 = 4(2P-1)(a-2) + 2(a-2).2 + (a-4)^2$$

(3) $\quad = KD.DB + BD^2 + DE^2$, [Eucl. II. 7]

$$= 4(2P-1)(a-2) + (a-2)^2 + 2^2$$

(4) $\quad = KB.BD + DE^2$. [Eucl. II. 1]

$$= \{4(2P-1) + a - 2\}(a-2) + 2^2$$

But, since $DK = 4(AB + BH)$,

$$DK > 4AH > 4,$$

and $\quad DC =$ half 4 or 2;

therefore $\quad CK > CD$.

Therefore, if DK be bisected at L, L falls between C and K.

$$DL = 2(2P-1)$$

$$[BL = 2(2P-1) + a - 2]$$

And, since DK is bisected at L,

$$KB.BD + LD^2 = LB^2,$$

whence $\quad KB.BD = LB^2 - LD^2$.

Therefore, by (4) above,

(5) $\quad FG^2 = BL^2 - LD^2 + DE^2$,

$$X^2 = \{2(2P-1) + a - 2\}^2 - \{2(2P-1)\}^2 + 2^2$$

or $\quad FG^2 + DL^2 = BL^2 + DE^2$,

(6) or $\quad LD^2 - DE^2 = LB^2 - FG^2$.

$$\{2(2P-1)\}^2 - 2^2 = \{2(2P-1) + a - 2\}^2 - X^2$$

Again, since $ED = DC$, and DC is produced to L,

$$EL.LC + CD^2 = DL^2;$$

$$[EL = 2(2P-1) + 2,$$
$$CL = 2(2P-1) - 2]$$

therefore $\quad EL.LC = DL^2 - DC^2$

$$= DL^2 - DE^2$$

(7) $\quad = LB^2 - FG^2$.

$$\{2(2P-1)+2\}\{2(2P-1)-2\} = \{2(2P-1) + a - 2\}^2 - X^2$$

Put $\quad FM = BL \quad$ (for $\quad BL > FG$, since $\quad FG^2 + DL^2 = BL^2 + ED^2$, while $\quad DL^2 > ED^2$).

$$FM = 2(2P-1) + a - 2$$

Therefore $FM^2 - FG^2 = EL.LC$.

Now, DK being bisected at L and being equal to $4(AB + BH)$,

$$DL = 2(AB + BH).$$

And $\quad DC = 2AH$.

Therefore $\quad CL = 4BH$,

$$CL = 4(P-1)$$

or $\quad BH = \frac{1}{4}CL$.

But $\quad AH (= 1) = \frac{1}{4}EC$;

therefore $\quad AB = \frac{1}{4}EL$,

$$EL = 4P$$

while $\quad BH = \frac{1}{4}CL$.

Therefore $AB.BH = \frac{1}{16}EL.LC$,

or $\quad EL.LC = 16AB.BH$.

(8) Therefore

$$16AB \,.\, BH = MF^2 - FG^2$$

(9) $$= GM^2 + 2FG \,.\, GM.$$

Therefore GM is even.

Let GM be bisected at N ....

....

$$16P(P-1)$$
$$= \{2(2P-1) + a - 2\}^2 - X^2$$
$$= \{2(2P-1) + a - 2 - X\}^2$$
$$+ 2X\{2(2P-1) + a - 2 - X\}$$
$$[= \{2(2P-2) - 2(n-1)(a-2)\}^2$$
$$+ 2\{2 + (2n-1)(a-2)\}$$
$$\times \{2(2P-2) - 2(n-1)(a-2)\}]$$

[Here the fragment ends, and the question of course arises whether Diophantus ever actually solved the problem of finding in how many different ways a given number can be a polygonal. Tannery went so far as to call the whole of the fragment, from and including the enunciation of the problem, the "vain attempt of a commentator" to solve it[1]. Wertheim[2] has however shown grounds for thinking that Diophantus did solve the problem and that the fragment is a genuine part of his argument leading to that result. The equation

$$8P(a-2) + (a-4)^2 = \{2 + (2n-1)(a-2)\}^2$$

easily reduces (by algebra) to

$$8P(a-2) = 4n(a-2)\{2 + (n-1)(a-2)\},$$

or $$2P = n\{2 + (n-1)(a-2)\}.$$

Wertheim has shown how this result can be obtained by a continuation of the work, from the point where the fragment leaves off, in the same geometrical form which is used up to that point[3], and how, when the

[1] Dioph. I. pp. 476–7, notes.

[2] *Zeitschrift für Math. u. Physik*, hist. litt. Abtheilung, 1897, pp. 121–6.

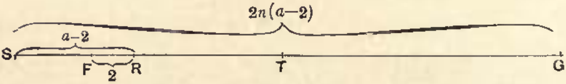
[3] The only thing, so far as I can see, tending to raise doubt as to the correctness of this restoration is the fact that, supposing it to be required to prove geometrically, from the geometrical equivalent of

$$8P(a-2) + (a-4)^2 = \{2 + (2n-1)(a-2)\}^2,$$

that $$2P = n\{2 + (n-1)(a-2)\},$$

it can be done much more easily than it is in Diophantus' proposition as extended by Wertheim.

For let $FG = 2 + (2n-1)(a-2)$. Cut off FR equal to 2, and produce RF to S so that $RS = a - 2$.



We have now $$8P \,.\, SR = FG^2 - SF^2$$
$$= (SG - SF)^2 - SF^2$$
$$= SG^2 - 2SG \,.\, SF.$$

Bisect SG at T, and divide out by 4;

therefore $$2P \,.\, SR = ST^2 - ST \,.\, SF$$
$$= ST(ST - SF)$$
$$= ST \,.\, FT$$
$$= ST \,.\, (FR + RT).$$

Now $ST = n \,.\, SR$, and $FR = 2$, while $RT = (n-1) \,.\, SR = (n-1)(a-2)$.

It follows that $$2P = n\{2 + (n-1)(a-2)\}.$$

formula is thus obtained, it can be used for the purpose of finding the number of ways in which P can be a polygonal number. The portion of the geometrical argument which has to be supplied is, it is true, somewhat long; and its length and difficulty may, as Wertheim suggests, account for the copyist having failed, as it were, to see his way through it and having stopped through discouragement when he had lost his bearings.

I shall now reproduce Wertheim's suggested restoration of the rest of the problem. The figure requires some extension, and I accordingly give a new one after Wertheim.

The last step in the above fragment is

(9) $2FG \cdot GM + GM^2 = 16AB \cdot BH$.	$2\{2+(2n-1)(a-2)\}$ $\{2(2P-2)-2(n-1)(a-2)\}$ $+\{2(2P-2)-2(n-1)(a-2)\}^2$ $=16P(P-1)$
Bisect GM in N, so that $GN = NM$.	$GN = NM$ $= 2(P-1)-(n-1)(a-2)$
Therefore, if we divide by 4, (10) $FG \cdot GN + GN^2 = 4AB \cdot BH$,	$\{2+(2n-1)(a-2)\}$ $\{2(P-1)-(n-1)(a-2)\}$ $+\{2(P-1)-(n-1)(a-2)\}^2$ $=4P(P-1)$
or (11) $FN \cdot NG = 4AB \cdot BH$.	$\{2P+n(a-2)\}$ $\{2(P-1)-(n-1)(a-2)\}$ $=4P(P-1)$
Put now $FR = 2AB$, and $RS = GN$, so that $GS = RN$, and we have $FS = FR - RS = 2AB - RS$,	$FS = 2P$ $-\{2(P-1)-(n-1)(a-2)\}$ $=2+(n-1)(a-2)$
$FN = FR + RN = 2AB + RN$, $GN = RS = 2AB - FS$. Substituting in (11), we have (12) $(2AB+RN)(2AB-FS) = 4AB \cdot BH$.	$FN = 2P+n(a-2)$, from above $GN = 2(P-1)-(n-1)(a-2)$ $RN = FN - 2AB = n(a-2)$ $\{2P+n(a-2)\}$ $\{2(P-1)-(n-1)(a-2)\}$ $=4P(P-1)$

or (13) $4AB^2 - 2AB(FS - RN) - RN \,.\, FS = 4AB^2 - 4AB \,.\, AH.$	$4P^2 - 2P\{(n-1)(a-2)+2-n(a-2)\} - n(a-2)\{(n-1)(a-2)+2\} = 4P^2 - 4P$
Therefore (14) $2AB(FS - RN) + RN \,.\, FS = 4AB \,.\, AH,$	$2P\{2-(a-2)\} + n(a-2)\{(n-1)(a-2)+2\} = 4P$
or (15) $2AB(2AH + RN - FS) = RN \,.\, FS.$	$2P(a-2) = n(a-2)\{(n-1)(a-2)+2\}$

Now
$$\begin{aligned} RN &= FN - FR = FM - NM - FR = FM - \tfrac{1}{2}GM - FR \\ &= BL - \tfrac{1}{2}GM - 2AB = BD + \tfrac{1}{2}DK - \tfrac{1}{2}GM - 2AB \\ &= BD + 2AB + 2BH - \tfrac{1}{2}GM - 2AB \\ &= BD + 2BH - \tfrac{1}{2}GM, \end{aligned}$$
and
$$\begin{aligned} FS &= FR - RS \\ &= 2AB - \tfrac{1}{2}GM. \end{aligned}$$
Therefore
$$\begin{aligned} RN - FS &= BD + 2BH - 2AB \\ &= BD - 2AH, \end{aligned}$$
and
$$RN - FS + 2AH = BD.$$
Again, we have
$$\begin{aligned} RN &= BD + 2BH - \tfrac{1}{2}GM = BD + 2BH - \tfrac{1}{2}BL + \tfrac{1}{2}FG \\ &= BD + 2BH - \tfrac{1}{2}BD - \tfrac{1}{2}DL + \tfrac{1}{2}FG \\ &= \tfrac{1}{2}BD + 2BH - \tfrac{1}{2}DL + \tfrac{1}{2}FG \\ &= \tfrac{1}{2}BD + 2BH - (AB + BH) + \tfrac{1}{2}FG \\ &= \tfrac{1}{2}BD + BH - AB + \tfrac{1}{2}FG \\ &= \tfrac{1}{2}BD - AH + \tfrac{1}{2}FG \\ &= \tfrac{1}{2}(BD + FG - 2AH). \end{aligned}$$
But, from the rule just preceding this proposition,
$$FG = BD(2n-1) + 2\,;$$
therefore
$$BD + FG = 2n \,.\, BD + 2,$$
or
$$BD + FG - 2AH = 2n \,.\, BD\,;$$
therefore
$$RN = n \,.\, BD.$$
Accordingly the equation (15) above becomes

(16) $2AB \,.\, BD = n \,.\, BD \,.\, FS,$	$2P(a-2) = n(a-2)\{(n-1)(a-2)+2\}$
or (17) $2AB = n \,.\, FS.$	$2P = n\{(n-1)(a-2)+2\}$

Thus the double of any polygonal number must be divisible by its side, and the quotient is the number arrived at by adding 2 to the product of (side $-$ 1) and (number of angles $-$ 2).

For a triangular number the quotient is $n+1$, and is therefore greater than the side; and, as the quotient increases by $n-1$ for every increase of 1 in the number of angles (a), it is always greater than the side.

We can therefore use the above formula (17) to find the number of ways in which a given number P can be a polygonal number. Separate $2P$ into two factors in all possible ways, excluding $1 \cdot 2P$. Take the smaller factor as the side (n). Then take the other factor, subtract 2 from it, and divide the remainder by $(n-1)$. If $(n-1)$ divides it without a remainder, the particular factors taken answer the purpose, and the quotient increased by 2 gives the number of angles (a). If the second factor diminished by 2 is not divisible by $(n-1)$ without a remainder, the particular division into factors is useless for the purpose. The number of ways in which P can be a polygonal is the number of pairs of factors which answer the purpose. There is always one pair of factors which will serve, namely 2 and P itself.

The process of finding pairs of factors is shortened by the following considerations.

$$2P = n\{(n-1)(a-2)+2\};$$

therefore
$$2P/n = 4 + an - a - 2n,$$

and
$$a = 2 + \frac{2(P-n)}{n(n-1)};$$

therefore not only $2P/n$ but also $\dfrac{2(P-n)}{n(n-1)}$ must be a whole number and, as a is not less than 3,

$$\frac{2(P-n)}{n(n-1)} > \text{or} = 1,$$

and consequently

$$n < \text{or} = \frac{-1+\sqrt{(1+8P)}}{2}.$$

Thus in choosing values for the factor n we need not go beyond that shown in the right-hand expression.

Example 1. In what ways is 325 a polygonal number?

Here $-1+\sqrt{(1+8P)} = -1+\sqrt{(2601)} = 50$. Therefore n cannot be greater than 25. Now $2 \cdot 325 = 2 \cdot 5 \cdot 5 \cdot 13$, and the only possible values for n are therefore 2, 5, 10, 13, 25. The corresponding values for a are shown in the following table.

n	2	5	10	13	25
a	325	34	9	6	3

Example 2. $P = 120$.

n	2	3	4	5	6	8	10	12	15
a	120	41	—	—	—	6	—	—	3